

Introduction to vectors and scalars



We come into contact with many physical quantities in the natural world on a daily basis. For example, things like time, mass, weight, force, and electric charge, are physical quantities with which we are all familiar. We know that time passes and physical objects have mass. Things have weight due to gravity. We exert forces when we open doors, walk along the street and kick balls. We experience electric charge directly through static shocks in winter and through using anything which runs on electricity.

There are many physical quantities in nature, and we can divide them up into two broad groups called **vectors** and **scalars**. See introductory video: (📺) Video: VPgao at www.everythingscience.co.za

Scalars and vectors



Scalars are physical quantities which have only a number value or a size (magnitude). A scalar tells you **how much** of something there is.

DEFINITION: Scalar

A scalar is a physical quantity that has only a magnitude (size).

For example, a person buys a tub of margarine which is labelled with a mass of 500 g. The mass of the tub of margarine is a scalar quantity. It only needs one number to describe it, in this case, 500 g.

Vectors are different because they are physical quantities which have a size *and* a direction. A vector tells you **how much** of something there is *and* **which direction** it is in.

DEFINITION: Vector

A vector is a physical quantity that has both a *magnitude* and a *direction*.

For example, a car is travelling east along a freeway at $100 \text{ km} \cdot \text{h}^{-1}$. What we have here is a vector called the velocity. The car is moving at $100 \text{ km} \cdot \text{h}^{-1}$ (this is the magnitude) and we know where it is going – east (this is the direction). These two quantities, the speed *and* direction of the car, (a magnitude and a direction) together form a vector we call velocity.

Examples of scalar quantities:

- **mass** has only a value, no direction
- **electric charge** has only a value, no direction

Examples of vector quantities:

- **force** has a value and a direction. You push or pull something with some strength (magnitude) in a particular direction
- **weight** has a value and a direction. Your weight is proportional to your mass (magnitude) and is always in the direction towards the centre of the earth.

Exercise 20 - 1

Classify the following as vectors or scalars

1. length
2. force
3. direction
4. height
5. time
6. speed
7. temperature



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Vector notation



Vectors are different to scalars and must have their own notation. There are many ways of writing the symbol for a vector. In this book vectors will be shown by symbols with an arrow pointing to the right above it. For example, $\vec{F}$, $\vec{W}$ and $\vec{v}$ represent the *vectors* of force, weight and velocity, meaning they have both a magnitude *and* a direction.

Sometimes just the magnitude of a vector is needed. In this case, the arrow is omitted. For the case of the force vector:

- $\vec{F}$ represents the force vector
- F represents the magnitude of the force vector

Graphical representation of vectors



Vectors are drawn as arrows. An arrow has both a magnitude (how long it is) and a direction (the direction in which it points). The starting point of a vector is known as the *tail* and the end point is known as the *head*.

▶ See video: VPgdw at www.everythingscience.co.za

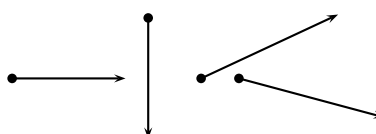


Figure 20.2: Examples of vectors

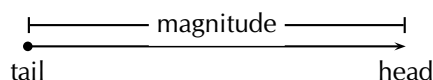


Figure 20.3: Parts of a vector

Directions



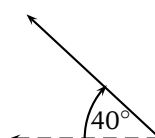
There are many acceptable methods of writing vectors. As long as the vector has a magnitude and a direction, it is most likely acceptable. These different methods come from the different methods of representing a direction for a vector.

Relative Directions

The simplest way to show direction is with relative directions: to the left, to the right, forward, backward, up and down.

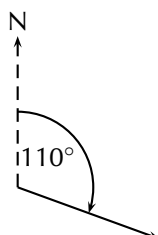
Compass Directions

Another common method of expressing directions is to use the points of a compass: North, South, East, and West. If a vector does not point exactly in one of the compass directions, then we use an angle. For example, we can have a vector pointing 40° North of West. Start with the vector pointing along the West direction (look at the dashed arrow below), then rotate the vector towards the north until there is a 40° angle between the vector and the West direction (the solid arrow below). The direction of this vector can also be described as: W 40° N (West 40° North); or N 50° W (North 50° West).



Bearing

A further method of expressing direction is to use a *bearing*. A bearing is a direction relative to a fixed point. Given just an angle, the convention is to define the angle clockwise with respect to North. So, a vector with a direction of 110° has been rotated clockwise 110° relative to North. A bearing is always written as a three digit number, for example 275° or 080° (for 80°).



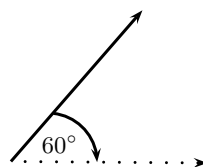
Exercise 20 - 2

- Classify the following quantities as scalars or vectors:
 - 12 km
 - 1 m south
 - $2 \text{ m} \cdot \text{s}^{-1}$, 45°
 - 075° , 2 cm
 - $100 \text{ k} \cdot \text{h}^{-1}$, 0°
- Use two different notations to write down the direction of the vector in each of the following diagrams:

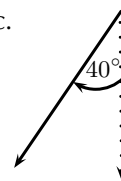
a.



b.



c.

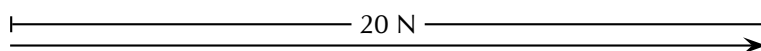


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Drawing Vectors



In order to draw a vector accurately we must represent its magnitude properly and include a reference direction in the diagram. A scale allows us to translate the length of the arrow into the vector's magnitude. For instance if one chooses a scale of $1 \text{ cm} = 2 \text{ N}$ (1 cm represents 2 N), a force of 20 N towards the East would be represented as an arrow 10 cm long pointing towards the right. The points of a compass are often used to show direction or alternatively an arrow pointing in the reference direction.



Method: Drawing Vectors

- Decide upon a scale and write it down.

2. Decide on a reference direction
3. Determine the length of the arrow representing the vector, by using the scale.
4. Draw the vector as an arrow. Make sure that you fill in the arrow head.
5. Fill in the magnitude of the vector.

Example 1: Drawing vectors I**QUESTION**

Draw the following vector quantity: $\vec{v} = 6 \text{ m} \cdot \text{s}^{-1} \text{ North}$

SOLUTION

Step 1 : **Decide on a scale and write it down.**

$$1 \text{ cm} = 2 \text{ m} \cdot \text{s}^{-1}$$

Step 2 : **Decide on a reference direction**



North will point up the page

Step 3 : **Determine the length of the arrow at the specific scale.**

$$\text{If } 1 \text{ cm} = 2 \text{ m} \cdot \text{s}^{-1}, \text{ then } 6 \text{ m} \cdot \text{s}^{-1} = 3 \text{ cm}$$

Step 4 : **Draw the vector as an arrow.**

$$\text{Scale used: } 1 \text{ cm} = 2 \text{ m} \cdot \text{s}^{-1}$$



$$6 \text{ m} \cdot \text{s}^{-1}$$

Example 2: Drawing vectors 2

QUESTION

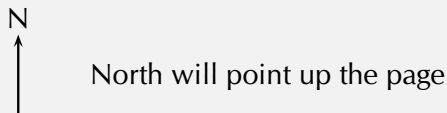
Draw the following vector quantity: $\vec{s} = 16 \text{ m east}$

SOLUTION

Step 1 : **Decide on a scale and write it down.**

$$1 \text{ cm} = 4 \text{ m}$$

Step 2 : **Decide on a reference direction**



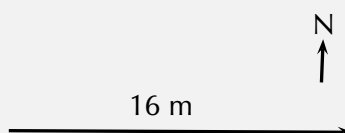
Step 3 : **Determine the length of the arrow at the specific scale.**

$$\text{If } 1 \text{ cm} = 4 \text{ m, then } 16 \text{ m} = 4 \text{ cm}$$

Step 4 : **Draw the vector as an arrow**

Scale used: $1 \text{ cm} = 4 \text{ m}$

Direction = East



Exercise 20 - 3

Draw each of the following vectors to scale. Indicate the scale that you have used:

1. 12 km south
2. 1,5 m N 45° W
3. $1 \text{ m} \cdot \text{s}^{-1}$, 20° East of North
4. $50 \text{ km} \cdot \text{h}^{-1}$, 085°
5. 5 mm, 225°

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Properties of vectors



Vectors are mathematical objects and we will now study some of their mathematical properties.

If two vectors have the same magnitude (size) *and* the same direction, then we call them equal to each other. For example, if we have two forces, $\vec{F}_1 = 20 \text{ N}$ *in the upward direction* and $\vec{F}_2 = 20 \text{ N}$ *in the upward direction*, then we can say that $\vec{F}_1 = \vec{F}_2$.

DEFINITION: Equality of vectors

Two vectors are equal if they have the **same** magnitude and the **same** direction.

Just like scalars which can have positive or negative values, vectors can also be positive or negative. A negative vector is a vector which points in the direction *opposite* to the **reference positive direction**. For example, if in a particular situation, we define the upward direction as the reference positive direction, then a force $\vec{F}_1 = 30 \text{ N}$ *downwards* would be a *negative vector* and could also be written as $\vec{F}_1 = -30 \text{ N}$. In this case, the negative (-) sign indicates that the direction of $\vec{F}_1$ is opposite to that of the reference positive direction.

DEFINITION: Negative vector

A negative vector is a vector that has the *opposite* direction to the reference positive direction.

Like scalars, vectors can also be added and subtracted. We will investigate how to do this next.

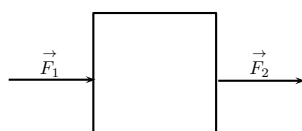
Addition and subtraction of vectors



Adding vectors

When vectors are added, we need to take into account *both* their magnitudes *and* directions. ▶ See video: VPgel at www.everythingscience.co.za

For example, imagine the following. You and a friend are trying to move a heavy box. You stand behind it and push forwards with a force $\vec{F}_1$ and your friend stands in front and pulls it towards them with a force $\vec{F}_2$. The two forces are in the *same* direction (i.e. forwards) and so the total force acting on the box is:



$$\vec{F}_{Tot} = \vec{F}_1 + \vec{F}_2$$

It is very easy to understand the concept of vector addition through an activity using the **displacement** vector.

Displacement is the vector which describes the change in an object's position. It is a vector that points from the initial position to the final position.

Activity:

Adding vectors

Materials: masking tape

Method:

Tape a line of masking tape horizontally across the floor. This will be your starting point.

Task 1:

Take 2 steps in the forward direction. Use a piece of masking tape to mark your end point and label it **A**. Then take another 3 steps in the forward direction. Use masking tape to mark your final position as **B**. Make sure you try to keep your steps all the same length!

Task 2:

Go back to your starting line. Now take 3 steps forward. Use a piece of masking tape to mark your end point and label it **B**. Then take another 2 steps forward and use a new piece of masking tape to mark your final position as **A**.

Discussion:

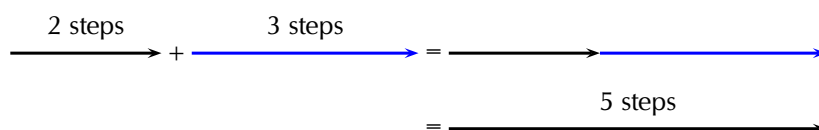
What do you notice?

1. In *Task 1*, the first 2 steps forward represent a displacement vector and the second 3 steps forward also form a displacement vector. If we did not stop after the first 2 steps, we would have taken 5 steps in the forward direction in

total. Therefore, if we add the displacement vectors for 2 steps and 3 steps, we should get a total of 5 steps in the forward direction.

2. It does not matter whether you take 3 steps forward and then 2 steps forward, or two steps followed by another 3 steps forward. Your final position is the same! The order of the addition does not matter!

We can represent vector addition graphically, based on the activity above. Draw the vector for the first two steps forward, followed by the vector with the next three steps forward.

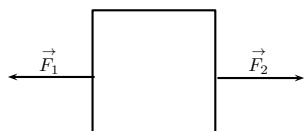


We add the second vector at the end of the first vector, since this is where we now are after the first vector has acted. The vector from the tail of the first vector (the starting point) to the head of the second vector (the end point) is then the sum of the vectors.

As you can convince yourself, the order in which you add vectors does not matter. In the example above, if you decided to first go 3 steps forward and then another 2 steps forward, the end result would still be 5 steps forward.

Subtracting vectors

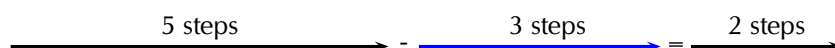
Let's go back to the problem of the heavy box that you and your friend are trying to move. If you didn't communicate properly first, you both might think that you should pull in your own directions! Imagine you stand behind the box and pull it towards you with a force $\vec{F}_1$ and your friend stands at the front of the box and pulls it towards them with a force $\vec{F}_2$. In this case the two forces are in *opposite* directions. If we define the direction your friend is pulling in as *positive* then the force you are exerting must be *negative* since it is in the opposite direction. We can write the total force exerted on the box as the sum of the individual forces:



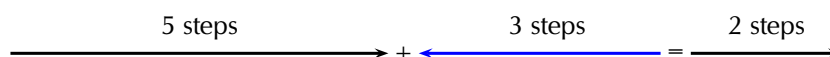
$$\begin{aligned}
 \vec{F}_{Tot} &= \vec{F}_2 + (-\vec{F}_1) \\
 &= \vec{F}_2 - \vec{F}_1
 \end{aligned}$$

What you have done here is actually to subtract two vectors! This is the same as adding two vectors which have opposite directions.

As we did before, we can illustrate vector subtraction nicely using displacement vectors. If you take 5 steps forward and then subtract 3 steps forward you are left with only two steps forward:



What did you physically do to subtract 3 steps? You originally took 5 steps forward but then you took 3 steps *backward* to land up back with only 2 steps forward. That backward displacement is represented by an arrow pointing to the left (backwards) with length 3. The net result of adding these two vectors is 2 steps forward:

**Tip**

Subtracting a vector from another is the same as adding a vector in the opposite direction.

Thus, subtracting a vector from another is the same as adding a vector in the opposite direction (i.e. subtracting 3 steps forwards is the same as adding 3 steps backwards).

The resultant vector

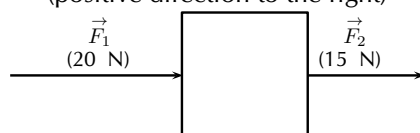
The final quantity you get when adding or subtracting vectors is called the **resultant vector**. In other words, the individual vectors can be replaced by the resultant – the overall effect is the same. ▶ See video: VPgem at www.everythingscience.co.za

DEFINITION: Resultant vector

The resultant vector is the single vector whose effect is the same as the individual vectors acting together.

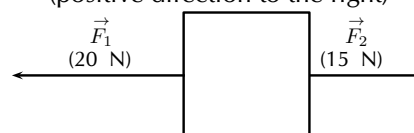
We can illustrate the concept of the resultant vector by considering our two situations in using forces to move the heavy box. In the first case (on the left), you and your friend are applying forces in the same direction. The resultant force will be the sum of your two applied forces in that direction. In the second case (on the right), the forces are applied in opposite directions. The resultant vector will again be the sum of your two applied forces, however after choosing a positive direction, one force will be positive and the other will be negative and the sign of the resultant force will just depend on which direction you chose as positive. For clarity look at the diagrams below.

Forces are applied in the same direction
(positive direction to the right)



$$\begin{aligned}
 \vec{F}_R &= \vec{F}_1 + \vec{F}_2 \\
 &= 20 \text{ N} + 15 \text{ N} \\
 &= 35 \text{ N to the right}
 \end{aligned}$$

Forces are applied in opposite directions
(positive direction to the right)



$$\begin{aligned}
 \vec{F}_R &= \vec{F}_2 + (-\vec{F}_1) \\
 &= \vec{F}_2 - \vec{F}_1 \\
 &= 15 \text{ N} - 20 \text{ N} \\
 &= -5 \text{ N} \\
 &= 5 \text{ N to the left}
 \end{aligned}$$

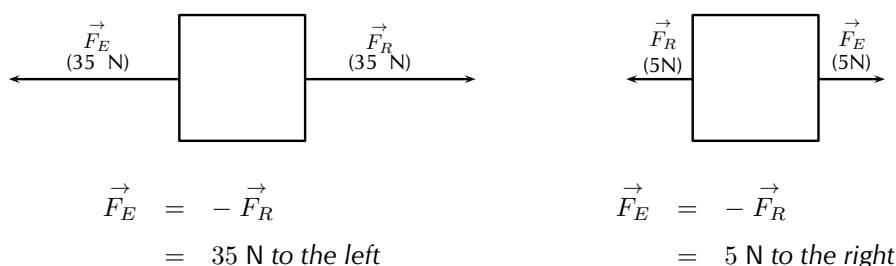


There is a special name for the vector which has the same magnitude as the resultant vector but the *opposite* direction: the **equilibrant**. If you add the resultant vector and the equilibrant vectors together, the answer is always zero because the equilibrant cancels the resultant out.

DEFINITION: Equilibrant

The equilibrant is the vector which has the *same magnitude* but *opposite direction* to the resultant vector.

If you refer to the pictures of the heavy box before, the equilibrant forces for the two situations would look like:



Techniques of vector addition



Now that you have learned about the mathematical properties of vectors, we return to vector addition in more detail. There are a number of techniques of vector addition. These techniques fall into two main categories - graphical and algebraic techniques.

Graphical techniques



Graphical techniques involve drawing accurate scale diagrams to denote individual vectors and their resultants. We will look at just one graphical method: the head-to-tail method.

Method: Head-to-Tail Method of Vector Addition

1. Draw a rough sketch of the situation.
2. Choose a scale and include a reference direction.
3. Choose any of the vectors and draw it as an arrow in the correct direction and of the correct length – remember to put an arrowhead on the end to denote its direction.
4. Take the next vector and draw it as an arrow starting from the arrowhead of the first vector in the correct direction and of the correct length.
5. Continue until you have drawn each vector – each time starting from the head of the previous vector. In this way, the vectors to be added are drawn one after the other head-to-tail.
6. The resultant is then the vector drawn from the tail of the first vector to the head of the last. Its magnitude can be determined from the length of its arrow using the scale. Its direction too can be determined from the scale diagram.

Let's consider some more examples of vector addition using displacements. The arrows tell you how far to move and in what direction. Arrows to the right correspond to steps forward, while arrows to the left correspond to steps backward. Look at all of the examples below and check them.

This example says 1 step forward and then another step forward is the same as an arrow twice as long – two steps forward.

$$\begin{array}{c} \text{1 step} \\ \longrightarrow \end{array} + \begin{array}{c} \text{1 step} \\ \longrightarrow \end{array} = \begin{array}{c} \text{2 steps} \\ \longrightarrow \end{array} = \begin{array}{c} \text{2 steps} \\ \longrightarrow \end{array}$$

This example says 1 step backward and then another step backward is the same as an arrow twice as long – two steps backward.

$$\begin{array}{c} \text{1 step} \\ \longleftarrow \end{array} + \begin{array}{c} \text{1 step} \\ \longleftarrow \end{array} = \begin{array}{c} \text{2 steps} \\ \longleftarrow \end{array} = \begin{array}{c} \text{2 steps} \\ \longleftarrow \end{array}$$

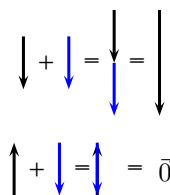
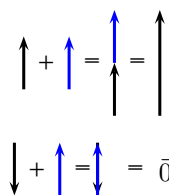
It is sometimes possible that you end up back where you started. In this case the net result of what you have done is that you have gone nowhere (your start and end points are at the same place). In this case, your resultant displacement is a vector with length zero units. We use the symbol $\vec{0}$ to denote such a vector:

$$\begin{array}{c} \text{1 step} \\ \longrightarrow \end{array} + \begin{array}{c} \text{1 step} \\ \longleftarrow \end{array} = \begin{array}{c} \text{1 step} \\ \longleftrightarrow \\ \text{1 step} \end{array} = \vec{0}$$

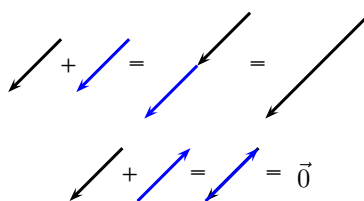
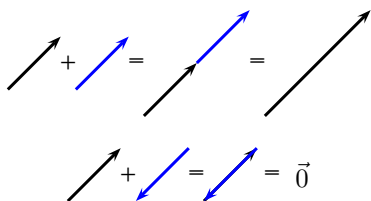
$$\begin{array}{c} \text{1 step} \\ \longleftarrow \end{array} + \begin{array}{c} \text{1 step} \\ \longrightarrow \end{array} = \begin{array}{c} \text{1 step} \\ \longleftrightarrow \\ \text{1 step} \end{array} = \vec{0}$$

Check the following examples in the same way. Arrows up the page can be seen as steps left and arrows down the page as steps right.

Try a couple to convince yourself!



It is important to realise that the directions are not special– ‘forward and backwards’ or ‘left and right’ are treated in the same way. The same is true of any set of parallel directions:



In the above examples the separate displacements were parallel to one another. However the same head-to-tail technique of vector addition can be applied to vectors in any direction.

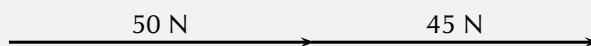
Example 3: Head-to-tail addition 1

QUESTION

A car breaks down in the road and you and your friend, who happen to be walking past, help the driver push-start it. You and your friend stand together at the rear of the car. If you push with a force of 50 N and your friend pushes with a force of 45 N, what is the resultant force on the car? Use the head-to-tail technique to calculate the answer graphically.

SOLUTION

Step 1 : **Draw a rough sketch of the situation**



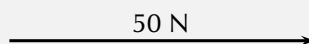
Step 2 : **Choose a scale and a reference direction**

Let's choose the direction to the right as the positive direction. The scale can be 1 cm = 10 N.

Step 3 : **Choose one of the vectors and draw it as an arrow of the correct**

length in the correct direction

Start with your force vector and draw an arrow pointing to the right which is 5 cm long (i.e. $50\text{ N} = 5 \times 10\text{ N}$, therefore, you must multiply your cm scale by 5 as well).



Step 4 : Take the next vector and draw it starting at the arrowhead of the previous vector.

Since your friend is pushing in the same direction as you, your force vectors must point in the same direction. Using the scale, this arrow should be 4,5 cm long.



Step 5 : Draw the resultant, measure its length and find its direction

There are only two vectors in this problem, so the resultant vector must be drawn from the tail (i.e. starting point) of the first vector to the head of the second vector.



The resultant vector measures 9,5 cm and points to the right. Therefore the resultant force must be 95 N in the positive direction (or to the right).

Example 4: Head-to-tail addition 2

QUESTION

Use the graphical head-to-tail method to determine the resultant force on a rugby player if two players on his team are pushing him forwards with forces of $\vec{F}_1 = 60\text{ N}$ and $\vec{F}_2 = 90\text{ N}$ respectively and two players from the opposing team are pushing him backwards with forces of $\vec{F}_3 = 100\text{ N}$ and $\vec{F}_4 = 65\text{ N}$ respectively.

SOLUTION

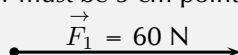
Step 1 : Choose a scale and a reference direction

Let's choose a scale of $0,5\text{ cm} = 10\text{ N}$ and for our diagram we will

define the positive direction as to the right.

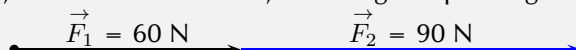
Step 2 : Choose one of the vectors and draw it as an arrow of the correct length in the correct direction

We will start with drawing the vector $\vec{F}_1 = 60 \text{ N}$, pointing in the positive direction. Using our scale of $0,5 \text{ cm} = 10 \text{ N}$, the length of the arrow must be 3 cm pointing to the right.



Step 3 : Take the next vector and draw it starting at the arrowhead of the previous vector

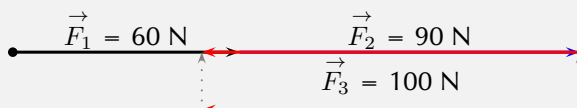
The next vector is $\vec{F}_2 = 90 \text{ N}$ in the same direction as $\vec{F}_1$. Using the scale, the arrow should be $4,5 \text{ cm}$ long and pointing to the right.



Step 4 : Take the next vector and draw it starting at the arrowhead of the previous vector

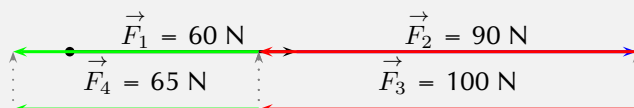
The next vector is $\vec{F}_3 = 100 \text{ N}$ in the opposite direction. Using the scale, this arrow should be 5 cm long and point to the left.

Note: We are working in one dimension so this arrow would be drawn on top of the first vectors to the left. This will get messy so we'll draw it next to the actual line as well to show you what it looks like.



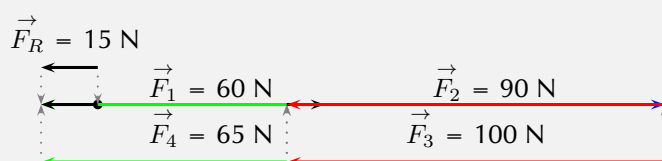
Step 5 : Take the next vector and draw it starting at the arrowhead of the previous vector

The fourth vector is $\vec{F}_4 = 65 \text{ N}$ also in the opposite direction. Using the scale, this arrow must be $3,25 \text{ cm}$ long and point to the left.



Step 6 : Draw the resultant, measure its length and find its direction

We have now drawn all the force vectors that are being applied to the player. The resultant vector is the arrow which starts at the tail of the first vector and ends at the head of the last drawn vector.



The resultant vector measures 0,75 cm which, using our scale is equivalent to 15 N and points to the left (or the negative direction or the direction the opposing team members are pushing in).

Algebraic techniques



Vectors in a Straight Line

Whenever you are faced with adding vectors acting in a straight line (i.e. some directed left and some right, or some acting up and others down) you can use a very simple algebraic technique:

Method: Addition/Subtraction of Vectors in a Straight Line

1. Choose a positive direction. As an example, for situations involving displacements in the directions west and east, you might choose west as your positive direction. In that case, displacements east are negative.
2. Next simply add (or subtract) the magnitude of the vectors using the appropriate signs.
3. As a final step the direction of the resultant should be included in words (positive answers are in the positive direction, while negative resultants are in the negative direction).

Let us consider a few examples.

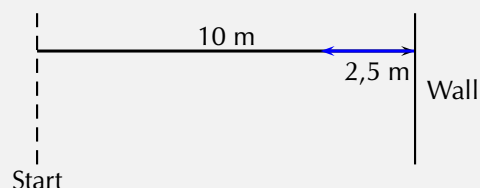
Example 5: Adding vectors algebraically I

QUESTION

A tennis ball is rolled towards a wall which is 10 m away from the ball. If after striking the wall the ball rolls a further 2,5 m along the ground away from the wall, calculate algebraically the ball's resultant displacement.

SOLUTION

Step 1 : **Draw a rough sketch of the situation**



Step 2 : **Decide which method to use to calculate the resultant**

We know that the resultant displacement of the ball ($\vec{x}_R$) is equal to the sum of the ball's separate displacements ($\vec{x}_1$ and $\vec{x}_2$):

$$\vec{x}_R = \vec{x}_1 + \vec{x}_2$$

Since the motion of the ball is in a straight line (i.e. the ball moves towards and away from the wall), we can use the method of algebraic addition just explained.

Step 3 : **Choose a positive direction**

Let's choose the **positive** direction to be towards the wall. This means that the **negative** direction is away from the wall.

Step 4 : **Now define our vectors algebraically**

With right positive:

$$\vec{x}_1 = +10, 0 \text{ m}$$

$$\vec{x}_2 = -2, 5 \text{ m}$$

Step 5 : **Add the vectors**

Next we simply add the two displacements to give the resultant:

$$\begin{aligned}\vec{x}_R &= (+10 \text{ m}) + (-2, 5 \text{ m}) \\ &= (+7, 5) \text{ m}\end{aligned}$$

Step 6 : **Quote the resultant**

Finally, in this case towards the wall is the positive direction, so: $\vec{x}_R = 7,5 \text{ m}$ towards the wall.

Example 6: Subtracting vectors algebraically I

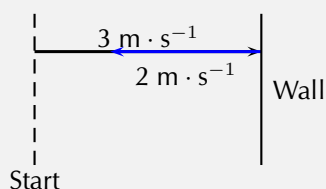
QUESTION

Suppose that a tennis ball is thrown horizontally towards a wall at an initial velocity of $3 \text{ m} \cdot \text{s}^{-1}$ to the right. After striking the wall, the ball returns to the thrower at $2 \text{ m} \cdot \text{s}^{-1}$. Determine the change in velocity of the ball.

SOLUTION

Step 1 : Draw a sketch

A quick sketch will help us understand the problem.



Step 2 : Decide which method to use to calculate the resultant

Remember that velocity is a vector. The change in the velocity of the ball is equal to the difference between the ball's initial and final velocities:

$$\Delta \vec{v} = \vec{v}_f - \vec{v}_i$$

Since the ball moves along a straight line (i.e. left and right), we can use the algebraic technique of vector subtraction just discussed.

Step 3 : Choose a positive direction

Choose the **positive** direction to be towards the wall. This means that the **negative** direction is away from the wall.

Step 4 : Now define our vectors algebraically

$$\vec{v}_i = +3 \text{ m} \cdot \text{s}^{-1}$$

$$\vec{v}_f = -2 \text{ m} \cdot \text{s}^{-1}$$

Step 5 : Subtract the vectors

Thus, the change in velocity of the ball is:

$$\begin{aligned}\Delta \vec{v} &= (-2 \text{ m} \cdot \text{s}^{-1}) - (+3 \text{ m} \cdot \text{s}^{-1}) \\ &= (-5) \text{ m} \cdot \text{s}^{-1}\end{aligned}$$

Step 6 : Quote the resultant

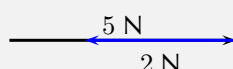
Remember that in this case towards the wall means a positive velocity,
so away from the wall means a negative velocity: $\Delta \vec{v} = 5 \text{ m} \cdot \text{s}^{-1}$
away from the wall.

Example 7: Adding vectors algebraically II**QUESTION**

A man applies a force of 5 N on a crate. The crate pushes back on the man with a force of 2 N. Calculate algebraically the resultant force that the man applies to the crate.

SOLUTION**Step 1 : Draw a sketch**

A quick sketch will help us understand the problem.

**Step 2 : Decide which method to use to calculate the resultant**

Remember that force is a vector. Since the crate moves along a straight line (i.e. left and right), we can use the algebraic technique of vector addition just discussed.

Step 3 : Choose a positive direction

Choose the **positive** direction to be towards the crate (i.e. in the same direction that the man is pushing). This means that the **negative** direction is away from the crate (i.e. against the direction that the man is pushing).

Step 4 : Now define our vectors algebraically

$$\vec{F}_{\text{man}} = +5 \text{ N}$$

$$\vec{F}_{\text{crate}} = -2 \text{ N}$$

Step 5 : **Subtract the vectors**

Thus, the resultant force is:

$$\begin{aligned}\vec{F}_{\text{man}} + \vec{F}_{\text{crate}} &= (5 \text{ N}) + (2 \text{ N}) \\ &= 7 \text{ N}\end{aligned}$$

Step 6 : **Quote the resultant**

Remember that in this case towards the crate means a positive force:
7 N towards the crate.

Remember that the technique of addition and subtraction just discussed can only be applied to vectors acting along a straight line. When vectors are not in a straight line, i.e. at an angle to each other then simple geometric and trigonometric techniques can be used to find resultant vectors.

Exercise 20 - 4

1. Harold walks to school by walking 600 m Northeast and then 500 m N 40° W. Determine his resultant displacement by using accurate scale drawings.
2. A dove flies from her nest, looking for food for her chick. She flies at a velocity of $2 \text{ m} \cdot \text{s}^{-1}$ on a bearing of 135° and then at a velocity of $1,2 \text{ m} \cdot \text{s}^{-1}$ on a bearing of 230°. Calculate her resultant velocity by using accurate scale drawings.
3. A squash ball is dropped to the floor with an initial velocity of $2,5 \text{ m} \cdot \text{s}^{-1}$. It rebounds (comes back up) with a velocity of $0,5 \text{ m} \cdot \text{s}^{-1}$.
 - (a) What is the change in velocity of the squash ball?
 - (b) What is the resultant velocity of the squash ball?
4. A frog is trying to cross a river. It swims at $3 \text{ m} \cdot \text{s}^{-1}$ in a northerly direction towards the opposite bank. The water is flowing in a westerly direction at $5 \text{ m} \cdot \text{s}^{-1}$. Find the frog's resultant velocity by using appropriate calculations. Include a rough sketch of the situation in your answer.

5. Mphlonhle walks to the shop by walking 500 m Northwest and then 400 m N 30° E. Determine her resultant displacement by doing appropriate calculations.

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(1.) 008v (2.) 008w (3.) 008x (4.) 008y (5.) 008z

Chapter 20 | Summary

See the summary presentation ( Presentation: VPgig at www.everythingscience.co.za)

- A scalar is a physical quantity with magnitude only.
- A vector is a physical quantity with magnitude and direction.
- Vectors may be represented as arrows where the length of the arrow indicates the magnitude and the arrowhead indicates the direction of the vector.
- The direction of a vector can be indicated by referring to another vector or a fixed point (e.g. 30° from the river bank); using a compass (e.g. N 30° W); or bearing (e.g. 053°).
- The resultant vector is the single vector whose effect is the same as the individual vectors acting together.

Chapter 20

End of chapter exercises

1. A point is acted on by two forces in equilibrium. The forces
 - a. have equal magnitudes and directions.
 - b. have equal magnitudes but opposite directions.
 - c. act perpendicular to each other.
 - d. act in the same direction.

2. Which of the following contains two vectors and a scalar?
 - a. distance, acceleration, speed
 - b. displacement, velocity, acceleration
 - c. distance, mass, speed
 - d. displacement, speed, velocity
3. Two vectors act on the same point. What should the angle between them be so that a maximum resultant is obtained?
 - a. 0°
 - b. 90°
 - c. 180°
 - d. cannot tell
4. Two forces, 4 N and 11 N, act on a point. Which one of the following cannot be the magnitude of a resultant?
 - a. 4 N
 - b. 7 N
 - c. 11 N
 - d. 15 N

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(1.) 0092 (2.) 0094 (3.) 0095 (4.) 0096

Motion in one dimension

21

Introduction

www ESAGS

This chapter is about how things move along a straight line or, more scientifically, how things move *in one dimension*. This is useful for learning how to describe the movement (motion) of cars along a straight road or of trains along straight railway tracks. There are three features of motion that we use to describe exactly how an object moves. They are:

Traffic often moves along a straight line.



Photograph by warrenski on Flickr.com

FACT

The **jerk** is the name we give to how fast the acceleration is changing.

1. **position** which tells us about an object's location or **displacement** which tells us about change of location
2. **speed** which tells us how fast the object is moving or **velocity** which tells us how fast the object is moving and where it is moving to, and
3. **acceleration** which tells us exactly how fast the object's speed and velocity are changing.

See introductory video: (📺 Video: VPgiq at www.everythingscience.co.za)

Reference frame

www ESAGT

The first thing to focus on when studying motion of an object or person is their position. The word *position* describes your location (where you are). However, saying that you are *here* or *there* is meaningless, you have to use known points (reference points) to help specify your position.

For example, if you were in a classroom and wanted to tell a classmate where you were standing you would first give them a reference point. The reference point might be the classroom door. You would then be able to say that you are 2 m from the doorway. This still doesn't give your position precisely. We need to provide a reference point and a coordinate system to use to define the location precisely.

Describing your location



Photograph by chokola on Flickr.com

Then you can say that you are, for example, 2 m from the door directly inside the classroom. The classroom door is a reference point and inside/outside is the coordinate system you have chosen. A frame of reference or **reference frame** is reference point which serves as the origin for a coordinate system. The coordinate system can be up or down, inside or outside, left or right or even forward or backward. These are all examples that define a 1-dimensional coordinate system. We choose one of the directions as the **positive** direction.

DEFINITION: Frame of reference

A frame of reference is a reference point combined with a set of directions.

A graphical representation of a 1-dimensional frame of reference:

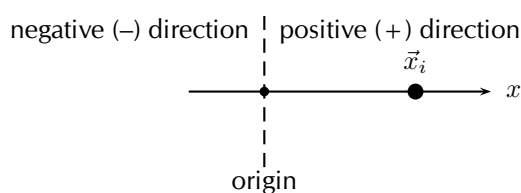
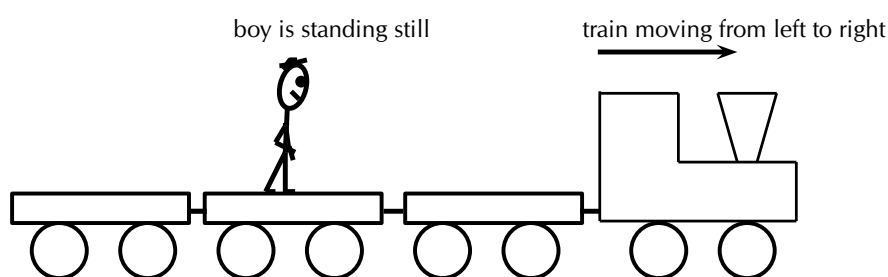


Figure 21.1: Frame of reference

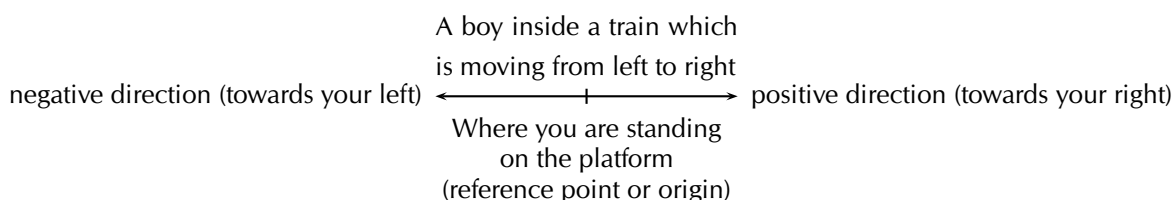
You can define different frames of reference for the same problem but the outcome, the physical results, will be the same. For example, a boy is standing still inside a train as it pulls out of a station. Both you and the boy define your location as the point of reference and the direction train is moving as a where you are standing as the point of reference and the direction the train is moving in as forward.

You are standing on the platform watching the train move from left to right. To you it looks as if the boy is moving from left to right, because relative to where you are standing (the platform), he is moving. According to the boy, and his *frame of reference* (the train), he is not moving.

A frame of reference must have an origin (where you are standing on the platform) and at least a positive direction. The train was moving from left to right, making to your right positive and to your left negative. If someone else was looking at the same boy, his frame of reference will be different. For example, if he was standing on the other side of the platform, the boy will be moving from right to left.



From your frame of reference the boy is moving from left to right.



For this chapter, we will only use frames of reference in the x -direction. By doing this we restrict ourselves to *one dimensional motion*. We can use the sign of the position value (positive or negative) to indicate the direction relative to the origin.

DEFINITION: *One dimensional motion*

An object is constrained to move back and forth along a line.

For example the blue dot in the figure below can only move along the x -axis.



Position



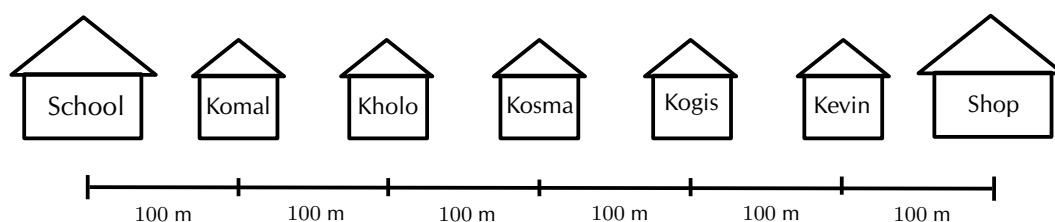
DEFINITION: Position

Position is a measurement of a location, with reference to an origin.

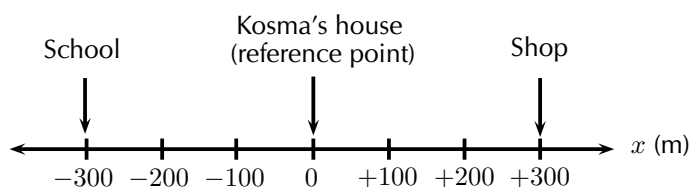
Quantity: Position (x) Unit name: metre Unit symbol: m

A position is a measurement of a location within a reference frame. This means that positions can be negative or positive depending on the choice for the reference frame's coordinate system. ▶ See video: VPgmf at www.everythingscience.co.za

Depending on which reference point we choose, we can say that the school is 300 m from Kosma's house (with Kosma's house as the reference point or origin) or 500 m from Kevin's house (with Kevin's house as the reference point or origin).

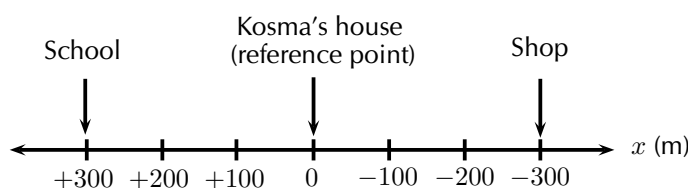


The shop is also 300 m from Kosma's house, but in the opposite direction as the school. When we choose a reference point, we have a positive direction and a negative direction. If we choose the direction **towards the school** as **negative**, then the direction towards the shop is positive. A negative direction is always opposite to the direction chosen as positive.



The origin is at Kosma's house and the position of the school is -300 m. Positions towards the left are defined as negative and positions towards the right are defined as positive.

Note that we could also choose the **positive** direction to be **towards the school**. In this case Kosma's house is still 300 m away from the school, but it is now in the positive direction.



The origin is at Kosma's house and the position of the school is +300 m. Positions towards the left are defined as positive and positions towards the right are defined as negative.

Group Discussion: Reference points

Divide into groups of 5 for this activity. On a straight line, choose a reference point. Since position can have both positive and negative values, discuss the advantages and disadvantages of choosing

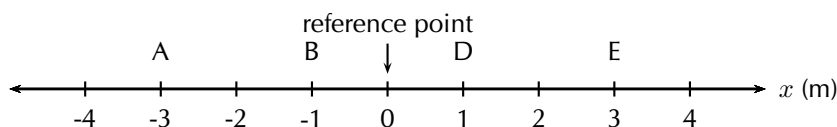
1. either end of the line,
2. the middle of the line. (This reference point can also be called "the origin".)

Stand in a straight line, take turns choosing a different group member as the origin. Let the group member choose which direction along the line is positive. Everyone should then try to define their position. You don't need to measure your position very precisely but can just approximate it. It is important to understand whether your position is positive or negative for each difference origin and choice of coordinate system.

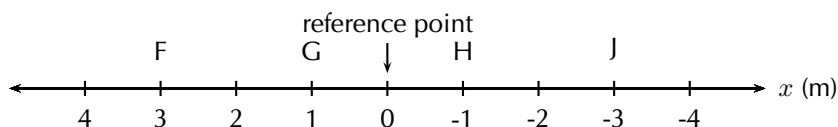
Notice that your position is different every time but that you didn't actually move. How you write an answer might be affected by the choice of a coordinate system but physical processes should never be affected.

Exercise 21 - 1

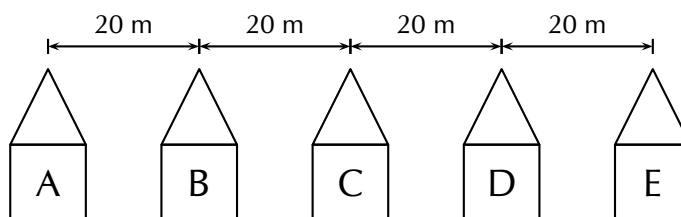
1. Write down the positions for objects at A, B, D and E. Do not forget the units.



2. Write down the positions for objects at F, G, H and J. Do not forget the units.



3. There are 5 houses on Newton Street, A, B, C, D and E. For all cases, assume that positions to the right are positive.



- Draw a frame of reference with house A as the origin and write down the positions of houses B, C, D and E.
- You live in house C. What is your position relative to house E?
- What are the positions of houses A, B and D, if house B is taken as the reference point?

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(1.) 009g (2.) 009h (3.) 009i

Displacement and distance

ESAGV

Tip

The symbol Δ is read out as *delta*. Δ is a letter of the Greek alphabet and is used in Mathematics and Science to indicate a change in a certain quantity, or a final value minus an initial value. For example, Δx means change in x while Δt means change in t .

DEFINITION: Distance

Distance is the total length of the path taken in going from the initial position, $\vec{x}_i$, to the final position, $\vec{x}_f$. Distance is a scalar.

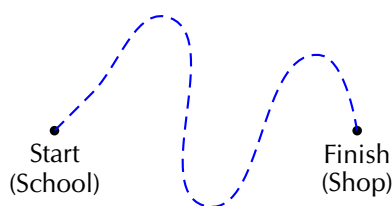
Quantity: Distance (D)

Unit name: metre

Unit symbol: m

In the simple map below you can see the path that winds because of a number of hills from a school to a nearby shop. The path is shown by a dashed line. The initial point, $\vec{x}_i$, is the school and the final point, $\vec{x}_f$, is the shop.

Distance is the length of dashed line. It is how far you have to walk along the path from the school to the shop.



DEFINITION: Displacement

Displacement is the change in an object's position. It is a vector that points from the initial position ($\vec{x}_i$) to the final position ($\vec{x}_f$).

Quantity: Displacement ($\Delta\vec{x}$) Unit name: metre Unit symbol: m

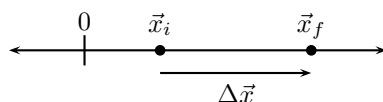
The displacement of an object is defined as its change in position (final position minus initial position). Displacement has a magnitude and direction and is therefore a vector. For example, if the initial position of a car is $\vec{x}_i$ and it moves to a final position of $\vec{x}_f$, then the displacement is:

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_i$$

To help visualise what the displacement vector looks like think back to the tail-to-head method. The displacement is the vector you add to the initial position vector to get a vector to the final position. However, subtracting an initial quantity from a final quantity happens often in Physics, so we use the shortcut Δ to mean *final - initial*. Therefore, displacement can be written:

$$\Delta\vec{x} = \vec{x}_f - \vec{x}_i$$

The following diagram illustrates the concept of displacement:



For example, if you roll a ball 5 m along a floor, in a straight line, then its displacement is 5 m, taking the direction of motion as positive, and the initial position as 0 m.

Displacement does not depend on the path travelled, but only on the initial and final positions. We use the word *distance* to describe how far an object travels along a particular path.

Tip

The words *initial* and *final* will be used very often in Physics. *Initial* refers to the situation in the beginning of the description/problem and *final* to the situation at the end. It will often happen that the final value is smaller than the initial value, such that the difference is negative. This is ok!

Tip

We will use D in this book, but you may see d used in other books.

Tip

We use the expression '*as the crow flies*' to mean a straight line between two points because birds can fly directly over many obstacles.

If we go back to the simple map repeated below you can see the path as before shown by a dashed line.

Distance is the length of dashed line. The displacement is different. Displacement is the straight-line distance from the starting point to the endpoint – from the school to the shop in the figure as shown by the solid arrow.

▶ See video: VPgmo at www.everythingscience.co.za

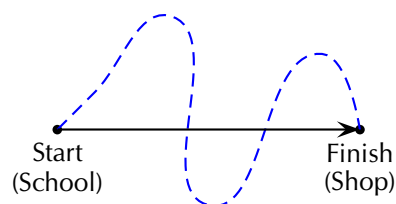
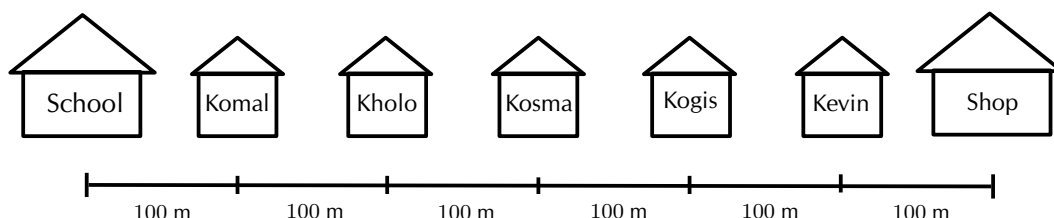


Illustration of distance and displacement

If we use the same situation as earlier we can explore the concepts in more detail. Consider our description of the location of the houses, school and the shop.



Komal walks to meet Kevin at his house before walking to school. What is Komal's displacement and what distance did he cover if he walks to school via Kevin's house?

Komal covers a distance of 400 m to Kevin's house and another 500 m from Kevin's house to the school. He covers a total distance of 900 m. His displacement, however, is only 100 m towards the school. This is because displacement only looks at the starting position (his house) and the end position (the school). It does not depend on the path he travelled.

To calculate his distance and displacement, we need to choose a reference point and a direction. Let's choose Komal's house as the reference point, and towards Kevin's house as the positive direction (which means that towards the school is negative). We would do the calculations as follows:

$$\begin{aligned}
 \text{Distance (D)} &= \text{path travelled} & \text{Displacement}(\Delta \vec{x}) &= \vec{x}_f - \vec{x}_i \\
 &= 400 \text{ m} + 500 \text{ m} & &= -100 \text{ m} + 0 \text{ m} \\
 &= 900 \text{ m} & &= -100 \text{ m} \\
 & & &= 100 \text{ m (in the negative } x \text{ direction)}
 \end{aligned}$$

Very often in calculations you will get a negative answer. For example, Komal's displacement in the example above, is calculated as -100 m . The minus sign in front of the answer means that his displacement is 100 m in the opposite direction (opposite to the direction chosen as positive in the beginning of the question). When we start a calculation we choose a frame of reference and a positive direction. In the first example above, the reference point is Komal's house and the positive direction is towards Kevin's house. Therefore Komal's displacement is 100 m towards the school. Notice that distance has no direction,

but displacement has a direction.
Kevin walks to school with Komal and after school walks back home. What is Kevin’s displacement and what distance did he cover? For this calculation we use Kevin’s house as the reference point. Let’s take towards the school as the positive direction.

Distance (D)

=

path travelled

=

500 m + 500 m

=

1000 m

Displacement($\Delta\vec{x}$)

=

$\vec{x}_f - \vec{x}_i$

=

0 m + 0 m

=

0 m

It is possible to have a displacement of 0 m and a distance that is not 0 m. This happens whenever you end at the same point you started.

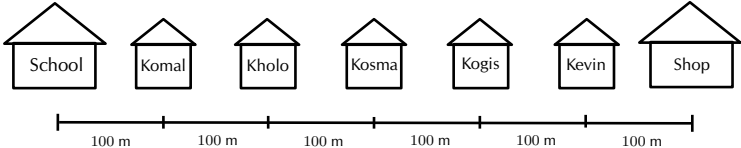
Differences between distance and displacement

The differences between distance and displacement can be summarised as:

Distance	Displacement
1. depends on the path	1. independent of path taken
2. always positive	2. can be positive or negative
3. is a scalar	3. is a vector

Exercise 21 - 2

1. Use this figure to answer the following questions.



- a. Kogis walks to Kosma’s house and then to school, what is her distance and displacement?

b. Kholo walks to Kosma’s house and then to school, what is her distance and displacement?

c. Komal walks to the shop and then to school, what is his distance and displacement?

d. What reference point did you use for each of the above questions?
2. You stand at the front door of your house (displacement, $\Delta\vec{x} = 0$ m). The street is 10 m away from the front door. You walk to the street and back again.

- a. What is the distance you have walked?
- b. What is your final displacement?
- c. Is displacement a vector or a scalar? Give a reason for your answer.

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(1.) 009j (2.) 009k

Speed and velocity

 ESAGW

DEFINITION: Average speed

Average speed is the **distance** (D) travelled divided by the time (Δt) taken for the journey.

Quantity: average speed (v_{av})

Unit name: metre per second

Unit

symbol: $\text{m} \cdot \text{s}^{-1}$

DEFINITION: Average velocity

Average velocity is the **change in position** of a body divided by the time it took for the displacement to occur.

Quantity: average velocity ($\vec{v}_{av}$)

Unit name: metre per second

Unit

symbol: $\text{m} \cdot \text{s}^{-1}$

Before moving on review the difference between distance and displacement. Sometimes the average speed can be a very big number while the average velocity is zero.  See video:

at www.everythingscience.co.za

Average velocity is the rate of change of position. It tells us how much an object's position changes per unit of time. Velocity is a vector. We use the symbol $\vec{v}_{av}$ for average velocity.

If we have a displacement of $\Delta\vec{x}$ and a time taken of Δt , $\vec{v}_{av}$ is then defined as:

$$\begin{aligned}\text{average velocity (in m} \cdot \text{s}^{-1}\text{)} &= \frac{\text{change in position (in m)}}{\text{change in time (in s)}} \\ \vec{v}_{av} &= \frac{\Delta\vec{x}}{\Delta t}\end{aligned}$$

Velocity can be positive or negative. A positive velocity points in the direction you chose as positive in your coordinate system. A negative velocity points in the direction opposite to the positive direction.

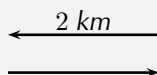
Average speed (symbol v_{av}) is the distance travelled (D) divided by the time taken (Δt) for the journey. Distance and time are scalars and therefore speed will also be a scalar. Speed is calculated as follows:

$$\begin{aligned}\text{average speed (in m} \cdot \text{s}^{-1}\text{)} &= \frac{\text{distance (in m)}}{\text{time (in s)}} \\ v_{av} &= \frac{D}{\Delta t}\end{aligned}$$

Example 1: Average speed and average velocity

QUESTION

James walks 2 km away from home in 30 minutes. He then turns around and walks back home along the same path, also in 30 minutes. Calculate James' average speed and average velocity.



SOLUTION

Step 1 : Identify what information is given and what is asked for
The question explicitly gives

- the distance and time out (2 km in 30 minutes)
- the distance and time back (2 km in 30 minutes)

Step 2 : Check that all units are SI units.

The information is not in SI units and must therefore be converted.

To convert km to m, we know that:

$$1 \text{ km} = 1\,000 \text{ m}$$

$$\therefore 2 \text{ km} = 2\,000 \text{ m} \quad (\text{multiply both sides by } 2.)$$

Similarly, to convert 30 minutes to seconds,

$$1 \text{ min} = 60 \text{ s}$$

$$\therefore 30 \text{ min} = 1\,800 \text{ s} \quad (\text{multiply both sides by } 30)$$

Step 3 : Determine James' displacement and distance.

James started at home and returned home, so his displacement is 0 m.

$$\Delta \vec{x} = 0 \text{ m}$$

James walked a total distance of 4000 m (2 000 m out and 2 000 m back).

$$D = 4\,000 \text{ m}$$

Step 4 : Determine his total time.

James took 1 800 s to walk out and 1 800 s to walk back.

$$\Delta t = 3\,600 \text{ s}$$

Step 5 : Determine his average speed

$$\begin{aligned} v_{av} &= \frac{D}{\Delta t} \\ &= \frac{4\,000 \text{ m}}{3\,600 \text{ s}} \\ &= 1,11 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

Step 6 : Determine his average velocity

$$\begin{aligned} \vec{v}_{av} &= \frac{\Delta \vec{x}}{\Delta t} \\ &= \frac{0 \text{ m}}{3\,600 \text{ s}} \\ &= 0 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

Differences between speed and velocity ESAGX

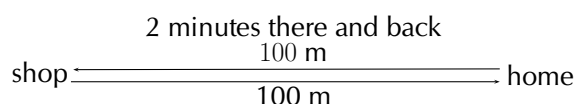
The differences between speed and velocity can be summarised as:

Speed	Velocity
1. depends on the path taken	1. independent of path taken
2. always positive	2. can be positive or negative
3. is a scalar	3. is a vector
4. no dependence on direction and so is only positive	4. direction can be determined from the sign convention used (i.e. positive or negative)

Additionally, an object that makes a round trip, i.e. travels away from its starting point and then returns to the same point has zero velocity but travels at a non-zero speed.

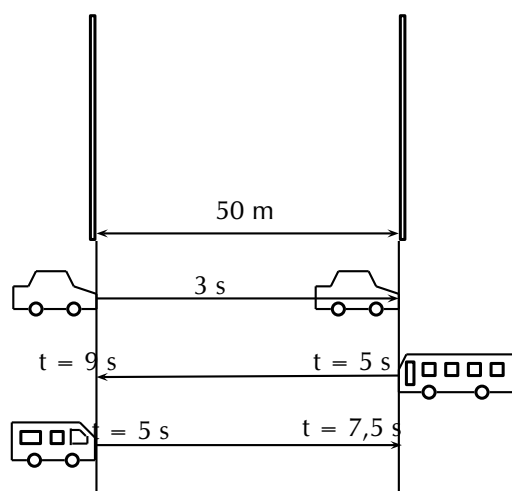
Exercise 21 - 3

1. Bongani has to walk to the shop to buy some milk. After walking 100 m, he realises that he does not have enough money, and goes back home. If it took him two minutes to leave and come back, calculate the following:
 - a. How long was he out of the house (the time interval Δt in seconds)?
 - b. How far did he walk (distance (D))?
 - c. What was his displacement ($\Delta \vec{x}$)?
 - d. What was his average velocity (in $\text{m} \cdot \text{s}^{-1}$)?
 - e. What was his average speed (in $\text{m} \cdot \text{s}^{-1}$)?

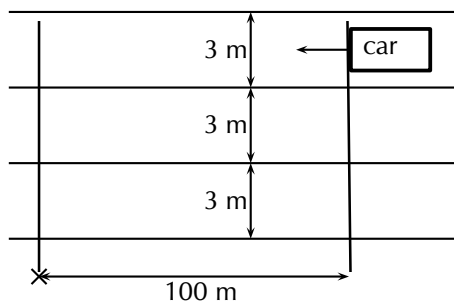


2. Bridget is watching a straight stretch of road from her classroom window. She can see two poles which she earlier measured to be 50 m apart. Using her stopwatch, Bridget notices that it takes 3 s for most cars to travel from the one pole to the other.

- a. Using the equation for velocity ($\vec{v}_{av} = \frac{\Delta \vec{x}}{\Delta t}$), show all the working needed to calculate the velocity of a car travelling from the left to the right.
- b. If Bridget measures the velocity of a red Golf to be $-16,67 \text{ m} \cdot \text{s}^{-1}$, in which direction was the Golf travelling? Bridget leaves her stopwatch running, and notices that at $t = 5,0 \text{ s}$, a taxi passes the left pole at the same time as a bus passes the right pole. At time $t = 7,5 \text{ s}$ the taxi passes the right pole. At time $t = 9,0 \text{ s}$, the bus passes the left pole.
- c. How long did it take the taxi and the bus to travel the distance between the poles? (Calculate the time interval (Δt) for both the taxi and the bus).
- d. What was the average velocity of the taxi and the bus?
- e. What was the average speed of the taxi and the bus?
- f. What was the average speed of taxi and the bus in $\text{km} \cdot \text{h}^{-1}$?



3. A rabbit runs across a freeway. There is a car, 100 m away travelling towards the rabbit.



- a. If the car is travelling at $120 \text{ km} \cdot \text{h}^{-1}$, what is the car's speed in $\text{m} \cdot \text{s}^{-1}$.
- b. How long will it take the a car to travel 100 m?
- c. If the rabbit is running at $10 \text{ km} \cdot \text{h}^{-1}$, what is its speed in $\text{m} \cdot \text{s}^{-1}$?

- d. If the freeway has 3 lanes, and each lane is 3 m wide, how long will it take for the rabbit to cross all three lanes?
- e. If the car is travelling in the furthestmost lane from the rabbit, will the rabbit be able to cross all 3 lanes of the freeway safely?

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(1.) 009m (2.) 009n (3.) 009p

Investigation: An exercise in safety

Divide into groups of 4 and perform the following investigation. Each group will be performing the same investigation, but the aim for each group will be different.

1. Choose an aim for your investigation from the following list and formulate a hypothesis:
 - Do cars travel at the correct speed limit?
 - Is it safe to cross the road outside of a pedestrian crossing?
 - Does the colour of your car determine the speed you are travelling at?
 - Any other relevant question that you would like to investigate.
2. On a road that you often cross, measure out 50 m along a straight section, far away from traffic lights or intersections.
3. Use a stopwatch to record the time each of 20 cars take to travel the 50 m section you measured.
4. Design a table to represent your results. Use the results to answer the question posed in the aim of the investigation. You might need to do some more measurements for your investigation. Plan in your group what else needs to be done.
5. Complete any additional measurements and write up your investigation under the following headings:
 - Aim and Hypothesis
 - Apparatus
 - Method
 - Results
 - Discussion

- Conclusion

6. Answer the following questions:

- How many cars took less than 3 s to travel 50 m?
- What was the shortest time a car took to travel 50 m?
- What was the average time taken by the 20 cars?
- What was the average speed of the 20 cars?
- Convert the average speed to $\text{km} \cdot \text{h}^{-1}$.



Acceleration



DEFINITION: Average acceleration

Average acceleration is the change in average velocity divided by the time taken.

Quantity: Average acceleration ($\vec{a}_{av}$)
squared Unit symbol: $\text{m} \cdot \text{s}^{-2}$

Unit name: metre per second

Acceleration is a measure of how fast the velocity of an object changes in time. If we have a change in velocity ($\Delta \vec{v}$) over a time interval (Δt), then the average acceleration ($\vec{a}_{av}$) is defined as:

$$\text{average acceleration (in } \text{m} \cdot \text{s}^{-2}) = \frac{\text{change in velocity (in } \text{m} \cdot \text{s}^{-1})}{\text{change in time (in s)}}$$

$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$$

We only deal with problems with constant acceleration. This means that the average acceleration and the instantaneous acceleration are the same. To make things simpler we will

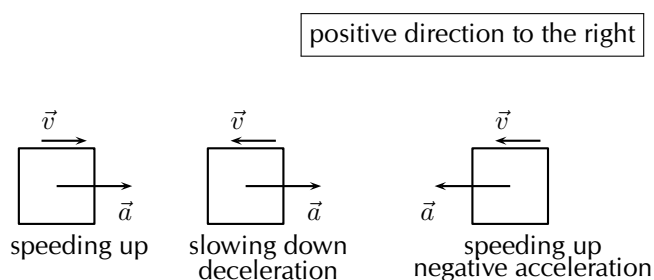
only talk about acceleration and not average or instantaneous. This is represented as $\vec{a}$. We can also have the magnitude of the acceleration. This is:

$$a = \frac{\Delta \vec{v}}{\Delta t}$$

Acceleration is a vector. Acceleration does not provide any information about the motion, but only about how the motion changes. It is not possible to tell how fast an object is moving or in which direction from the acceleration alone.

▶ See video: VPgly at www.everythingscience.co.za

Like velocity, acceleration can be negative or positive. We see that when the sign of the acceleration and the velocity are the same, the object is speeding up. If both velocity and acceleration are positive, the object is speeding up in a positive direction. If both velocity and acceleration are negative, the object is speeding up in a negative direction. We can see this in the following diagram:



If velocity is positive and acceleration is negative, then the object is slowing down. Similarly, if the velocity is negative and the acceleration is positive the object is slowing down. This is illustrated in the following worked example.

Example 2: Acceleration

QUESTION

A car accelerates uniformly from an initial velocity of $2 \text{ m}\cdot\text{s}^{-1}$ to a final velocity of $10 \text{ m}\cdot\text{s}^{-1}$ in 8 seconds. It then slows down uniformly to a final velocity of $4 \text{ m}\cdot\text{s}^{-1}$ in 6 seconds. Calculate the acceleration of the car during the first 8 seconds and during the last 6 seconds.

SOLUTION

Tip

Avoid the use of the word *deceleration* to refer to a negative acceleration. This word usually means *slowing down* and it is possible for an object to slow down with both a positive and negative acceleration, because the sign of the velocity of the object must also be taken into account to determine whether the body is slowing down or not.

Step 1 : Choose a reference frame

We choose the point where the car starts to accelerate as the origin and the direction in which the car is already moving as the positive direction.

Step 2 : Identify what information is given and what is asked for:

Consider the motion of the car in two parts: the first 8 seconds and the last 6 seconds.

For the first 8 seconds:

$$\vec{v}_i = 2 \text{ m} \cdot \text{s}^{-1}$$

$$\vec{v}_f = 10 \text{ m} \cdot \text{s}^{-1}$$

$$t_i = 0 \text{ s}$$

$$t_f = 8 \text{ s}$$

For the last 6 seconds:

$$\vec{v}_i = 10 \text{ m} \cdot \text{s}^{-1}$$

$$\vec{v}_f = 4 \text{ m} \cdot \text{s}^{-1}$$

$$t_i = 8 \text{ s}$$

$$t_f = 14 \text{ s}$$

Step 3 : Calculate the acceleration.

For the first 8 seconds:

$$\begin{aligned} a &= \frac{\Delta v}{\Delta t} \\ &= \frac{10}{8 \text{ s} - 0 \text{ s}} \\ &= 1 \text{ m} \cdot \text{s}^{-2} \end{aligned}$$

For the next 6 seconds:

$$\begin{aligned} a &= \frac{\Delta v}{\Delta t} \\ &= \frac{4}{14 \text{ s} - 8 \text{ s}} \\ &= -1 \text{ m} \cdot \text{s}^{-2} \end{aligned}$$

During the first 8 seconds the car had a positive acceleration. This means that its velocity increased. The velocity is positive so the car is speeding up. During the next 6 seconds the car had a negative acceleration. This means that its velocity decreased. The velocity is positive so the car is slowing down.

Exercise 21 - 4

1. An athlete is accelerating uniformly from an initial velocity of $0 \text{ m} \cdot \text{s}^{-1}$ to a final velocity of $4 \text{ m} \cdot \text{s}^{-1}$ in 2 seconds. Calculate his acceleration. Let the direction that the athlete is running in be the positive direction.
2. A bus accelerates uniformly from an initial velocity of $15 \text{ m} \cdot \text{s}^{-1}$ to a final velocity of $7 \text{ m} \cdot \text{s}^{-1}$ in 4 seconds. Calculate the acceleration of the bus. Let the direction of motion of the bus be the positive direction.
3. An aeroplane accelerates uniformly from an initial velocity of $100 \text{ m} \cdot \text{s}^{-1}$ to a velocity of $200 \text{ m} \cdot \text{s}^{-1}$ in 10 seconds. It then accelerates uniformly to a final velocity of $240 \text{ m} \cdot \text{s}^{-1}$ in 20 seconds. Let the direction of motion of the aeroplane be the positive direction.
 - a. Calculate the acceleration of the aeroplane during the first 10 seconds

of the motion.

- b. Calculate the acceleration of the aeroplane during the next 20 seconds of its motion.

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(1.) 009q (2.) 009r (3.) 009s

Instantaneous velocity and speed

 ESAGZ

Sprinters taking off



Sprinters finishing race



Photographs by wwarby on Flickr.com

We have looked at the average velocity and speed but sometimes we want to be more precise about what is happening between the initial and final times in a problem.

Instantaneous velocity is the velocity at a specific instant in time. This can be different to the average velocity if the velocity isn't constant.

Look at the photographs of the sprinters in a race. Their velocity is different as they take off and as they end. Their average velocity for the race doesn't change but their instantaneous velocity, as captured in the "snapshots" of an instant in time does change. Their velocity in the photograph would be an instantaneous velocity.

Tip

An instant in time is different from the time taken or the time interval. It is therefore useful to use the symbol t for an instant in time (for example during the 4th second) and the symbol Δt for the time taken (for example during the first 5 seconds of the motion).

DEFINITION: Instantaneous velocity

Instantaneous velocity is the change in position over the change in a very small time interval ($\Delta t \approx 0$).

Quantity: Instantaneous velocity ($\vec{v}$)

Unit name: metre per second

Unit name: $\text{m} \cdot \text{s}^{-1}$

DEFINITION: *Instantaneous speed*

Instantaneous speed is the magnitude of instantaneous velocity.

Quantity: Instantaneous speed (v)

Unit name: metre per second

Unit symbol: $\text{m} \cdot \text{s}^{-1}$

Instantaneous velocity is a vector. Instantaneous speed is the magnitude of instantaneous velocity. It has the same value but is not a vector so it has no direction.

Description of motion



The purpose of this chapter is to describe motion, and now that we understand the definitions of displacement, distance, velocity, speed and acceleration, we are ready to start using these ideas to describe how an object or person is moving. We will look at three ways of describing motion:

1. words
2. diagrams
3. graphs

These methods will be described in this section.

We will consider three types of motion: when the object is not moving (stationary object), when the object is moving at a constant velocity (uniform motion) and when the object is moving at a constant acceleration (motion at constant acceleration).

Stationary Object



The simplest motion that we can come across is that of a stationary object. A stationary object does not move and so its position does not change.

Vivian stands at a stop sign.

Consider an example, Vivian is waiting for a taxi. She is standing two metres from a stop street at $t = 0$ s. After one minute, at $t = 60$ s, she is still 2 metres from the stop street and after two minutes, at $t = 120$ s, also 2 metres from the stop street. Her position has not changed. Her displacement is zero (because her position is the same), her velocity is zero (because her displacement is zero) and her acceleration is also zero (because her velocity is not changing).

We can now draw graphs of position vs. time ($\vec{x}$ vs. t), velocity vs. time ($\vec{v}$ vs. t) and acceleration vs. time ($\vec{a}$ vs. t) for a stationary object. The graphs are shown below.



Photograph by Rob Boudon on Flickr.com

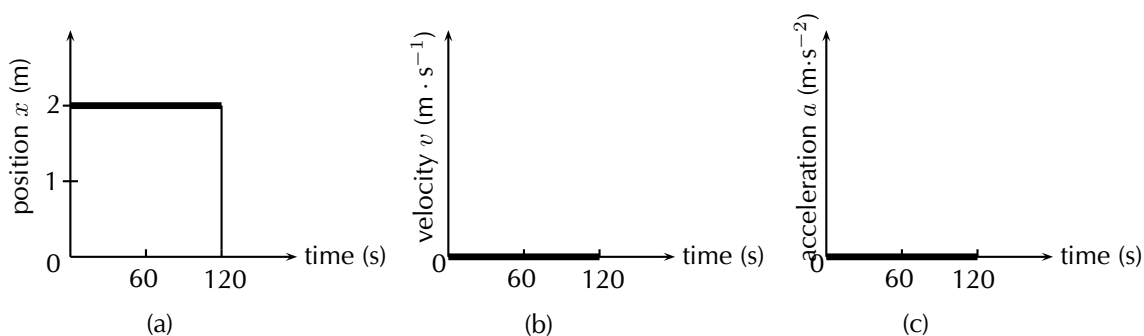


Figure 21.2: Graphs for a stationary object (a) position vs. time (b) velocity vs. time (c) acceleration vs. time.

Vivian's position is 2 metres in the positive direction from the stop street. If the stop street is taken as the reference point, her position remains at 2 metres for 120 seconds. The graph is a horizontal line at 2 m. The velocity and acceleration graphs are also shown. They are both horizontal lines on the x -axis. Since her position is not changing, her velocity is $0 \text{ m} \cdot \text{s}^{-1}$ and since velocity is not changing, acceleration is $0 \text{ m} \cdot \text{s}^{-2}$.

DEFINITION: Gradient

(Recall from Mathematics) The gradient, m , of a line can be calculated by dividing the change in the y -value (dependent variable) by the change in the x -value (independent variable). $m = \frac{\Delta y}{\Delta x}$

Tip

The gradient of a position vs. time graph gives the average velocity, while the tangent of a position vs. time graph gives the instantaneous velocity.

Tip

The gradient of a velocity vs. time graph gives the average acceleration, while the tangent of a velocity vs. time graph gives the instantaneous acceleration.

Tip

The area under the velocity vs. time graph gives the displacement.

Since we know that velocity is the rate of change of position, we can confirm the value for the velocity vs. time graph, by calculating the gradient of the $\vec{x}$ vs. t graph.

If we calculate the gradient of the $\vec{x}$ vs. t graph for a stationary object we get:

$$\begin{aligned} v &= \frac{\Delta \vec{x}}{\Delta t} \\ &= \frac{\vec{x}_f - \vec{x}_i}{t_f - t_i} \\ &= \frac{2 \text{ m} - 2 \text{ m}}{120 \text{ s} - 60 \text{ s}} \text{ (initial position = final position)} \\ &= 0 \text{ m} \cdot \text{s}^{-1} \text{ (for the time that Vivian is stationary)} \end{aligned}$$

Similarly, we can confirm the value of the acceleration by calculating the gradient of the velocity vs. time graph.

If we calculate the gradient of the $\vec{v}$ vs. t graph for a stationary object we get:

$$\begin{aligned} a &= \frac{\Delta v}{\Delta t} \\ &= \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} \\ &= \frac{0 \text{ m} \cdot \text{s}^{-1} - 0 \text{ m} \cdot \text{s}^{-1}}{120 \text{ s} - 60 \text{ s}} \\ &= 0 \text{ m} \cdot \text{s}^{-2} \end{aligned}$$

Additionally, because the velocity vs. time graph is related to the position vs. time graph, we can use the area under the velocity vs. time graph to calculate the displacement of an object.

The displacement of the object is given by the area under the graph, which is 0 m. This is obvious, because the object is not moving.

Motion at Constant Velocity



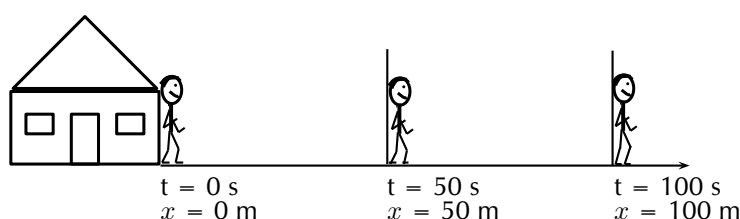
Motion at a constant velocity or *uniform motion* means that the position of the object is changing at the same rate.

Assume that Vivian takes 100 s to walk the 100 m to the taxi-stop every morning. If we assume that Vivian's house is the origin and the direction to the taxi is positive, then

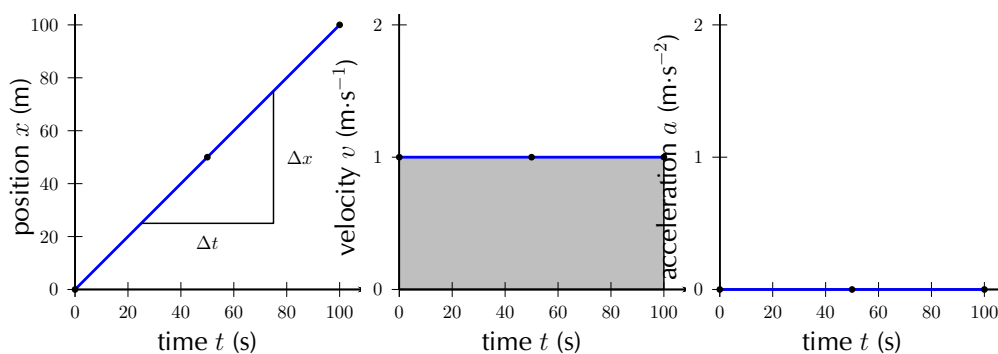
Vivian's velocity is:

$$\begin{aligned}
 v &= \frac{\Delta \vec{x}}{\Delta t} \\
 &= \frac{x_f - x_i}{t_f - t_i} \\
 &= \frac{100 \text{ m} - 0 \text{ m}}{100 \text{ s} - 0 \text{ s}} \\
 &= 1 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Vivian's velocity is $1 \text{ m} \cdot \text{s}^{-1}$. This means that she walked 1 m in the first second, another metre in the second second, and another in the third second, and so on. For example, after 50 s she will be 50 m from home. Her position increases by 1 m every 1 s. A diagram of Vivian's position is shown below:

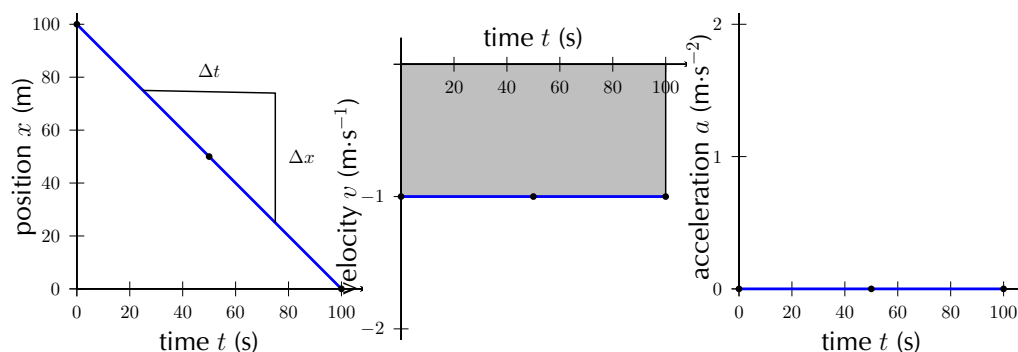


We can now draw graphs of position vs. time ($\vec{x}$ vs. t), velocity vs. time ($\vec{v}$ vs. t) and acceleration vs. time ($\vec{a}$ vs. t) for Vivian moving at a constant velocity. The graphs are shown here:



Graphs for motion at constant velocity (a) position vs. time (b) velocity vs. time (c) acceleration vs. time. The area of the shaded portion in the v vs. t graph corresponds to the object's displacement.

In the evening Vivian walks 100 m from the bus stop to her house in 100 s. Assume that Vivian's house is the origin. The following graphs can be drawn to describe the motion.



Graphs for motion with a constant negative velocity. The area of the shaded portion in the v vs. t graph corresponds to the object's displacement.

We see that the $\vec{v}$ vs. t graph is a horizontal line. If the velocity vs. time graph is a horizontal line, it means that the velocity is *constant* (not changing). Motion at a constant velocity is known as *uniform motion*. We can use the $\vec{x}$ vs. t to calculate the velocity by finding the gradient of the line.

$$\begin{aligned}
 v &= \frac{\Delta \vec{x}}{\Delta t} \\
 &= \frac{\vec{x}_f - \vec{x}_i}{t_f - t_i} \\
 &= \frac{0 \text{ m} - 100 \text{ m}}{100 \text{ s} - 0 \text{ s}} \\
 &= -1 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Vivian has a velocity of $-1 \text{ m} \cdot \text{s}^{-1}$, or $1 \text{ m} \cdot \text{s}^{-1}$ towards her house. You will notice that the $\vec{v}$ vs. t graph is a horizontal line corresponding to a velocity of $-1 \text{ m} \cdot \text{s}^{-1}$. The horizontal line means that the velocity stays the same (remains constant) during the motion. This is uniform velocity.

We can use the $\vec{v}$ vs. t to calculate the acceleration by finding the gradient of the line.

$$\begin{aligned}
 a &= \frac{\Delta \vec{v}}{\Delta t} \\
 &= \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} \\
 &= \frac{1 \text{ m} \cdot \text{s}^{-1} - 1 \text{ m} \cdot \text{s}^{-1}}{100 \text{ s} - 0 \text{ s}} \\
 &= 0 \text{ m} \cdot \text{s}^{-2}
 \end{aligned}$$

Vivian has an acceleration of $0 \text{ m} \cdot \text{s}^{-2}$. You will notice that the graph of $\vec{a}$ vs. t is a horizontal line corresponding to an acceleration value of $0 \text{ m} \cdot \text{s}^{-2}$. There is no acceleration during the motion because his velocity does not change.

We can use the $\vec{v}$ vs. t graph to calculate the displacement by finding the area under the graph.

$$\begin{aligned}
 \Delta \vec{x} &= \text{Area under graph} \\
 &= \ell \times b \\
 &= 100 \text{ } (-1) \\
 &= -100 \text{ m}
 \end{aligned}$$

This means that Vivian has a displacement of 100 m towards her house.

Exercise 21 - 5

1. Use the graphs in Figure 21.6 to calculate each of the following:
 - a. Calculate Vivian's velocity between 50 s and 100 s using the x vs. t graph. Hint: Find the gradient of the line.
 - b. Calculate Vivian's acceleration during the whole motion using the v vs. t graph.
 - c. Calculate Vivian's displacement during the whole motion using the v vs. t graph.
2. Thandi takes 200 s to walk 100 m to the bus stop every morning. In the evening Thandi takes 200 s to walk 100 m from the bus stop to her home.
 - a. Draw a graph of Thandi's position as a function of time for the morning (assuming that Thandi's home is the reference point). Use the gradient of the x vs. t graph to draw the graph of velocity vs. time. Use the gradient of the v vs. t graph to draw the graph of acceleration vs. time.
 - b. Draw a graph of Thandi's position as a function of time for the evening (assuming that Thandi's home is the origin). Use the gradient of the x vs. t graph to draw the graph of velocity vs. time. Use the gradient of the v vs. t graph to draw the graph of acceleration vs. time.
 - c. Discuss the differences between the two sets of graphs in questions 2 and 3.

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(1.) 009t (2.) 009u

General experiment: Motion at constant velocity

Aim:

To measure the position and time during motion at constant velocity and determine the average velocity as the gradient of a "Position vs. Time" graph.

Apparatus:

A battery operated toy car, stopwatch, meter stick or measuring tape.

Method:

1. Work with a friend. Copy the table below into your workbook.
2. Complete the table by timing the car as it travels each distance.
3. Time the car twice for each distance and take the average value as your accepted time.
4. Use the distance and average time values to plot a graph of "Distance vs. Time" *onto graph paper*. Stick the graph paper into your workbook. (Remember that "A vs. B" always means "y vs. x").
5. Insert all axis labels and units onto your graph.
6. Draw the best straight line through your data points.
7. Find the gradient of the straight line. This is the average velocity.

Results:

Distance (m)	Time (s)		
	1	2	Ave.
0			
0,5			
1,0			
1,5			
2,0			
2,5			
3,0			

Conclusions:

Answer the following questions in your workbook:

1. Did the car travel with a constant velocity?

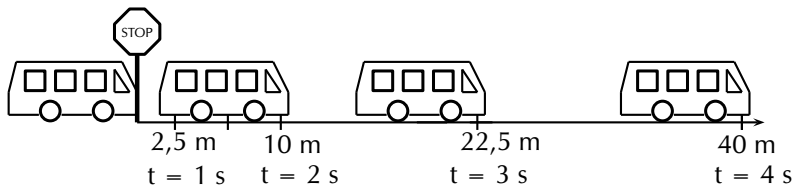
2. How can you tell by looking at the “Distance vs. Time” graph if the velocity is constant?
3. How would the “Distance vs. Time” graph look for a car with a faster velocity?
4. How would the “Distance vs. Time” graph look for a car with a slower velocity?

Motion at constant acceleration



The final situation we will be studying is motion at *constant acceleration*. We know that acceleration is the rate of change of velocity. So, if we have a constant acceleration, this means that the velocity changes at a constant rate.

Let's look at our first example of Vivian waiting at the taxi stop again. A taxi arrived and Vivian got in. The taxi stopped at the stop street and then accelerated in the positive direction as follows: After 1 s the taxi covered a distance of 2,5 m, after 2 s it covered 10 m, after 3 s it covered 22,5 m and after 4 s it covered 40 m. The taxi is covering a larger distance every second. This means that it is accelerating.



To calculate the velocity of the taxi you need to calculate the gradient of the line at each second:

$$\begin{aligned}
 \vec{v}_{1s} &= \frac{\Delta \vec{x}}{\Delta t} \\
 &= \frac{\vec{x}_f - \vec{x}_i}{t_f - t_i} \\
 &= \frac{5 \text{ m} - 0 \text{ m}}{1,5 \text{ s} - 0,5 \text{ s}} \\
 &= 5 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \vec{v}_{2s} &= \frac{\Delta \vec{x}}{\Delta t} \\
 &= \frac{\vec{x}_f - \vec{x}_i}{t_f - t_i} \\
 &= \frac{15 \text{ m} - 5 \text{ m}}{2,5 \text{ s} - 1,5 \text{ s}} \\
 &= 10 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 \vec{v}_{3s} &= \frac{\Delta \vec{x}}{\Delta t} \\
 &= \frac{\vec{x}_f - \vec{x}_i}{t_f - t_i} \\
 &= \frac{30 \text{ m} - 15 \text{ m}}{3,5 \text{ s} - 2,5 \text{ s}} \\
 &= 15 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

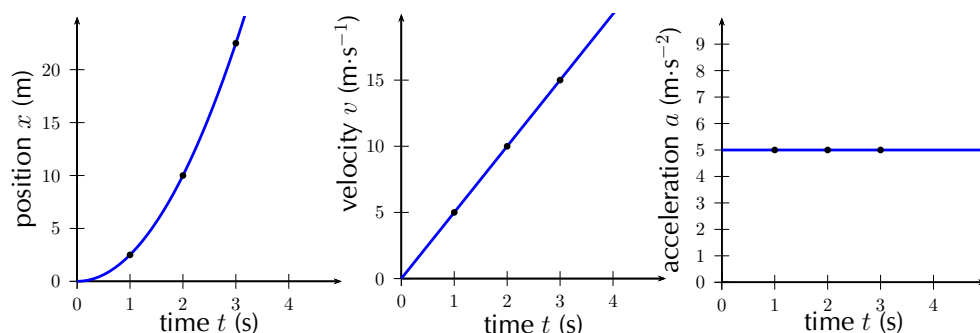
From these velocities, we can draw the velocity-time graph which forms a straight line.

The acceleration is the gradient of the v vs. t graph and can be calculated as follows:

$$\begin{aligned}
 a &= \frac{\Delta \vec{v}}{\Delta t} \\
 &= \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i} \\
 &= \frac{15 \text{ m} \cdot \text{s}^{-1} - 5 \text{ m} \cdot \text{s}^{-1}}{3 \text{ s} - 1 \text{ s}} \\
 &= 5 \text{ m} \cdot \text{s}^{-2}
 \end{aligned}$$

The acceleration does not change during the motion (the gradient stays constant). This is motion at constant or uniform acceleration.

The graphs for this situation are shown below:



Graphs for motion with a constant acceleration starting from rest.

Velocity from acceleration vs. time graphs

Just as we used velocity vs. time graphs to find displacement, we can use acceleration vs. time graphs to find the velocity of an object at a given moment in time. We simply calculate the area under the acceleration vs. time graph, at a given time. In the graph below, showing an object at a constant positive acceleration, the increase in velocity of the object after 2 seconds corresponds to the shaded portion.

$$\begin{aligned}
 v &= \text{area of rectangle} = a \times \Delta t \\
 &= 5 \text{ m} \cdot \text{s}^{-2} \times 2 \text{ s} \\
 &= 10 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

The velocity of the object at $t = 2 \text{ s}$ is therefore $10 \text{ m} \cdot \text{s}^{-1}$.

Summary of Graphs

The relation between graphs of position, velocity and acceleration as functions of time is summarised in the next figure.

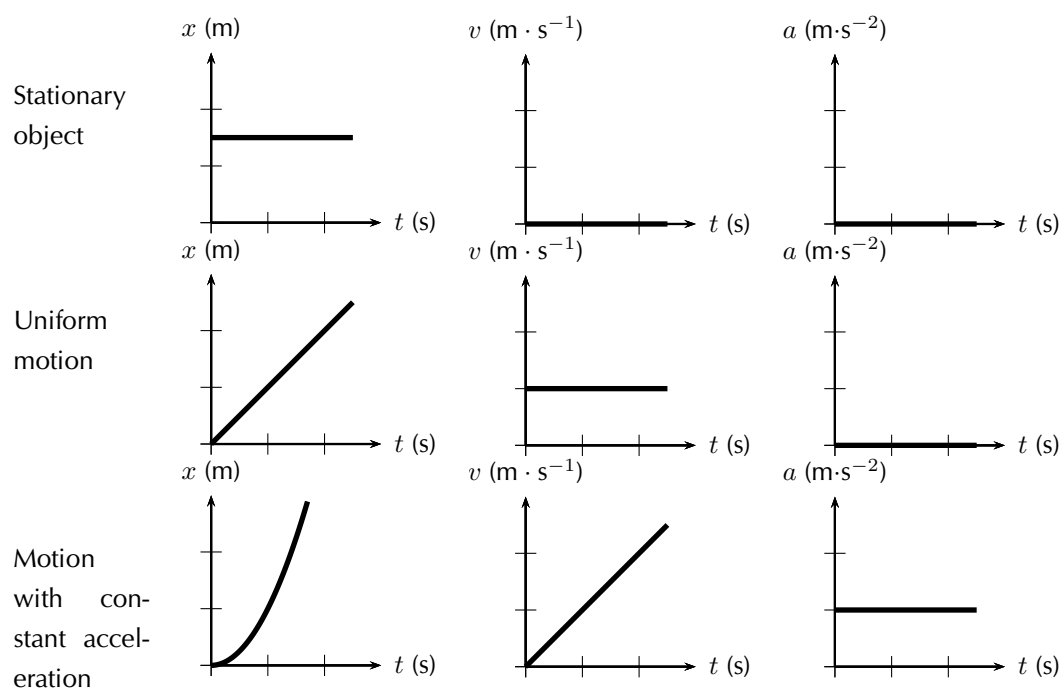


Figure 21.3: Position-time, velocity-time and acceleration-time graphs.

Tip

The description of the motion represented by a graph should include the following (where possible):

1. whether the object is moving in the positive or negative direction
2. whether the object is at rest, moving at constant velocity or moving at constant positive acceleration (speeding up) or constant negative acceleration (slowing down)

You will also often be required to draw graphs based on a description of the motion in words or from a diagram. Remember that these are just different methods of presenting the same information. If you keep in mind the general shapes of the graphs for the different types of motion, there should not be any difficulty with explaining what is happening.

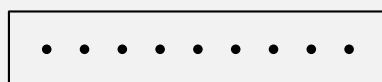
Formal experiment: *Position versus time using a ticker timer*

Aim:

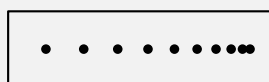
To measure the position and time during motion and to use that data to plot a "Position vs. Time" graph.

Apparatus:

Trolley, ticker tape apparatus, tape, graph paper, ruler, ramp



Motion at constant velocity



Motion with increasing velocity

Method:

1. Work with a friend. Copy the table below into your workbook.
2. Attach a length of tape to the trolley.
3. Run the other end of the tape through the ticker timer.
4. Start the ticker timer going and roll the trolley down the ramp.
5. Repeat steps 1 - 3.
6. On each piece of tape, measure the distance between successive dots. Note these distances in the table below.
7. Use the frequency of the ticker timer to work out the time intervals between successive dots. Note these times in the table below.
8. Work out the average values for distance and time.
9. Use the average distance and average time values to plot a graph of "Distance vs. Time" *onto graph paper*. Stick the graph paper into your workbook. (Remember that "A vs. B" always means "y vs. x").
10. Insert all axis labels and units onto your graph.
11. Draw the best straight line through your data points.

Results:

Distance (m)			Time (s)		
1	2	Ave.	1	2	Ave.

Discussion:
Describe the motion of the trolley down the ramp.

Worked examples

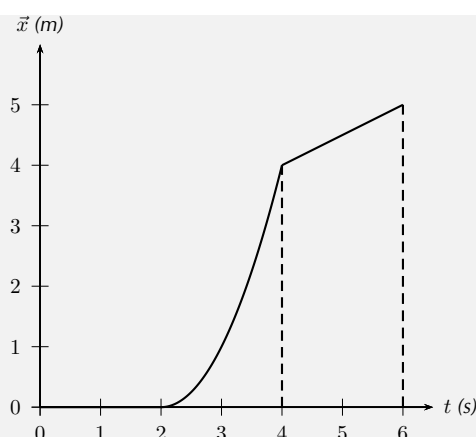


The worked examples in this section demonstrate the types of questions that can be asked about graphs.

Example 3: *Description of motion based on a position-time graph*

QUESTION

The position vs. time graph for the motion of a car is given below. Draw the corresponding velocity vs. time and acceleration vs. time graphs, and then describe the motion of the car.



SOLUTION

Step 1 : Identify what information is given and what is asked for

The question gives a position vs. time graph and the following three things are required:

1. Draw a v vs. t graph.
2. Draw an a vs. t graph.
3. Describe the motion of the car.

To answer these questions, break the motion up into three sections: 0 – 2 seconds, 2 – 4 seconds and 4 – 6 seconds.

Step 2 : Velocity vs. time graph for 0 – 2 seconds

For the first 2 seconds we can see that the position (and hence the displacement) remains constant - so the object is not moving, thus it has zero velocity during this time. We can reach this conclusion by another path too: remember that the gradient of a displacement vs. time graph is the velocity. For the first 2 seconds we can see that the displacement vs. time graph is a horizontal line, i.e.. it has a gradient of zero. Thus the velocity during this time is zero and the object is stationary.

Step 3 : Velocity vs. time graph for 2 – 4 seconds

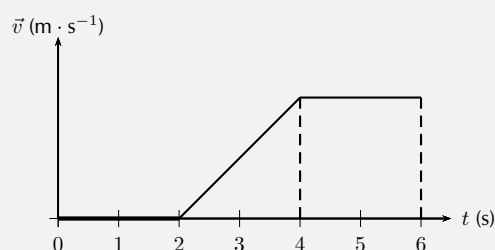
For the next 2 seconds, displacement is increasing with time so the object is moving. Looking at the gradient of the displacement graph we can see that it is not constant. In fact, the slope is getting steeper (the gradient is increasing) as time goes on. Thus, remembering that the gradient of a displacement vs. time graph is the velocity, the velocity must be increasing with time during this phase.

Step 4 : Velocity vs. time graph for 4 – 6 seconds

For the final 2 seconds we see that displacement is still increasing with time, but this time the gradient is constant, so we know that the object

is now travelling at a constant velocity, thus the velocity vs. time graph will be a horizontal line during this stage. We can now draw the graphs:

So our velocity vs. time graph looks like this one below. Because we haven't been given any values on the vertical axis of the displacement vs. time graph, we cannot figure out what the exact gradients are and therefore what the values of the velocities are. In this type of question it is just important to show whether velocities are positive or negative, increasing, decreasing or constant.



Once we have the velocity vs. time graph its much easier to get the acceleration vs. time graph as we know that the gradient of a velocity vs. time graph is the just the acceleration.

Step 5 : Acceleration vs. time graph for 0 – 2 seconds

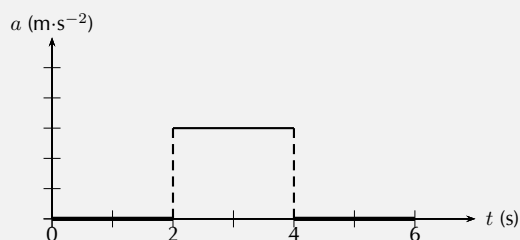
For the first 2 seconds the velocity vs. time graph is horizontal and has a value of zero, thus it has a gradient of zero and there is no acceleration during this time. (This makes sense because we know from the displacement time graph that the object is stationary during this time, so it can't be accelerating).

Step 6 : Acceleration vs. time graph for 2 – 4 seconds

For the next 2 seconds the velocity vs. time graph has a positive gradient. This gradient is not changing (i.e. its constant) throughout these 2 seconds so there must be a constant positive acceleration.

Step 7 : Acceleration vs. time graph for 4 – 6 seconds

For the final 2 seconds the object is travelling with a constant velocity. During this time the gradient of the velocity vs. time graph is once again zero, and thus the object is not accelerating. The acceleration vs. time graph looks like this:



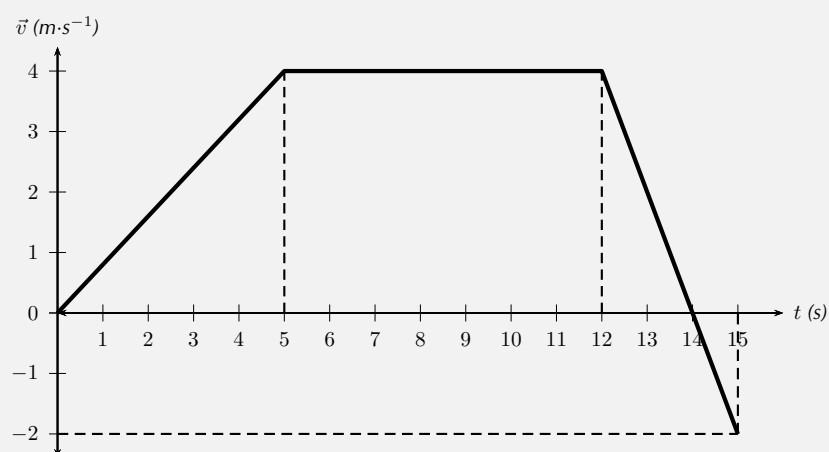
Step 8 : A description of the object's motion

A brief description of the motion of the object could read something like this: At $t = 0$ s and object is stationary at some position and remains stationary until $t = 2$ s when it begins accelerating. It accelerates in a positive direction for 2 seconds until $t = 4$ s and then travels at a constant velocity for a further 2 seconds.

Example 4: Calculations from a velocity vs. time graph

QUESTION

The velocity vs. time graph of a truck is plotted below. Calculate the distance and displacement of the truck after 15 seconds.



SOLUTION

Step 1 : **Decide how to tackle the problem**

We are asked to calculate the distance and displacement of the car. All we need to remember here is that we can use the area between the velocity vs. time graph and the time axis to determine the distance and displacement.

Step 2 : **Determine the area under the velocity vs. time graph**

Break the motion up: 0 – 5 seconds, 5 – 12 seconds, 12 – 14 seconds and 14 – 15 seconds.

For 0 – 5 seconds: The displacement is equal to the area of the triangle on the left:

$$\begin{aligned}\text{Area}_{\triangle} &= \frac{1}{2} b \times h \\ &= \frac{1}{2} \times 5 \text{ s} \times 4 \text{ m} \cdot \text{s}^{-1} \\ &= 10 \text{ m}\end{aligned}$$

For 12 – 14 seconds the displacement is equal to the area of the triangle above the time axis on the right:

$$\begin{aligned}\text{Area}_{\triangle} &= \frac{1}{2} b \times h \\ &= \frac{1}{2} \times 2 \text{ s} \times 4 \text{ m} \cdot \text{s}^{-1} \\ &= 4 \text{ m}\end{aligned}$$

For 5 – 12 seconds: The displacement is equal to the area of the rectangle:

$$\begin{aligned}\text{Area}_{\square} &= \ell \times b \\ &= 7 \text{ s} \times 4 \text{ m} \cdot \text{s}^{-1} \\ &= 28 \text{ m}^2\end{aligned}$$

For 14 – 15 seconds the displacement is equal to the area of the triangle below the time axis:

$$\begin{aligned}\text{Area}_{\triangle} &= \frac{1}{2} b \times h \\ &= \frac{1}{2} \times 1 \text{ s} \times 2 \text{ m} \cdot \text{s}^{-1} \\ &= 1 \text{ m}\end{aligned}$$

Step 3 : Determine the total distance of the car

Now the total distance of the car is the sum of all of these areas:

$$\begin{aligned}D &= 10 \text{ m} + 28 \text{ m} + 4 \text{ m} + 1 \text{ m} \\ &= 43 \text{ m}\end{aligned}$$

Step 4 : Determine the total displacement of the car

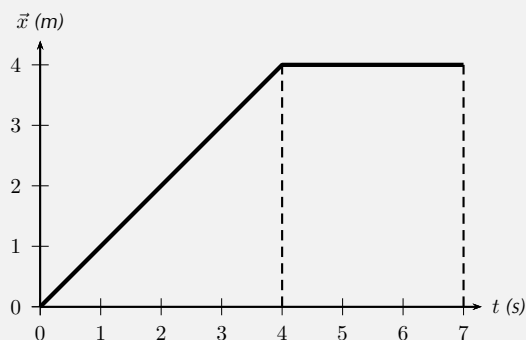
Now the total displacement of the car is just the sum of all of these areas. HOWEVER, because in the last second (from $t = 14 \text{ s}$ to $t = 15 \text{ s}$) the velocity of the car is negative, it means that the car was going in the opposite direction, i.e. back where it came from! So, to find the total displacement, we have to add the first 3 areas (those with positive displacements) and subtract the last one (because it is a displacement in the opposite direction).

$$\begin{aligned}\Delta \vec{x} &= 10 \text{ m} + 28 \text{ m} + 4 \text{ m} - 1 \text{ m} \\ &= 41 \text{ m in the positive direction}\end{aligned}$$

Example 5: Velocity from a position vs. time graph

QUESTION

The position vs. time graph below describes the motion of an athlete.



1. What is the velocity of the athlete during the first 4 seconds?
2. What is the velocity of the athlete from $t = 4$ s to $t = 7$ s?

SOLUTION

Step 1 : The velocity during the first 4 seconds

The velocity is given by the gradient of a position vs. time graph. During the first 4 seconds, this is

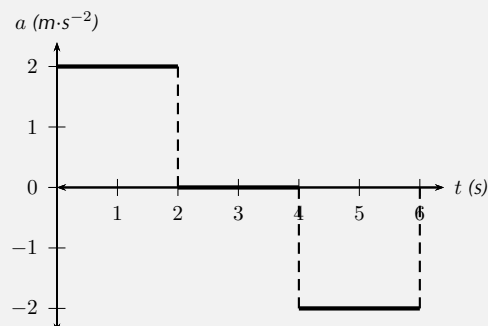
$$\begin{aligned}\vec{v} &= \frac{\Delta \vec{x}}{\Delta t} \\ &= \frac{4 \text{ m} - 0 \text{ m}}{4 \text{ s} - 0 \text{ s}} \\ &= 1 \text{ m} \cdot \text{s}^{-1}\end{aligned}$$

Step 2 : The velocity during the last 3 seconds

For the last 3 seconds we can see that the displacement stays constant. The graph shows a horizontal line and therefore the gradient is zero. Thus $v = 0 \text{ m} \cdot \text{s}^{-1}$.

Example 6: Drawing a v vs. t graph from an a vs. t graph
QUESTION

The acceleration vs. time graph for a car starting from rest, is given below. Calculate the velocity of the car and hence draw the velocity vs. time graph.


SOLUTION

Step 1 : Calculate the velocity values by using the area under each part of the graph.

The motion of the car can be divided into three time sections: 0 – 2 seconds; 2 – 4 seconds and 4 – 6 seconds. To be able to draw the velocity vs. time graph, the velocity for each time section needs to be calculated. The velocity is equal to the area of the square under the graph:

For 0 – 2 seconds:

$$\begin{aligned} \text{Area}_{\square} &= \ell \times b \\ &= 2 \text{ s} \times 2 \text{ m} \cdot \text{s}^{-2} \\ &= 4 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

The velocity of the car is $4 \text{ m} \cdot \text{s}^{-1}$ at $t = 2 \text{ s}$.

For 2 – 4 seconds:

$$\begin{aligned} \text{Area}_{\square} &= \ell \times b \\ &= 2 \text{ s} \times 0 \text{ m} \cdot \text{s}^{-2} \\ &= 0 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

The velocity of the car is $0 \text{ m} \cdot \text{s}^{-1}$ from $t = 2 \text{ s}$ to $t = 4 \text{ s}$.

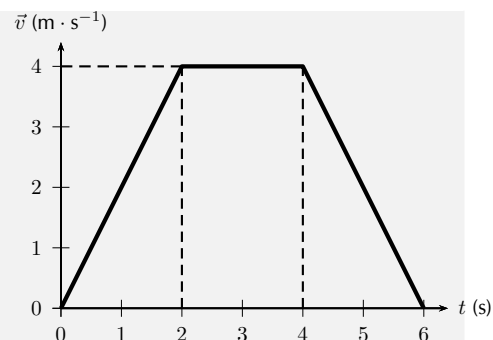
For 4 – 6 seconds:

$$\begin{aligned} \text{Area}_{\square} &= \ell \times b \\ &= 2 \text{ s} \times -2 \text{ m} \cdot \text{s}^{-2} \\ &= -4 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

The acceleration had a negative value, which means that the velocity is decreasing. It starts at a velocity of $4 \text{ m} \cdot \text{s}^{-1}$ and decreases to $0 \text{ m} \cdot \text{s}^{-1}$.

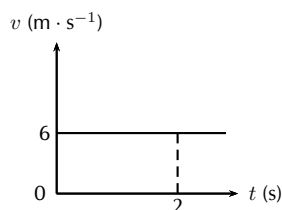
Step 2 : Now use the values to draw the velocity vs. time graph.

The velocity vs. time graph looks like this:

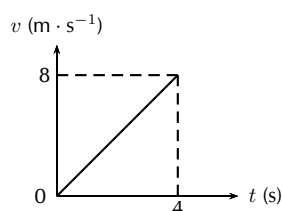


Exercise 21 - 6

1. A car is parked 10 m from home for 10 minutes. Draw a displacement-time, velocity-time and acceleration-time graphs for the motion. Label all the axes.
2. A bus travels at a constant velocity of $12 \text{ m} \cdot \text{s}^{-1}$ for 6 seconds. Draw the displacement-time, velocity-time and acceleration-time graph for the motion. Label all the axes.
3. An athlete runs with a constant acceleration of $1 \text{ m} \cdot \text{s}^{-2}$ for 4 s. Draw the acceleration-time, velocity-time and displacement time graphs for the motion. Accurate values are only needed for the acceleration-time and velocity-time graphs.
4. The following velocity-time graph describes the motion of a car. Draw the displacement-time graph and the acceleration-time graph and explain the motion of the car according to the three graphs.



5. The following velocity-time graph describes the motion of a truck. Draw the displacement-time graph and the acceleration-time graph and explain the motion of the truck according to the three graphs.



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(1.) 009v (2.) 009w (3.) 009x (4.) 009y (5.) 009z

See simulation: Simulation: VPrwb at www.everythingscience.co.za

Equations of motion



In this section we will look at the third way to describe motion. We have looked at describing motion in terms of words and graphs. In this section we examine equations that can be used to describe motion.

This section is about solving problems relating to uniformly accelerated motion. In other words, motion at constant acceleration.

The following are the variables that will be used in this section:

$$\begin{aligned}
 \vec{v}_i &= \text{initial velocity (m} \cdot \text{s}^{-1}\text{) at } t = 0 \text{ s} \\
 \vec{v}_f &= \text{final velocity (m} \cdot \text{s}^{-1}\text{) at time } t \\
 \Delta \vec{x} &= \text{displacement (m)} \\
 t &= \text{time (s)} \\
 \Delta t &= \text{time interval (s)} \\
 \vec{a} &= \text{acceleration (m} \cdot \text{s}^{-2}\text{)}
 \end{aligned}$$

An alternate convention for some of the variables exists that you will likely encounter so

FACT

Galileo Galilei of Pisa, Italy, was the first to determine the correct mathematical law for acceleration: the total distance covered, starting from rest, is proportional to the square of the time. He also concluded that objects retain their velocity unless a force – often friction – acts upon them, refuting the accepted Aristotelian hypothesis that objects “naturally” slow down and stop unless a force acts upon them. This principle was incorporated into Newton’s laws of motion (1st law).

here is a list for reference purposes:

$$\vec{u} = \text{initial velocity (m} \cdot \text{s}^{-1}\text{) at } t = 0 \text{ s}$$

$$\vec{v} = \text{final velocity (m} \cdot \text{s}^{-1}\text{) at time } t$$

$$\vec{s} = \text{displacement (m)}$$

In this book we will use the first convention.

$$\vec{v}_f = \vec{v}_i + \vec{a}t \quad (21.1)$$

$$\Delta\vec{x} = \frac{(\vec{v}_i + \vec{v}_f)}{2}t \quad (21.2)$$

$$\Delta\vec{x} = \vec{v}_i t + \frac{1}{2}\vec{a}t^2 \quad (21.3)$$

$$v_f^2 = v_i^2 + 2\vec{a}\Delta\vec{x} \quad (21.4)$$

The questions can vary a lot, but the following method for answering them will always work. Use this when attempting a question that involves motion with constant acceleration. You need any three known quantities ($\vec{v}_i$, $\vec{v}_f$, $\Delta\vec{x}$, t or $\vec{a}$) to be able to calculate the fourth one.

Problem solving strategy:

1. Read the question carefully to identify the quantities that are given. Write them down.
2. Identify the equation to use. *Write it down!!!*
3. Ensure that all the values are in the correct units and fill them in your equation.
4. Calculate the answer and check your units.

Example 7: Equations of motion**QUESTION**

A racing car is travelling North. It accelerates uniformly covering a distance of 725 m in 10 s. If it has an initial velocity of $10 \text{ m} \cdot \text{s}^{-1}$, find its acceleration.

SOLUTION

Step 1 : **Identify what information is given and what is asked for**

We are given:

$$\begin{aligned}\vec{v}_i &= 10 \text{ m} \cdot \text{s}^{-1} \\ \Delta \vec{x} &= 725 \text{ m} \\ t &= 10 \text{ s} \\ \vec{a} &= ?\end{aligned}$$

Step 2 : Find an equation of motion relating the given information to the acceleration

If you struggle to find the correct equation, find the quantity that is not given and then look for an equation that has this quantity in it.

We can use Equation 21.3

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$

Step 3 : Substitute your values in and find the answer

$$\begin{aligned}\Delta \vec{x} &= \vec{v}_i t + \frac{1}{2} \vec{a} t^2 \\ 725 \text{ m} &= (10 \text{ m} \cdot \text{s}^{-1} \times 10 \text{ s}) + \frac{1}{2} \vec{a} \times (10 \text{ s})^2 \\ 725 \text{ m} - 100 \text{ m} &= (50 \text{ s}^2) \vec{a} \\ \vec{a} &= 12,5 \text{ m} \cdot \text{s}^{-2}\end{aligned}$$

Step 4 : Quote the final answer

The racing car is accelerating at $12,5 \text{ m} \cdot \text{s}^{-2}$ North.

Example 8: Equations of motion I

QUESTION

A motorcycle, travelling East, starts from rest, moves in a straight line with a constant acceleration and covers a distance of 64 m in 4 s. Calculate

- its acceleration
- its final velocity
- at what time the motorcycle had covered half the total distance
- what distance the motorcycle had covered in half the total time.

SOLUTION

Step 1 : **Identify what information is given and what is asked for**

We are given:

$$\vec{v}_i = 0 \text{ m} \cdot \text{s}^{-1} \text{ (because the object starts from rest.)}$$

$$\Delta \vec{x} = 64 \text{ m}$$

$$t = 4 \text{ s}$$

$$\vec{a} = ?$$

$$\vec{v}_f = ?$$

$$t = ? \text{ at half the distance } \Delta \vec{x} = 32 \text{ m.}$$

$$\Delta \vec{x} = ? \text{ at half the time } t = 2 \text{ s.}$$

All quantities are in SI units.

Step 2 : **Acceleration: Find a suitable equation to calculate the acceleration**

We can use Equations 21.3

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$

Step 3 : **Substitute the values and calculate the acceleration**

$$\Delta \vec{x} = \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$

$$64 \text{ m} = (0 \text{ m} \cdot \text{s}^{-1} \times 4 \text{ s}) + \frac{1}{2} \vec{a} \times (4 \text{ s})^2$$

$$64 \text{ m} = (8 \text{ s}^2) \vec{a}$$

$$\vec{a} = 8 \text{ m} \cdot \text{s}^{-2} \text{ East}$$

Step 4 : **Final velocity: Find a suitable equation to calculate the final velocity**

We can use Equation 21.1 - remember we now also know the acceleration of the object.

$$\vec{v}_f = \vec{v}_i + \vec{a} t$$

Step 5 : **Substitute the values and calculate the final velocity**

$$\begin{aligned}\vec{v}_f &= \vec{v}_i + at \\ \vec{v}_f &= 0 \text{ m} \cdot \text{s}^{-1} + (8 \text{ m} \cdot \text{s}^{-2})(4 \text{ s}) \\ &= 32 \text{ m} \cdot \text{s}^{-1} \text{ East}\end{aligned}$$

Step 6 : **Time at half the distance: Find an equation to calculate the time**

We can use Equation 21.3:

$$\begin{aligned}\Delta \vec{x} &= \vec{v}_i t + \frac{1}{2} \vec{a} t^2 \\ 32 \text{ m} &= (0 \text{ m} \cdot \text{s}^{-1})t + \frac{1}{2}(8 \text{ m} \cdot \text{s}^{-2})(t)^2 \\ 32 \text{ m} &= 0 + (4 \text{ m} \cdot \text{s}^{-2})t^2 \\ 8 \text{ s}^2 &= t^2 \\ t &= 2,83 \text{ s}\end{aligned}$$

Step 7 : **Distance at half the time: Find an equation to relate the distance and time**

Half the time is 2 s, thus we have $\vec{v}_i$, $\vec{a}$ and t - all in the correct units. We can use Equation 21.3 to get the distance:

$$\begin{aligned}\Delta \vec{x} &= \vec{v}_i t + \frac{1}{2} \vec{a} t^2 \\ &= (0 \text{ m} \cdot \text{s}^{-1})(2 \text{ s}) + \frac{1}{2}(8 \text{ m} \cdot \text{s}^{-2})(2 \text{ s})^2 \\ &= 16 \text{ m East}\end{aligned}$$

Exercise 21 - 7

1. A car starts off at $10 \text{ m} \cdot \text{s}^{-1}$ and accelerates at $1 \text{ m} \cdot \text{s}^{-2}$ for 10 s. What is its final velocity?

2. A train starts from rest, and accelerates at $1 \text{ m}\cdot\text{s}^{-2}$ for 10 s. How far does it move?
3. A bus is going $30 \text{ m}\cdot\text{s}^{-1}$ and stops in 5 s. What is its stopping distance for this speed?
4. A racing car going at $20 \text{ m}\cdot\text{s}^{-1}$ stops in a distance of 20 m. What is its acceleration?
5. A ball has a uniform acceleration of $4 \text{ m}\cdot\text{s}^{-1}$. Assume the ball starts from rest. Determine the velocity and displacement at the end of 10 s.
6. A motorcycle has a uniform acceleration of $4 \text{ m}\cdot\text{s}^{-1}$. Assume the motorcycle has an initial velocity of $20 \text{ m}\cdot\text{s}^{-1}$. Determine the velocity and displacement at the end of 12 s.
7. An aeroplane accelerates uniformly such that it goes from rest to $144 \text{ km}\cdot\text{hr}^{-1}$ in 8 s. Calculate the acceleration required and the total distance that it has travelled in this time.

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(1.) 00a0 (2.) 00a1 (3.) 00a2 (4.) 00a3 (5.) 00a4 (6.) 00a5
(7.) 00a6

Extension: Finding the equations of motion



The following does not form part of the syllabus and can be considered additional information.

Derivation of 21.1

According to the definition of acceleration:

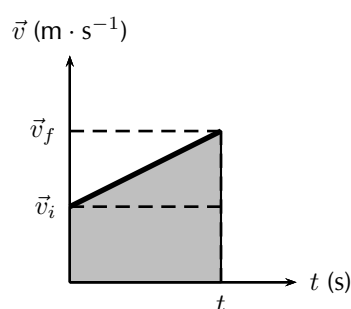
$$\vec{a} = \frac{\Delta \vec{v}}{t}$$

where $\Delta \vec{v}$ is the change in velocity, i.e. $\Delta v = \vec{v}_f - \vec{v}_i$. Thus we have

$$\begin{aligned}\vec{a} &= \frac{\vec{v}_f - \vec{v}_i}{t} \\ \vec{v}_f &= \vec{v}_i + \vec{a}t\end{aligned}$$

Derivation of 21.2

We have seen that displacement can be calculated from the area under a velocity vs. time graph. For *uniformly accelerated motion* the most complicated velocity vs. time graph we can have is a straight line. Look at the graph below - it represents an object with a starting velocity of $\vec{v}_i$, accelerating to a final velocity $\vec{v}_f$ over a total time t .



To calculate the final displacement we must calculate the area under the graph - this is just the area of the rectangle added to the area of the triangle. This portion of the graph has been shaded for clarity.

$$\begin{aligned}\text{Area}_{\triangle} &= \frac{1}{2}b \times h \\ &= \frac{1}{2}t \times (v_f - v_i) \\ &= \frac{1}{2}v_f t - \frac{1}{2}v_i t\end{aligned}$$

$$\begin{aligned}\text{Area}_{\square} &= \ell \times b \\ &= t \times v_i \\ &= v_i t\end{aligned}$$

$$\begin{aligned}
 \text{Displacement} &= \text{Area}_{\square} + \text{Area}_{\triangle} \\
 \Delta \vec{x} &= v_i t + \frac{1}{2} v_f t - \frac{1}{2} v_i t \\
 \Delta \vec{x} &= \frac{(v_i + v_f)}{2} t
 \end{aligned}$$

Derivation of 21.3

This equation is simply derived by eliminating the final velocity v_f in 21.2. Remembering from 21.1 that

$$\vec{v}_f = \vec{v}_i + \vec{a}t$$

then 21.2 becomes

$$\begin{aligned}
 \Delta \vec{x} &= \frac{\vec{v}_i + \vec{v}_i + \vec{a}t}{2} t \\
 &= \frac{2\vec{v}_i t + \vec{a}t^2}{2} \\
 \Delta \vec{x} &= \vec{v}_i t + \frac{1}{2} \vec{a}t^2
 \end{aligned}$$

Derivation of 21.4

This equation is just derived by eliminating the time variable in the above equation. From 21.1 we know

$$t = \frac{\vec{v}_f - \vec{v}_i}{a}$$

Substituting this into 21.3 gives

$$\begin{aligned}
 \Delta \vec{x} &= \vec{v}_i \left(\frac{\vec{v}_f - \vec{v}_i}{a} \right) + \frac{1}{2} a \left(\frac{\vec{v}_f - \vec{v}_i}{a} \right)^2 \\
 &= \frac{\vec{v}_i \vec{v}_f}{a} - \frac{\vec{v}_i^2}{a} + \frac{1}{2} a \left(\frac{\vec{v}_f^2 - 2\vec{v}_i \vec{v}_f + \vec{v}_i^2}{a^2} \right) \\
 &= \frac{\vec{v}_i \vec{v}_f}{a} - \frac{\vec{v}_i^2}{a} + \frac{\vec{v}_f^2}{2a} - \frac{\vec{v}_i \vec{v}_f}{a} + \frac{\vec{v}_i^2}{2a} \\
 2a\Delta \vec{x} &= -2\vec{v}_i^2 + \vec{v}_f^2 + \vec{v}_i^2 \\
 \vec{v}_f^2 &= \vec{v}_i^2 + 2a\Delta \vec{x}
 \end{aligned}$$

This gives us the final velocity in terms of the initial velocity, acceleration and displacement and is independent of the time variable.

Applications in the real-world



What we have learnt in this chapter can be directly applied to road safety. We can analyse the relationship between speed and stopping distance. The following worked example illustrates this application.

Example 9: Stopping distance

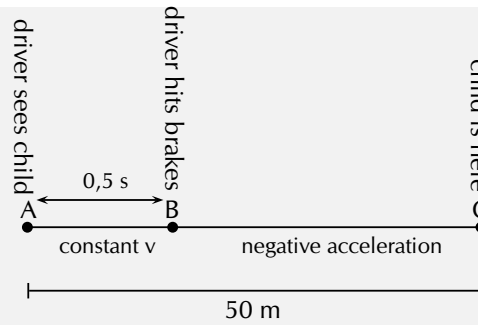
QUESTION

A truck is travelling at a constant velocity of $10 \text{ m} \cdot \text{s}^{-1}$ when the driver sees a child 50 m in front of him in the road. He hits the brakes to stop the truck. The truck accelerates at a rate of $-1.25 \text{ m} \cdot \text{s}^{-2}$. His reaction time to hit the brakes is 0,5 seconds. Will the truck hit the child?

SOLUTION

Step 1 : **Analyse the problem and identify what information is given**

It is useful to draw a time-line like this one:



We need to know the following:

- What distance the driver covers before hitting the brakes.
- How long it takes the truck to stop after hitting the brakes.
- What total distance the truck covers to stop.

Step 2 : Calculate the distance AB

Before the driver hits the brakes, the truck is travelling at constant velocity. There is no acceleration and therefore the equations of motion are not used. To find the distance travelled, we use:

$$v = \frac{d}{t}$$

$$10 = \frac{d}{0,5}$$

$$d = 5 \text{ m}$$

The truck covers 5 m before the driver hits the brakes.

Step 3 : Calculate the time BC

We have the following for the motion between B and C:

$$\vec{v}_i = 10 \text{ m} \cdot \text{s}^{-1}$$

$$\vec{v}_f = 0 \text{ m} \cdot \text{s}^{-1}$$

$$a = -1,25 \text{ m} \cdot \text{s}^{-2}$$

$$t = ?$$

We can use Equation 21.1

$$\vec{v}_f = \vec{v}_i + at$$

$$0 = 10 \text{ m} \cdot \text{s}^{-1} + (-1,25 \text{ m} \cdot \text{s}^{-2})t$$

$$-10 \text{ m} \cdot \text{s}^{-1} = (-1,25 \text{ m} \cdot \text{s}^{-2})t$$

$$t = 8 \text{ s}$$

Step 4 : Calculate the distance BC

For the distance we can use Equation 21.2 or Equation 21.3. We will

use Equation 21.2:

$$\begin{aligned}\Delta \vec{x} &= \frac{(\vec{v}_i + \vec{v}_f)}{2} t \\ \Delta \vec{x} &= \frac{10 + 0}{s}(8) \\ \Delta \vec{x} &= 40 \text{ m}\end{aligned}$$

Step 5 : Write the final answer

The total distance that the truck covers is $d_{AB} + d_{BC} = 5 + 40 = 45$ meters. The child is 50 meters ahead. The truck will not hit the child.

Chapter 21 | Summary

See the summary presentation (📺 Presentation: VPgjl at www.everythingscience.co.za)

- A reference point is a point from where you take your measurements.
- A frame of reference is a reference point with a set of directions.
- Your position is where you are located with respect to your reference point.
- The displacement of an object is how far it is from the reference point. It is the shortest distance between the object and the reference point. It has magnitude and direction because it is a vector.
- The distance of an object is the length of the path travelled from the starting point to the end point. It has magnitude only because it is a scalar.
- Speed (v) is the distance covered (D) divided by the time taken (Δt):

$$v = \frac{D}{\Delta t}$$

- Average velocity ($\vec{v}_{av}$) is the displacement ($\Delta \vec{x}$) divided by the time taken (Δt):

$$\vec{v}_{av} = \frac{\Delta \vec{x}}{\Delta t}$$

- Instantaneous speed is the speed at a specific instant in time.
- Instantaneous velocity is the velocity at a specific instant in time.

- Acceleration ($\vec{a}$) is the change in velocity ($\Delta\vec{v}$) over a time interval (Δt):

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$$

- The gradient of a position - time graph (x vs. t) gives the velocity.
- The gradient of a velocity - time graph (v vs. t) gives the acceleration.
- The area under a velocity - time graph (v vs. t) gives the displacement.
- The area under an acceleration - time graph (a vs. t) gives the velocity.
- The graphs of motion are summarised in Figure 21.3.
- The equations of motion are used where constant acceleration takes place:

$$v_f = v_i + at$$

$$\Delta\vec{x} = \frac{(v_i + v_f)}{2}t$$

$$\Delta\vec{x} = v_i t + \frac{1}{2}at^2$$

$$\vec{v}_f^2 = \vec{v}_i^2 + 2a\Delta\vec{x}$$

Physical Quantities			
Quantity	Vector	Unit name	Unit symbol
Position (x)	-	metre	m
Distance (D)	-	metre	m
Displacement ($\Delta\vec{x}$)	✓	metre	m
Speed (v_{av})	-	metre per second	$\text{m} \cdot \text{s}^{-1}$
Average velocity ($\vec{v}_{av}$)	✓	metre per second	$\text{m} \cdot \text{s}^{-1}$
Instantaneous velocity ($\vec{v}$)	✓	metre per second	$\text{m} \cdot \text{s}^{-1}$
Instantaneous speed (v)	-	metre per second	$\text{m} \cdot \text{s}^{-1}$
Instantaneous acceleration ($\vec{a}$)	✓	metre per second per second	$\text{m} \cdot \text{s}^{-2}$
Average acceleration ($\vec{a}_{av}$)	✓	metre per second per second	$\text{m} \cdot \text{s}^{-2}$
Magnitude of acceleration (a)	-	metre per second per second	$\text{m} \cdot \text{s}^{-2}$

Table 21.1: Units used in **motion in one dimension**

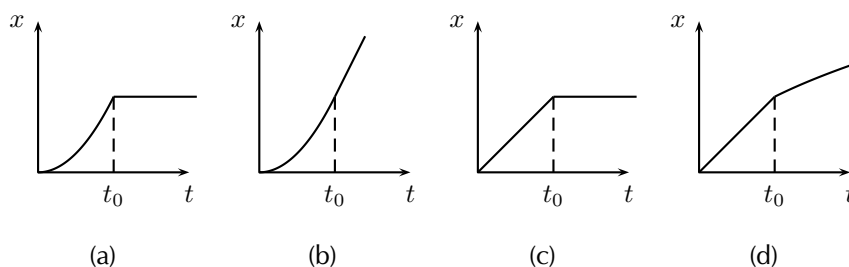
Chapter 21

End of chapter exercises

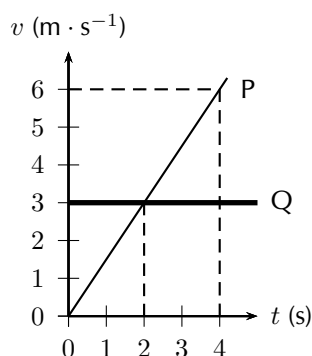
1. Give one word/term for the following descriptions.
 - a. The shortest path from start to finish.
 - b. A physical quantity with magnitude and direction.
 - c. The quantity defined as a change in velocity over a time period.
 - d. The point from where you take measurements.
 - e. The distance covered in a time interval.
 - f. The velocity at a specific instant in time.
2. Choose an item from column B that match the description in column A. Write down only the letter next to the question number. You may use an item from column B more than once.

Column A	Column B
a. The area under a velocity - time graph	gradient
b. The gradient of a velocity - time graph	area
c. The area under an acceleration - time graph	velocity
d. The gradient of a position - time graph	displacement
	acceleration
	slope

3. Indicate whether the following statements are TRUE or FALSE. Write only "true" or "false". If the statement is false, write down the correct statement.
 - a. A scalar is the displacement of an object over a time interval.
 - b. The position of an object is where it is located.
 - c. The sign of the velocity of an object tells us in which direction it is travelling.
 - d. The acceleration of an object is the change of its displacement over a period in time.
4. [SC 2003/11] A body accelerates uniformly from rest for t_0 seconds after which it continues with a constant velocity. Which graph is the correct representation of the body's motion?

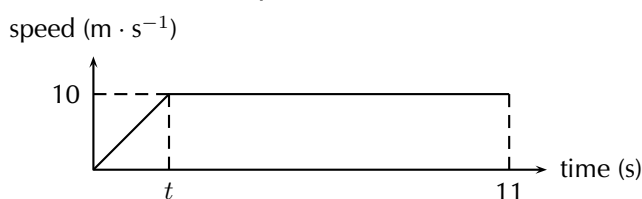


5. [SC 2003/11] The velocity-time graphs of two cars are represented by P and Q as shown



The difference in the distance travelled by the two cars (in m) after 4 s is ...

- (a) 12
(b) 6
(c) 2
(d) 0
6. [IEB 2005/11 HG] The graph that follows shows how the speed of an athlete varies with time as he sprints for 100 m.

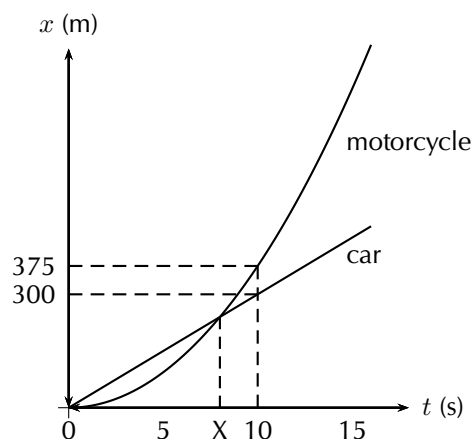


Which of the following equations can be used to correctly determine the time t for which he accelerates?

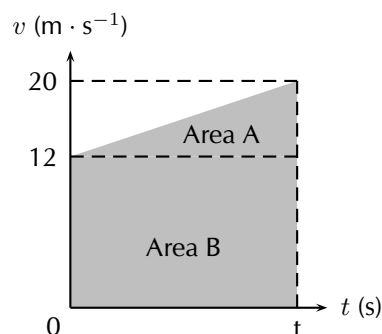
- (a) $100 = (10)(11) - \frac{1}{2}(10)t$
(b) $100 = (10)(11) + \frac{1}{2}(10)t$
(c) $100 = 10t + \frac{1}{2}(10)t^2$
(d) $100 = \frac{1}{2}(0)t + \frac{1}{2}(10)t^2$
7. [SC 2002/03 HG1] In which one of the following cases will the distance covered and the magnitude of the displacement be the same?
- (a) A girl climbs a spiral staircase.
(b) An athlete completes one lap in a race.
(c) A raindrop falls in still air.

(d) A passenger in a train travels from Cape Town to Johannesburg.

8. [SC 2003/11] A car, travelling at constant velocity, passes a stationary motor cycle at a traffic light. As the car overtakes the motorcycle, the motorcycle accelerates uniformly from rest for 10 s. The following displacement-time graph represents the motions of both vehicles from the traffic light onwards.



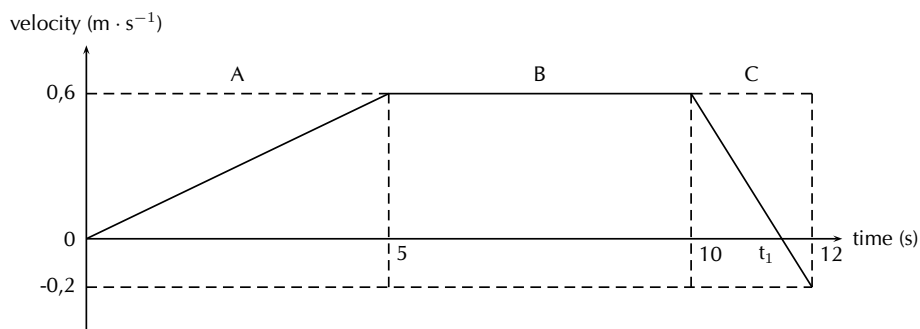
- Use the graph to find the magnitude of the constant velocity of the car.
 - Use the information from the graph to show by means of calculation that the magnitude of the acceleration of the motorcycle, for the first 10 s of its motion is $7,5 \text{ m} \cdot \text{s}^{-2}$.
 - Calculate how long (in seconds) it will take the motorcycle to catch up with the car (point X on the time axis).
 - How far behind the motorcycle will the car be after 15 seconds?
9. [IEB 2005/11 HG] Which of the following statements is *true* of a body that accelerates uniformly?
- Its rate of change of position with time remains constant.
 - Its position changes by the same amount in equal time intervals.
 - Its velocity increases by increasing amounts in equal time intervals.
 - Its rate of change of velocity with time remains constant.
10. [IEB 2003/11 HG1] The velocity-time graph for a car moving along a straight horizontal road is shown below.



Which of the following expressions gives the magnitude of the average velocity of the car?

- (a) $\frac{\text{Area A}}{t}$
- (b) $\frac{\text{Area A} + \text{Area B}}{t}$
- (c) $\frac{\text{Area B}}{t}$
- (d) $\frac{\text{Area A} - \text{Area B}}{t}$

11. [SC 2002/11 SG] A car is driven at $25 \text{ m} \cdot \text{s}^{-1}$ in a municipal area. When the driver sees a traffic officer at a speed trap, he realises he is travelling too fast. He immediately applies the brakes of the car while still 100 m away from the speed trap.
- (a) Calculate the magnitude of the minimum acceleration which the car must have to avoid exceeding the speed limit, if the municipal speed limit is $16.6 \text{ m} \cdot \text{s}^{-1}$.
 - (b) Calculate the time from the instant the driver applied the brakes until he reaches the speed trap. Assume that the car's velocity, when reaching the trap, is $16.6 \text{ m} \cdot \text{s}^{-1}$.
12. A traffic officer is watching his speed trap equipment at the bottom of a valley. He can see cars as they enter the valley 1 km to his left until they leave the valley 1 km to his right. Nelson is recording the times of cars entering and leaving the valley for a school project. Nelson notices a white Toyota enter the valley at 11:01:30 and leave the valley at 11:02:42. Afterwards, Nelson hears that the traffic officer recorded the Toyota doing $140 \text{ km} \cdot \text{hr}^{-1}$.
- (a) What was the time interval (Δt) for the Toyota to travel through the valley?
 - (b) What was the average speed of the Toyota?
 - (c) Convert this speed to $\text{km} \cdot \text{hr}^{-1}$.
 - (d) Discuss whether the Toyota could have been travelling at $140 \text{ km} \cdot \text{hr}^{-1}$ at the bottom of the valley.
 - (e) Discuss the differences between the instantaneous speed (as measured by the speed trap) and average speed (as measured by Nelson).
13. [IEB 2003/11HG] A velocity-time graph for a ball rolling along a track is shown below. The graph has been divided up into 3 sections, A, B and C for easy reference. (Disregard any effects of friction.)



- (a) Use the graph to determine the following:
 - i. the speed 5 s after the start
 - ii. the distance travelled in Section A
 - iii. the acceleration in Section C
 - (b) At time t_1 the velocity-time graph intersects the time axis. Use an appropriate equation of motion to calculate the value of time t_1 (in s).
 - (c) Sketch a displacement-time graph for the motion of the ball for these 12 s. (You do not need to calculate the actual values of the displacement for each time interval, but do pay attention to the general shape of this graph during each time interval.)
- 14.** In towns and cities, the speed limit is $60 \text{ km}\cdot\text{hr}^{-1}$. The length of the average car is 3.5 m, and the width of the average car is 2 m. In order to cross the road, you need to be able to walk further than the width of a car, before that car reaches you. To cross safely, you should be able to walk at least 2 m further than the width of the car (4 m in total), before the car reaches you.
- (a) If your walking speed is $4 \text{ km}\cdot\text{hr}^{-1}$, what is your walking speed in $\text{m}\cdot\text{s}^{-1}$?
 - (b) How long does it take you to walk a distance equal to the width of the average car?
 - (c) What is the speed in $\text{m}\cdot\text{s}^{-1}$ of a car travelling at the speed limit in a town?
 - (d) How many metres does a car travelling at the speed limit travel, in the same time that it takes you to walk a distance equal to the width of car?
 - (e) Why is the answer to the previous question important?
 - (f) If you see a car driving toward you, and it is 28 m away (the same as the length of 8 cars), is it safe to walk across the road?
 - (g) How far away must a car be, before you think it might be safe to cross? How many car-lengths is this distance?
- 15.** A bus on a straight road starts from rest at a bus stop and accelerates at $2 \text{ m}\cdot\text{s}^{-2}$ until it reaches a speed of $20 \text{ m}\cdot\text{s}^{-1}$. Then the bus travels for 20 s at a constant speed until the driver sees the next bus stop in the distance. The driver applies the brakes, stopping the bus in a uniform manner in 5 s.
- (a) How long does the bus take to travel from the first bus stop to the second bus stop?
 - (b) What is the average velocity of the bus during the trip?

- (1.) 00a7 (2.) 00a8 (3.) 00a9 (4.) 00aa (5.) 00ab (6.) 00ac
(7.) 00ad (8.) 00ae (9.) 00af (10.) 00ag (11.) 00ah (12.) 00ai
(13.) 00aj (14.) 00ak (15.) 00am

Mechanical energy

22

Introduction

 ESAHJ

All objects have energy. The word energy comes from the Greek word *energeia* (ἐνέργεια), meaning activity or operation. Energy is closely linked to mass and cannot be created or destroyed. In this chapter we will consider gravitational potential and kinetic energy. See introductory video: (📺 Video: VPgjm at www.everythingscience.co.za)

Potential energy

 ESAHK

The potential energy of an object is generally defined as the energy an object has because of its position relative to other objects that it interacts with. There are different kinds of potential energy such as gravitational potential energy, chemical potential energy, electrical potential energy, to name a few. In this section we will be looking at gravitational potential energy. (📺 See video: VPfhw at www.everythingscience.co.za)

DEFINITION: Potential energy

Potential energy is the energy an object has due to its position or state.

DEFINITION: Gravitational potential energy

Gravitational potential energy is the energy an object has due to its position in a gravitational field relative to some reference point.

Quantity: Gravitational potential energy (E_P)
symbol: J

Unit name: Joule

Unit

In the case of Earth, *gravitational* potential energy is the energy of an object due to its position above the surface of the Earth. The symbol E_P is used to refer to gravitational potential energy. You will often find that the words potential energy are used where *gravitational* potential energy is meant. We can define gravitational potential energy as:

$$E_P = mgh$$

where E_P = potential energy (measured in joules, J)

m = mass of the object (measured in kg)

g = gravitational acceleration ($9,8 \text{ m} \cdot \text{s}^{-2}$)

h = *perpendicular* height from the reference point (measured in m)

You can treat the gravitational acceleration, g, as a constant and you will learn more about it in grade 11 and 12.

Let's look at the case of a suitcase, with a mass of 1 kg, which is placed at the top of a 2 m high cupboard. By lifting the suitcase against the force of gravity, we give the suitcase potential energy. We can calculate its gravitational potential energy using the equation defined above as:

$$\begin{aligned} E_P &= mgh \\ &= (1\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(2\text{m}) = 19,6 \text{ J} \end{aligned}$$

If the suitcase falls off the cupboard, it will lose its potential energy. Halfway down to the floor, the suitcase will have lost half its potential energy and will have only 9,8 J left.

$$\begin{aligned} E_P &= mgh \\ &= (1\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(1\text{m}) = 9,8 \text{ J} \end{aligned}$$

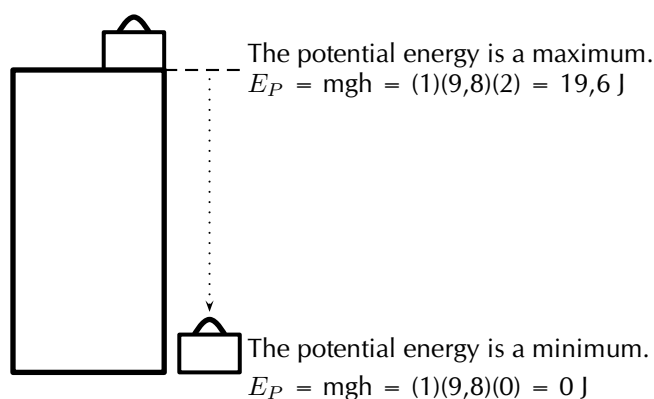
At the bottom of the cupboard the suitcase will have lost all its potential energy and its potential energy will be equal to zero.

$$\begin{aligned} E_P &= mgh \\ &= (1\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(0\text{m}) = 0 \text{ J} \end{aligned}$$

This example shows us that objects have maximum potential energy at a maximum height and will lose their potential energy as they fall.

Tip

You may sometimes see potential energy written as PE. We will not use this notation in this book, but you may see it in other books.

**Example 1: Gravitational potential energy****QUESTION**

A brick with a mass of 1 kg is lifted to the top of a 4 m high roof. It slips off the roof and falls to the ground. Calculate the gravitational potential energy of the brick at the top of the roof and on the ground once it has fallen.

SOLUTION

Step 1 : Analyse the question to determine what information is provided

- The mass of the brick is $m = 1 \text{ kg}$
- The height lifted is $h = 4 \text{ m}$

All quantities are in SI units.

Step 2 : Analyse the question to determine what is being asked

- We are asked to find the gain in potential energy of the brick as it is lifted onto the roof.
- We also need to calculate the potential energy once the brick is on the ground again.

Step 3 : Use the definition of gravitational potential energy to calculate the answer

$$\begin{aligned}E_P &= mgh \\&= (1\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(4\text{m}) \\&= 39,2 \text{ J}\end{aligned}$$

Example 2: More gravitational potential energy**QUESTION**

A netball player, who is 1,7 m tall, holds a 0,5 kg netball 0,5 m above her head and shoots for the goal net which is 2,5 m above the ground. What is the gravitational potential energy of the ball:

1. when she is about to shoot it into the net?
2. when it gets right into the net?
3. when it lands on the ground after the goal is scored?

SOLUTION

Step 1 : Analyse the question to determine what information is provided

- the netball net is 2,5 m above the ground
- the girl has a height of 1,7 m
- the ball is 0,5 m above the girl's head when she shoots for goal
- the mass of the ball is 0,5 kg

Step 2 : Analyse the question to determine what is being asked

We need to find the gravitational potential energy of the netball at three different positions:

- when it is above the girl's head as she starts to throw it into the net
- when it reaches the net
- when it reaches the ground

Step 3 : Use the definition of gravitational potential energy to calculate the value for the ball when the girl shoots for goal

$$E_P = mgh$$

First we need to calculate h . The height of the ball above the ground when the girl shoots for goal is $h = (1,7 + 0,5) = 2,2$ m.

Now we can use this information in the equation for gravitational potential energy:

$$\begin{aligned} E_P &= mgh \\ &= (0,5\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(2,2\text{m}) \\ &= 10,78 \text{ J} \end{aligned}$$

Step 4 : Calculate the potential energy of the ball at the height of the net

Again we use the definition of gravitational potential energy to solve this:

$$\begin{aligned} E_P &= mgh \\ &= (0,5\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(2,5\text{m}) \\ &= 12,25 \text{ J} \end{aligned}$$

Step 5 : Calculate the potential energy of the ball on the ground

$$\begin{aligned} E_P &= mgh \\ &= (0,5\text{kg})(9,8\text{m} \cdot \text{s}^{-2})(0\text{m}) \\ &= 0 \text{ J} \end{aligned}$$

Exercise 22 - 1

1. Describe the relationship between an object's gravitational potential energy and its:
 - a. mass and
 - b. height above a reference point.
2. A boy, of mass 30 kg, climbs onto the roof of a garage. The roof is 2,5 m from the ground.

- a. How much potential energy did the boy gain by climbing onto the roof?
 - b. The boy now jumps down. What is the potential energy of the boy when he is 1m from the ground?
 - c. What is the potential energy of the boy when he lands on the ground?
3. A hiker, of mass 70 kg, walks up a mountain, 800 m above sea level, to spend the night at the top in the first overnight hut. The second day she walks to the second overnight hut, 500 m above sea level. The third day she returns to her starting point, 200 m above sea level.
- a. What is the potential energy of the hiker at the first hut (relative to sea level)?
 - b. How much potential energy has the hiker lost during the second day?
 - c. How much potential energy did the hiker have when she started her journey (relative to sea level)?
 - d. How much potential energy did the hiker have at the end of her journey when she reached her original starting position?

 More practice
  video solutions
  or help at www.everythingscience.co.za

(1.) 00an (2.) 00ap (3.) 00aq

Kinetic energy

 ESAHL

DEFINITION: Kinetic energy

Kinetic energy is the energy an object has due to its motion.

Quantity: Kinetic energy E_K
J

Unit name: Joule

Unit symbol:

Kinetic energy is the energy an object has because of its motion. This means that any moving object has kinetic energy. Kinetic energy is defined as:

$$E_K = \frac{1}{2}mv^2$$

where E_K is the kinetic energy (measured in joules, J)

m = mass of the the object (measured in kg)

v = velocity of the object (measured in $\text{m} \cdot \text{s}^{-1}$).

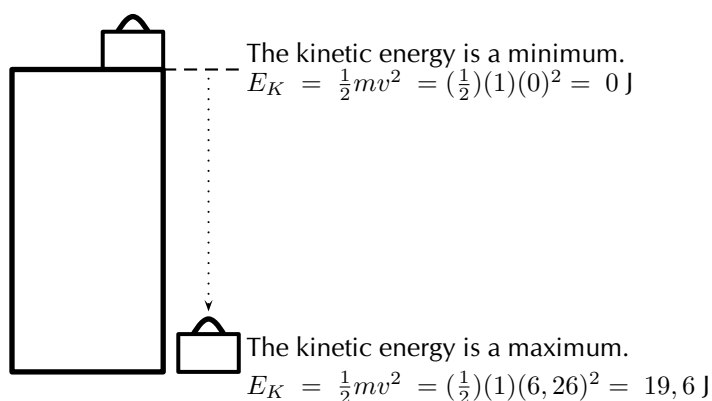
Therefore the kinetic energy E_K depends on the mass and velocity of an object. The faster it moves, and the more massive it is, the more kinetic energy it has. A truck of 2000 kg, moving at $100 \text{ km} \cdot \text{hr}^{-1}$ will have more kinetic energy than a car of 500 kg, also moving at $100 \text{ km} \cdot \text{hr}^{-1}$.

Consider the 1 kg suitcase on the cupboard that was discussed earlier. When it is on the top of the cupboard, it will not have any kinetic energy because it is not moving:

$$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(1 \text{ kg})(0 \text{ m} \cdot \text{s}^{-1})^2 = 0 \text{ J}. \end{aligned}$$

When the suitcase falls, its velocity increases (falls faster), until it reaches the ground with a maximum velocity. As its velocity increases, it will gain kinetic energy. Its kinetic energy will increase until it is a maximum when the suitcase reaches the ground. If it has a velocity of $6,26 \text{ m} \cdot \text{s}^{-1}$ when it reaches the ground, its kinetic energy will be:

$$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(1 \text{ kg})(6,26 \text{ m} \cdot \text{s}^{-1})^2 = 19,6 \text{ J}. \end{aligned}$$



Tip

You may sometimes see kinetic energy written as KE. This is simply another way to write kinetic energy. We will not use this form in this book, but you may see it written like this in other books.

Example 3: Calculation of Kinetic Energy**QUESTION**

A 1 kg brick falls off a 4 m high roof. It reaches the ground with a velocity of $8,85 \text{ m} \cdot \text{s}^{-1}$. What is the kinetic energy of the brick when it starts to fall and when it reaches the ground?

SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- The mass of the brick $m = 1 \text{ kg}$
- The velocity of the brick at the bottom $v = 8,85 \text{ m} \cdot \text{s}^{-1}$

These are both in the correct units so we do not have to worry about unit conversions.

Step 2 : **Analyse the question to determine what is being asked**

We are asked to find the kinetic energy of the brick at the top and the bottom. From the definition we know that to work out E_K , we need to know the mass and the velocity of the object and we are given both of these values.

Step 3 : **Calculate the kinetic energy at the top**

Since the brick is not moving at the top, its kinetic energy is zero.

Step 4 : **Substitute and calculate the kinetic energy**

$$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(1 \text{ kg})(8,85 \text{ m} \cdot \text{s}^{-1})^2 \\ &= 39,2 \text{ J} \end{aligned}$$

Example 4: Kinetic energy of 2 moving objects**QUESTION**

A herder is herding his sheep into the kraal. A mother sheep and its lamb are both running at $2,7 \text{ m} \cdot \text{s}^{-1}$ towards the kraal. The sheep has a mass of 80 kg and the lamb has a mass of 25 kg. Calculate the kinetic energy for each of the sheep and the lamb.

SOLUTION

Step 1 : Analyse the question to determine what information is provided

- the mass of the mother sheep is 80 kg
- the mass of the lamb is 25 kg
- both the sheep and the lamb have a velocities of $2,7 \text{ m} \cdot \text{s}^{-1}$

Step 2 : Analyse the question to determine what is being asked

We need to find the kinetic energy of the sheep and the kinetic energy of its lamb

Step 3 : Use the definition to calculate the sheep's kinetic energy

$$\begin{aligned}E_K &= \frac{1}{2}mv^2 \\&= \frac{1}{2}(80 \text{ kg})(2,7 \text{ m} \cdot \text{s}^{-1})^2 \\&= 291,6 \text{ J}\end{aligned}$$

Step 4 : Use the definition to calculate the lamb's kinetic energy

$$\begin{aligned}E_K &= \frac{1}{2}mv^2 \\&= \frac{1}{2}(25 \text{ kg})(2,7 \text{ m} \cdot \text{s}^{-1})^2 \\&= 91,13 \text{ J}\end{aligned}$$

Note: Even though the sheep and the lamb are running at the same velocity, due to their different masses, they have different amounts of kinetic energy. The sheep has more than the lamb because it has a

higher mass.

Checking units



According to the equation for kinetic energy, the unit should be $\text{kg} \cdot \text{m}^2 \cdot \text{s}^{-2}$. We can prove that this unit is equal to the joule, the unit for energy.

$$\begin{aligned}
 (\text{kg}) (\text{m} \cdot \text{s}^{-1})^2 &= (\text{kg} \cdot \text{m} \cdot \text{s}^{-2}) \cdot \text{m} \\
 &= \text{N} \cdot \text{m} && \text{(because Force (N) = mass (kg) } \times \text{ acceleration (m} \cdot \text{s}^{-2}\text{))} \\
 &= \text{J} && \text{(Work (J) = Force (N) } \times \text{ distance (m))}
 \end{aligned}$$

We can do the same to prove that the unit for potential energy is equal to the joule:

$$\begin{aligned}
 (\text{kg}) (\text{m} \cdot \text{s}^{-2}) (\text{m}) &= \text{N} \cdot \text{m} \\
 &= \text{J}
 \end{aligned}$$

Example 5: Mixing units & energy calculations

QUESTION

A bullet, having a mass of 150 g, is shot with a muzzle velocity of $960 \text{ m} \cdot \text{s}^{-1}$. Calculate its kinetic energy.

SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- We are given the mass of the bullet $m = 150 \text{ g}$. This is not the unit we want mass to be in. We need to convert to kg.

$$\text{Mass in grams} \div 1000 = \text{Mass in kg}$$

$$150 \text{ g} \div 1000 = 0,150 \text{ kg}$$

- We are given the initial velocity with which the bullet leaves the barrel, called the muzzle velocity, and it is $v = 960 \text{ m} \cdot \text{s}^{-1}$.

Step 2 : Analyse the question to determine what is being asked

- We are asked to find the kinetic energy.

Step 3 : Substitute and calculate

We just substitute the mass and velocity (which are known) into the equation for kinetic energy:

$$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ &= \frac{1}{2}(0,150 \text{ kg})(960 \text{ m} \cdot \text{s}^{-1})^2 \\ &= 69\,120 \text{ J} \end{aligned}$$

Exercise 22 - 2

1. Describe the relationship between an object's kinetic energy and its:
 - a. mass and
 - b. velocity
2. A stone with a mass of 100 g is thrown up into the air. It has an initial velocity of $3 \text{ m} \cdot \text{s}^{-1}$. Calculate its kinetic energy:
 - a. as it leaves the thrower's hand.
 - b. when it reaches its turning point.
3. A car with a mass of 700 kg is travelling at a constant velocity of $100 \text{ km} \cdot \text{hr}^{-1}$. Calculate the kinetic energy of the car.

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Mechanical energy

 ESAHN

DEFINITION: Mechanical energy

Mechanical energy is the sum of the gravitational potential energy and the kinetic energy of a system.

Quantity: Mechanical energy (E_M)

Unit name: Joule

Unit

symbol: J

Mechanical energy, E_M , is simply the sum of gravitational potential energy (E_P) and the kinetic energy (E_K). Mechanical energy is defined as:

Tip

You may see mechanical energy written as U . We will not use this notation in this book, but you should be aware that this notation is sometimes used.

$$E_M = E_P + E_K$$

$$E_M = mgh + \frac{1}{2}mv^2$$

Example 6: Mechanical energy

QUESTION

Calculate the total mechanical energy for a ball of mass 0,15 kg which has a kinetic energy of 20 J and is 2 m above the ground.

SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- The ball has a mass $m = 0,15 \text{ kg}$
- The ball is at a height $h = 2 \text{ m}$
- The ball has a kinetic energy $E_K = 20 \text{ J}$

Step 2 : **Analyse the question to determine what is being asked**

We need to find the total mechanical energy of the ball

Step 3 : **Use the definition to calculate the total mechanical energy**

$$\begin{aligned}
 E_M &= E_P + E_K \\
 &= mgh + \frac{1}{2}mv^2 \\
 &= mgh + 20 \\
 &= (0,15 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-1})(2 \text{ m}) + 20 \text{ J} \\
 &= 2,94 \text{ J} + 20 \text{ J} \\
 &= 22,94 \text{ J}
 \end{aligned}$$

Conservation of mechanical energy



DEFINITION: Conservation of Energy

The Law of Conservation of Energy: Energy cannot be created or destroyed, but is merely changed from one form into another.

So far we have looked at two types of energy: gravitational potential energy and kinetic energy. The sum of the gravitational potential energy and kinetic energy is called the mechanical energy. In a closed system, one where there are no external dissipative forces acting, the mechanical energy will remain constant. In other words, it will not change (become more or less). This is called the Law of Conservation of Mechanical Energy.

Tip

In problems involving the use of conservation of energy, the path taken by the object can be ignored. The only important quantities are the object's velocity (which gives its kinetic energy) and height above the reference point (which gives its gravitational potential energy).

DEFINITION: *Conservation of mechanical energy*

Law of Conservation of Mechanical Energy: The total amount of mechanical energy, in a closed system in the absence of dissipative forces (e.g. friction, air resistance), remains constant.

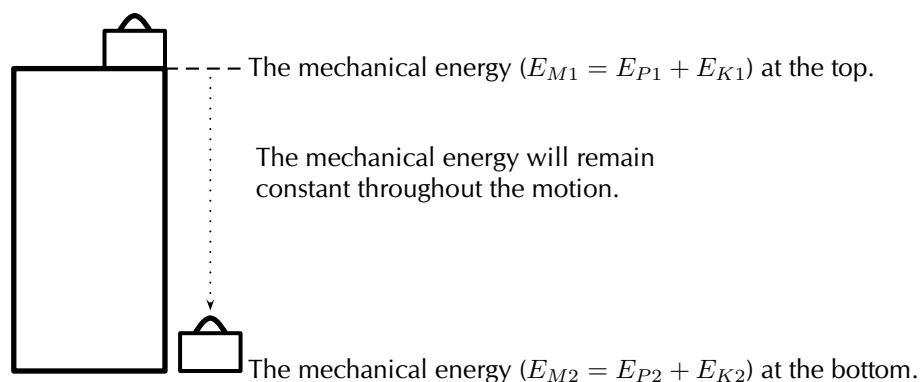
This means that potential energy can become kinetic energy, or vice versa, but energy cannot “disappear”. For example, in the absence of air resistance, the mechanical energy of an object moving through the air in the Earth’s gravitational field, remains constant (is conserved).

See simulation: (🎥 Simulation: VPgoo at www.everythingscience.co.za)

Using the law of conservation of energy ESAHP

Mechanical energy is conserved (in the absence of friction). Therefore we can say that the sum of the E_P and the E_K anywhere during the motion must be equal to the sum of the E_P and the E_K anywhere else in the motion.

We can now apply this to the example of the suitcase on the cupboard. Consider the mechanical energy of the suitcase at the top and at the bottom. We can say:



$$\begin{aligned}
 E_{M1} &= E_{M2} \\
 E_{P1} + E_{K1} &= E_{P2} + E_{K2} \\
 mgh + \frac{1}{2}mv^2 &= mgh + \frac{1}{2}mv^2 \\
 (1 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-1})(2 \text{ m}) + 0 &= 0 + \frac{1}{2}(1 \text{ kg})(v^2) \\
 19,6 &= \frac{1}{2}(v^2) \\
 v^2 &= 39,2 \text{ m}^2 \cdot \text{s}^{-2} \\
 v &= 6,26 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

The suitcase will strike the ground with a velocity of $6,26 \text{ m} \cdot \text{s}^{-1}$.

From this we see that when an object is lifted, like the suitcase in our example, it gains potential energy. As it falls back to the ground, it will lose this potential energy, but gain kinetic energy. We know that energy cannot be created or destroyed, but only changed from one form into another. In our example, the potential energy that the suitcase loses is changed to kinetic energy.

The suitcase will have maximum potential energy at the top of the cupboard and maximum kinetic energy at the bottom of the cupboard. Halfway down it will have half kinetic energy and half potential energy. As it moves down, the potential energy will be converted (changed) into kinetic energy until all the potential energy is gone and only kinetic energy is left. The $19,6 \text{ J}$ of potential energy at the top will become $19,6 \text{ J}$ of kinetic energy at the bottom.

Activity:
Conversion of energy
Materials:

A length of plastic pipe with diameter approximately 20 mm, a marble, some masking tape and a measuring tape.

To do (1):

First put one end of the pipe on the table top so that it is parallel to the top of the table and tape it in position with the masking tape.

Lift the other end of the pipe upwards and hold it at a steady height not too high above the table.

Measure the vertical height from the table top to the top opening of the pipe.

Now put the marble at the top of the pipe and let it go so that it travels through the pipe and out the other end.

Questions:

- What is the velocity (i.e. fast, slow, not moving) of the marble when you first put it into the top of the pipe and what does this mean for its gravitational potential and kinetic energy?

- What is the velocity (i.e. fast, slow, not moving) of the marble when it reaches the other end of the pipe and rolls onto the desk? What does this mean for its gravitational potential and kinetic energy?

To do (2):

Now lift the top of the pipe as high as it will go.

Measure the vertical height of the top of the pipe above the table top.

Put the marble into the top opening and let it roll through the pipe onto the table.

Questions:

- What is the velocity (i.e. fast, slow, not moving) of the marble when you put it into the top of the pipe, and what does this mean for its gravitational potential and kinetic energy?
- Compared to the first attempt, what was different about the height of the top of the tube? How do you think this affects the gravitational potential energy of the marble?
- Compared to your first attempt, was the marble moving faster or slower when it came out of the bottom of the pipe the second time? What does this mean for the kinetic energy of the marble?

The activity with the marble rolling down the pipe shows very nicely the conversion between gravitational potential energy and kinetic energy. In the first instance, the pipe was held relatively low and therefore the gravitational potential energy was also relatively low. The kinetic energy at this point was zero since the marble wasn't moving yet. When the marble rolled out of the other end of the pipe, it was moving relatively slowly, and therefore its kinetic energy was also relatively low. At this point its gravitational potential energy was zero since it was at zero height above the table top.

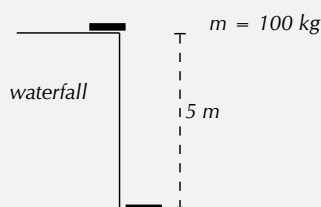
In the second instance, the marble started off higher up and therefore its gravitational potential energy was higher. By the time it got to the bottom of the pipe, its gravitational potential energy was zero (zero height above the table) but its kinetic energy was high since it was moving much faster than the first time. Therefore, the gravitational potential energy was converted completely to kinetic energy (if we ignore friction with the pipe).

In the case of the pipe being held higher, the gravitational potential energy at the start was higher, and the kinetic energy (and velocity) of the marble was higher at the end. In other words, the total mechanical energy was higher and only depended on the height you held the pipe above the table top and not on the distance the marble had to travel through the pipe.

Example 7: Using the Law of Conservation of Mechanical Energy

QUESTION

During a flood a tree trunk of mass 100 kg falls down a waterfall. The waterfall is 5 m high.



If air resistance is ignored, calculate:

1. the potential energy of the tree trunk at the top of the waterfall.
2. the kinetic energy of the tree trunk at the bottom of the waterfall.
3. the magnitude of the velocity of the tree trunk at the bottom of the waterfall.

SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- The mass of the tree trunk $m = 100 \text{ kg}$
 - The height of the waterfall $h = 5 \text{ m}$.
- These are all in SI units so we do not have to convert.

Step 2 : **Analyse the question to determine what is being asked**

- Potential energy at the top
- Kinetic energy at the bottom
- Velocity at the bottom

Step 3 : **Calculate the potential energy at the top of the waterfall.**

$$\begin{aligned}
 E_P &= mgh \\
 &= (100 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-1})(5 \text{ m}) \\
 &= 4900 \text{ J}
 \end{aligned}$$

Step 4 : **Calculate the kinetic energy at the bottom of the waterfall.**

The total mechanical energy must be conserved.

$$E_{K1} + E_{P1} = E_{K2} + E_{P2}$$

Since the trunk's velocity is zero at the top of the waterfall, $E_{K1} = 0$.

At the bottom of the waterfall, $h = 0$ m, so $E_{P2} = 0$.

Therefore $E_{P1} = E_{K2}$ or in words:

The kinetic energy of the tree trunk at the bottom of the waterfall is equal to the potential energy it had at the top of the waterfall. Therefore $E_K = 4900$ J.

Step 5 : Calculate the velocity at the bottom of the waterfall.

To calculate the velocity of the tree trunk we need to use the equation for kinetic energy.

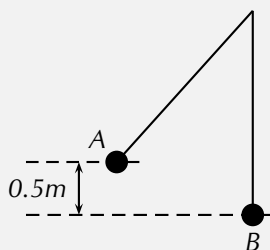
$$\begin{aligned} E_K &= \frac{1}{2}mv^2 \\ 4900 &= \frac{1}{2}(100 \text{ kg})(v^2) \\ 98 &= v^2 \\ v &= 9,899 \dots \text{m} \cdot \text{s}^{-1} \\ v &= 9,90 \text{ m} \cdot \text{s}^{-1} \end{aligned}$$

Example 8: Pendulum

QUESTION

A 2 kg metal ball is suspended from a rope as a pendulum. If it is released from point A and swings down to the point B (the bottom of its arc):

1. show that the velocity of the ball is independent of its mass,
2. calculate the velocity of the ball at point B.



SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- The mass of the metal ball is $m = 2 \text{ kg}$
- The change in height going from point A to point B is $h = 0,5 \text{ m}$
- The ball is released from point A so the velocity at point, $v_A = 0 \text{ m} \cdot \text{s}^{-1}$.

All quantities are in SI units.

Step 2 : **Analyse the question to determine what is being asked**

- Prove that the velocity is independent of mass.
- Find the velocity of the metal ball at point B .

Step 3 : **Apply the Law of Conservation of Mechanical Energy to the situation**

Since there is no friction, mechanical energy is conserved. Therefore:

$$\begin{aligned}
 E_{M1} &= E_{M2} \\
 E_{P1} + E_{K1} &= E_{P2} + E_{K2} \\
 mgh_1 + \frac{1}{2}m(v_1)^2 &= mgh_2 + \frac{1}{2}m(v_2)^2 \\
 mgh_1 + 0 &= 0 + \frac{1}{2}m(v_2)^2 \\
 mgh_1 &= \frac{1}{2}m(v_2)^2
 \end{aligned}$$

The mass of the ball m appears on both sides of the equation so it can be eliminated so that the equation becomes:

$$\begin{aligned}
 gh_1 &= \frac{1}{2}(v_2)^2 \\
 2gh_1 &= (v_2)^2
 \end{aligned}$$

This proves that the velocity of the ball is independent of its mass. It does not matter what its mass is, it will always have the same velocity when it falls through this height.

Step 4 : **Calculate the velocity of the ball at point B**

We can use the equation above, or do the calculation from “first prin-

ciples”:

$$\begin{aligned}
 (v_2)^2 &= 2gh_1 \\
 (v_2)^2 &= (2)(9.8 \text{ m} \cdot \text{s}^{-1})(0,5 \text{ m}) \\
 (v_2)^2 &= 9,8 \text{ m} \cdot \text{s}^{-1} \cdot \text{m} \\
 v_2 &= \sqrt{9,8 \text{ m}^2 \cdot \text{s}^{-2}} \\
 v_2 &= 3,13 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Alternatively you can do:

$$\begin{aligned}
 E_{K1} + E_{P1} &= E_{K2} + E_{P2} \\
 mgh_1 + \frac{1}{2}m(v_1)^2 &= mgh_2 + \frac{1}{2}m(v_2)^2 \\
 mgh_1 + 0 &= 0 + \frac{1}{2}m(v_2)^2 \\
 (v_2)^2 &= \frac{2mgh_1}{m} \\
 (v_2)^2 &= \frac{2(2 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-2})(0,5 \text{ m})}{2 \text{ kg}} \\
 v_2 &= \sqrt{9,8 \text{ m}^2 \cdot \text{s}^{-2}} \\
 v_2 &= 3,13 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Example 9: The roller coaster

QUESTION

A roller coaster ride at an amusement park starts from rest at a height of 50 m above the ground and rapidly drops down along its track. At some point, the track does a full 360 degree loop which has a height of 20 m, before finishing off at ground level. The roller coaster train itself with a full load of people on it has a mass of 850 kg.

If the roller coaster and its track are frictionless, calculate:

Roller coaster



Photograph by Upsilon Andromedae on Flickr.com

1. the velocity of the roller coaster when it reaches the top of the loop
2. the velocity of the roller coaster at the bottom of the loop (i.e. ground level)

SOLUTION

Step 1 : **Analyse the question to determine what information is provided**

- The mass of the roller coaster is $m = 850 \text{ kg}$
- The initial height of the roller coaster at its starting position is $h_1 = 50 \text{ m}$
- The roller coaster starts from rest, so its initial velocity $v_1 = 0 \text{ m} \cdot \text{s}^{-1}$
- The height of the loop is $h_2 = 20 \text{ m}$
- The height at the bottom of the loop is at ground level, $h_3 = 0 \text{ m}$

We do not need to convert units as they are in the correct form already.

Step 2 : **Analyse the question to determine what is being asked**

- the velocity of the roller coaster at the top of the loop
- the velocity of the roller coaster at the bottom of the loop

Step 3 : **Calculate the velocity at the top of the loop**

From the conservation of mechanical energy, We know that at any two points in the system, the total mechanical energy must be the same. Let's compare the situation at the start of the roller coaster to the situation at the top of the loop:

$$\begin{aligned}
 E_{M1} &= E_{M2} \\
 E_{K1} + E_{P1} &= E_{K2} + E_{P2} \\
 0 + mgh_1 &= \frac{1}{2}m(v_2)^2 + mgh_2
 \end{aligned}$$

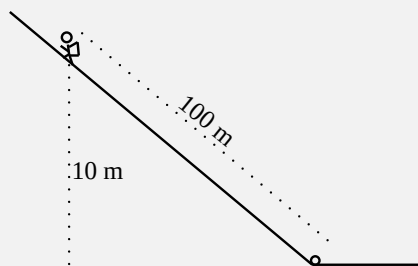
We can eliminate the mass, m , from the equation by dividing both sides by m .

$$\begin{aligned}
 gh_1 &= \frac{1}{2}(v_2)^2 + gh_2 \\
 (v_2)^2 &= 2(gh_1 - gh_2) \\
 (v_2)^2 &= 2((9,8 \text{ m} \cdot \text{s}^{-2})(50 \text{ m}) - (9,8 \text{ m} \cdot \text{s}^{-2})(20 \text{ m})) \\
 v_2 &= 24,25 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Step 4 : Calculate the velocity at the bottom of the loop

Again we can use the conservation of energy and the total mechanical energy at the bottom of the loop should be the same as the total mechanical energy of the system at any other position. Let's compare the situations at the start of the roller coaster's trip and the bottom of the loop:

$$\begin{aligned}
 E_{M1} &= E_{M3} \\
 E_{K1} + E_{P1} &= E_{K3} + E_{P3} \\
 \frac{1}{2}m_1(0)^2 + mgh_1 &= \frac{1}{2}m(v_3)^2 + mg(0) \\
 mgh_1 &= \frac{1}{2}m(v_3)^2 \\
 (v_3)^2 &= 2gh_1 \\
 (v_3)^2 &= 2(9,8 \text{ m} \cdot \text{s}^{-2})(50 \text{ m}) \\
 v_3 &= 31,30 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Example 10: An inclined plane**QUESTION**

A mountain climber who is climbing a mountain in the Drakensberg during winter, by mistake drops her water bottle which then slides 100 m down the side of a steep icy slope to a point which is 10 m lower than the climber's position. The mass of the climber is 60 kg and her water bottle has a mass of 500 g.

1. If the bottle starts from rest, how fast is it travelling by the time it reaches the bottom of the slope? (Neglect friction.)
2. What is the total change in the climber's potential energy as she climbs down the mountain to fetch her fallen water bottle? i.e. what is the difference between her potential energy at the top of the slope and the bottom of the slope?

SOLUTION

Step 1 : Analyse the question to determine what information is provided

- the distance travelled by the water bottle down the slope,
 $d = 100 \text{ m}$
- the difference in height between the starting position and the final position of the water bottle is $h = 10 \text{ m}$
- the bottle starts sliding from rest, so its initial velocity
 $v_1 = 0 \text{ m} \cdot \text{s}^{-1}$
- the mass of the climber is 60 kg
- the mass of the water bottle is 500 g . We need to convert this mass into kg: $500 \text{ g} = 0,5 \text{ kg}$

Step 2 : Analyse the question to determine what is being asked

- What is the velocity of the water bottle at the bottom of the slope?
- What is the difference between the climber's potential energy when she is at the top of the slope compared to when she reaches the bottom?

Step 3 : Calculate the velocity of the water bottle when it reaches the bottom of the slope

$$\begin{aligned}
 E_{M1} &= E_{M2} \\
 E_{K1} + E_{P1} &= E_{K2} + E_{P2} \\
 \frac{1}{2}m(v_1)^2 + mgh_1 &= \frac{1}{2}m(v_2)^2 + mgh_2 \\
 0 + mgh_1 &= \frac{1}{2}m(v_2)^2 + 0 \\
 (v_2)^2 &= \frac{2mgh}{m} \\
 (v_2)^2 &= 2gh \\
 (v_2)^2 &= (2)(9,8 \text{ m} \cdot \text{s}^{-2})(10 \text{ m}) \\
 v_2 &= 14 \text{ m} \cdot \text{s}^{-1}
 \end{aligned}$$

Note: the distance that the bottle travelled (i.e. 100 m) does not play any role in calculating the energies. It is only the height difference that is important in calculating potential energy.

Step 4 : Calculate the difference between the climber's potential energy at the top of the slope and her potential energy at the bottom of the slope

At the top of the slope, her potential energy is:

$$\begin{aligned}
 E_{P1} &= mgh_1 \\
 &= (60 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-2})(10 \text{ m}) \\
 &= 5880 \text{ J}
 \end{aligned}$$

At the bottom of the slope, her potential energy is:

$$\begin{aligned}E_{P2} &= mgh_1 \\&= (60 \text{ kg})(9,8 \text{ m} \cdot \text{s}^{-1})(0) \\&= 0 \text{ J}\end{aligned}$$

Therefore the difference in her potential energy when moving from the top of the slope to the bottom is:

$$E_{P1} - E_{P2} = 5880 - 0 = 5880 \text{ J}$$

Exercise 22 - 3

1. A tennis ball, of mass 120 g, is dropped from a height of 5 m. Ignore air friction.
 - a. What is the potential energy of the ball when it has fallen 3 m?
 - b. What is the velocity of the ball when it hits the ground?
2. A ball rolls down a hill which has a vertical height of 15 m. Ignoring friction, what would be the
 - a. gravitational potential energy of the ball when it is at the top of the hill?
 - b. velocity of the ball when it reaches the bottom of the hill?
3. A bullet, mass 50 g, is shot vertically up in the air with a muzzle velocity of $200 \text{ m} \cdot \text{s}^{-1}$. Use the Principle of Conservation of Mechanical Energy to determine the height that the bullet will reach. Ignore air friction.
4. A skier, mass 50 kg, is at the top of a 6,4 m ski slope.
 - a. Determine the maximum velocity that she can reach when she skis to the bottom of the slope.
 - b. Do you think that she will reach this velocity? Why/Why not?
5. A pendulum bob of mass 1,5 kg, swings from a height A to the bottom of its arc at B. The velocity of the bob at B is $4 \text{ m} \cdot \text{s}^{-1}$. Calculate the height A from which the bob was released. Ignore the effects of air friction.

6. Prove that the velocity of an object, in free fall, in a closed system, is independent of its mass.

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Chapter 22 | Summary

See the summary presentation (Presentation: VPfvd at www.everythingscience.co.za)

- The gravitational potential energy of an object is the energy the object has because of its position in the gravitational field relative to some reference point.
- The kinetic energy of an object is the energy the object has due to its motion.
- The mechanical energy of an object is the sum of the potential energy and kinetic energy of the object.
- The unit for energy is the joule (J).
- The Law of Conservation of Energy states that energy cannot be created or destroyed, but can only be changed from one form into another.
- The Law of Conservation of Mechanical Energy states that the total mechanical energy of an isolated system (i.e. no friction or air resistance) remains constant.
- The table below summarises the most important equations:

Potential Energy	$E_P = mgh$
Kinetic Energy	$E_K = \frac{1}{2}mv^2$
Mechanical Energy	$E_M = E_K + E_P$

Physical Quantities		
Quantity	Unit name	Unit symbol
Potential energy (E_P)	joule	J
Kinetic energy (E_K)	joule	J
Mechanical energy (E_M)	joule	J

Table 22.1: Units used in **mechanical energy**

Chapter 22

End of chapter exercises

1. Give one word/term for the following descriptions.
 - a. The force with which the Earth attracts a body.
 - b. The unit for energy.
 - c. The movement of a body in the Earth's gravitational field when no other forces act on it.

- d. The sum of the potential and kinetic energy of a body.
 - e. The amount of matter an object is made up of.
2. Consider the situation where an apple falls from a tree. Indicate whether the following statements regarding this situation are TRUE or FALSE. Write only “true” or “false”. If the statement is false, write down the correct statement.
- a. The potential energy of the apple is a maximum when the apple lands on the ground.
 - b. The kinetic energy remains constant throughout the motion.
 - c. To calculate the potential energy of the apple we need the mass of the apple and the height of the tree.
 - d. The mechanical energy is a maximum only at the beginning of the motion.
 - e. The apple falls at an acceleration of $9,8 \text{ m} \cdot \text{s}^{-2}$.
3. A man fires a rock out of a slingshot directly upward. The rock has an initial velocity of $15 \text{ m} \cdot \text{s}^{-1}$.
- a. What is the maximum height that the rock will reach?
 - b. Draw graphs to show how the potential energy, kinetic energy and mechanical energy of the rock changes as it moves to its highest point.
4. A metal ball of mass 200 g is tied to a light string to make a pendulum. The ball is pulled to the side to a height (A), 10 cm above the lowest point of the swing (B). Air friction and the mass of the string can be ignored. The ball is let go to swing freely.
- a. Calculate the potential energy of the ball at point A.
 - b. Calculate the kinetic energy of the ball at point B.
 - c. What is the maximum velocity that the ball will reach during its motion?
5. A truck of mass 1,2 tons is parked at the top of a hill, 150 m high. The truck driver lets the truck run freely down the hill to the bottom.
- a. What is the maximum velocity that the truck can achieve at the bottom of the hill?
 - b. Will the truck achieve this velocity? Why/why not?
6. A stone is dropped from a window, 6 m above the ground. The mass of the stone is 25 g. Use the Principle of Conservation of Energy to determine the speed with which the stone strikes the ground.

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 00az (2.) 00b0 (3.) 00b1 (4.) 00b2 (5.) 00b3 (6.) 00b4

Introduction

www ESAHQ

As far as we know, the Earth we live on is the only planet that is able to support life. Amongst other factors, the Earth is just the right distance from the sun to have temperatures that are suitable for life to exist. Also, the Earth's atmosphere has exactly the right type of gases in the right amounts for life to survive. Our planet also has **water** on its surface, which is something very unique. In fact, Earth is often called the "Blue Planet" because most of it is covered in water. This water is made up of *freshwater* in rivers and lakes, the *saltwater* of the oceans and estuaries, *groundwater* and *water vapour*. Together, all these water bodies are called the **hydrosphere**.

See introductory video: (▶) Video: VPbzp at www.everythingscience.co.za

The Earth



Photo by NASA on
Flickr.com

Interactions of the hydrosphere

www ESAHR

FACT

The total mass of the hydrosphere is approximately $1,4 \times 10^{18}$ tonnes! (The volume of one tonne of water is approximately 1 cubic meter (this is about 900 A4 textbooks!))

It is important to realise that the hydrosphere is not an isolated system, but rather interacts with other global systems, including the *atmosphere*, *lithosphere* and *biosphere*. These interactions are sometimes known collectively as the water cycle.

▶ See video: VPbzy at www.everythingscience.co.za

- **Atmosphere** When water is heated (e.g. by energy from the sun), it evaporates and forms water vapour. When water vapour cools again, it condenses to form liquid water which eventually returns to the surface by precipitation e.g. rain or snow. This cycle of water moving through the atmosphere and the energy changes that accompany it, is what drives weather patterns on earth.
- **Lithosphere**
In the lithosphere (the ocean and continental crust at the Earth's surface), water is an important *weathering* agent, which means that it helps to break rock down into rock fragments and then soil. These fragments may then be transported by water to another place, where they are deposited. These two processes (weathering and the

transporting of fragments) are collectively called *erosion*. Erosion helps to shape the earth's surface. For example, you can see this in rivers. In the upper streams, rocks are eroded and sediments are transported down the river and deposited on the wide flood plains lower down.

On a bigger scale, river valleys in mountains have been carved out by the action of water, and cliffs and caves on rocky beach coastlines are also the result of weathering and erosion by water. The processes of weathering and erosion also increase the content of dissolved minerals in the water. These dissolved minerals are important for the plants and animals that live in the water.



- **Biosphere**

In the biosphere, land plants absorb water through their roots and then transport this through their vascular (transport) system to stems and leaves. This water is needed in *photosynthesis*, the food production process in plants. Transpiration (evaporation of water from the leaf surface) then returns water back to the atmosphere.

Exploring the hydrosphere

www ESAHS

The large amount of water on our planet is something quite unique. In fact, about 71% of the earth is covered by water. Of this, almost 97% is found in the oceans as saltwater, about 2,2% occurs as a solid in ice sheets, while the remaining amount (less than 1%) is available as freshwater. So from a human perspective, despite the vast amount of water on the planet, only a very small amount is actually available for human consumption (e.g. drinking water).

In chapter 18 we looked at some of the reactions that occur in aqueous solution and saw some of the chemistry of water, in this section we are going to spend some time exploring a part of the hydrosphere in order to start appreciating what a complex and beautiful part of the world it is. After completing the following investigation, you should start to see just how important it is to know about the chemistry of water.



Photo by Duncan Brown (Cradlehall) on Flickr.com

Investigation: *Investigating the hydrosphere*

In groups of 3 - 4 choose one of the following as a study site: rock pool, lake, river, wetland or small pond. When choosing your study site, consider how accessible it is (how easy is it to get to?) and the problems you may experience (e.g. tides, rain).

Your teacher will provide you with the equipment you need to collect the following data. You can pick more than one study site and compare your data for each site.

- Measure and record data such as temperature, pH, conductivity and dissolved oxygen at each of your sites.
- Collect a water sample in a clear bottle, hold it to the light and see whether the water is clear or whether it has particles in it (i.e. what is the clarity of the water).
- What types of animals and plants are found in or near this part of the hydrosphere? Are they specially adapted to their environment?

Record your data in a table like the one shown below:

	Site 1	Site 2	Site 3
Temperature			
pH			
Conductivity			
Dissolved oxygen			
Clarity			
Animals			
Plants			

Interpreting the data Once you have collected and recorded your data, think about the following questions:

- How does the data you have collected vary at different sites?
- Can you explain these differences?
- What effect do you think *temperature*, *dissolved oxygen* and *pH* have on animals and plants that are living in the hydrosphere?
- Water is seldom "pure". It usually has lots of things dissolved (e.g.

Mg²⁺, Ca²⁺ and NO₃⁻ ions) or suspended (e.g. soil particles, debris) in it. Where do these substances come from?

- Are there any human activities near this part of the hydrosphere? What effect could these activities have on the hydrosphere?

The importance of the hydrosphere



It is so easy sometimes to take our hydrosphere for granted and we seldom take the time to really think about the role that this part of the planet plays in keeping us alive. Below are just some of the important functions of water in the hydrosphere:

- *Water is a part of living cells* Each cell in a living organism is made up of almost 75% water, and this allows the cell to function normally. In fact, most of the chemical reactions that occur in life, involve substances that are dissolved in water. Without water, cells would not be able to carry out their normal functions and life could not exist.
- *Water provides a habitat* The hydrosphere provides an important place for many animals and plants to live. Many gases (e.g. CO₂, O₂), nutrients e.g. nitrate (NO₃⁻), nitrite (NO₂⁻) and ammonium (NH₄⁺) ions, as well as other ions (e.g. Mg²⁺ and Ca²⁺) are dissolved in water. The presence of these substances is critical for life to exist in water.
- *Regulating climate* One of water's unique characteristics is its high *specific heat*. This means that water takes a long time to heat up and also a long time to cool down. This is important in helping to regulate temperatures on earth so that they stay within a range that is acceptable for life to exist. *Ocean currents* also help to disperse heat.
- *Human needs* Humans use water in a number of ways. Drinking water is obviously very important, but water is also used domestically (e.g. washing and cleaning) and in industry. Water can also be used to generate electricity through hydropower.

These are just a few of the functions that water plays on our planet. Many of the functions of water relate to its chemistry and to the way in which it is able to dissolve substances in it.

Threats to the hydrosphere



It should be clear by now that the hydrosphere plays an extremely important role in the survival of life on Earth and that the unique properties of water allow various important chemical processes to take place which would otherwise not be possible. Unfortunately for us however, there are a number of factors that threaten our hydrosphere and most of these threats are because of human activities. We are going to focus on two of these issues: **pollution** and **overuse** and look at ways in which these problems can possibly be overcome.

► See video: VPcbf at www.everythingscience.co.za

1. Pollution

Pollution of the hydrosphere is a major problem. When we think of pollution, we sometimes only think of things like plastic, bottles, oil and so on. But any chemical that is present in the hydrosphere in an amount that is not what it should be is a pollutant. Animals and plants that live in the Earth's water bodies are specially adapted to surviving within a certain range of conditions. If these conditions are changed (e.g. through pollution), these organisms may not be able to survive. Pollution then, can affect entire aquatic ecosystems. The most common forms of pollution in the hydrosphere are *waste products* from humans and from industries, *nutrient pollution* e.g. fertiliser runoff which causes eutrophication (an excess of nutrients in the water leading to excessive plant growth) and toxic trace elements such as aluminium, mercury and copper to name a few. Most of these elements come from mines or from industries.

2. Overuse of water

We mentioned earlier that only a very small percentage of the hydrosphere's water is available as freshwater. However, despite this, humans continue to use more and more water to the point where water *consumption* is fast approaching the amount of water that is *available*. The situation is a serious one, particularly in countries such as South Africa which are naturally dry and where water resources are limited. It is estimated that between 2020 and 2040, water supplies in South Africa will no longer be able to meet the growing demand for water in this country. This is partly due to population growth, but also because of the increasing needs of industries as they expand and develop. For each of us, this should be a very scary thought. Try to imagine a day without water... difficult isn't it? Water is so much a part of our lives, that we are hardly aware of the huge part that it plays in our daily lives.

As populations grow, so do the demands that are placed on dwindling water resources. While many people argue that building dams helps to solve this water-shortage problem, there is evidence that dams are only a temporary solution and that they often end up doing far more ecological damage than good. The only sustainable solution is to reduce the *demand* for water, so that water supplies are sufficient to meet this. The more important

question then is how to do this.

Group Discussion: Creative water conservation

Divide the class into groups, so that there are about five people in each. Each group is going to represent a different sector within society. Your teacher will tell you which sector you belong to from the following: farming, industry, city management, water conservation, tourism or civil society (i.e. you will represent the ordinary “man on the street”). In your groups, discuss the following questions as they relate to the group of people you represent: (Remember to take notes during your discussions and nominate a spokesperson to give feedback to the rest of the class on behalf of your group)

Dry landscape

Photo by flowcomm on Flickr.com

- What steps could be taken by your group to conserve water?
 - Why do you think these steps are *not* being taken?
 - What incentives do you think could be introduced to encourage this group to conserve water more efficiently?
-

Investigation: Building of dams

In the previous discussion, we mentioned that there is evidence that dams are only a temporary solution to the water crisis. In this investigation you will look at why dams are a potentially bad solution to the problem.

For this investigation you will choose a dam that has been built in your area, or an area close to you. Make a note of which rivers are in the area.

Try to answer the following questions:

A dam

Photo by Redeo on Flickr.com

- If possible talk to people who have lived in the area for a long time and try to get their opinion on how life changed since the dam was built. If it is not possible to talk to people in the area, then look for relevant literature on the area.
- Try to find out if any environmental impact assessments (this is where people study the environment and see what effect the proposed project has on the environment) were done before the dam was built. Why do you think this is important?

- Look at how the ecology has changed. What was the ecology of the river before the dam was built? What is the current ecology? Do you think it has changed in a good way or a bad way? You could interview people in the community who lived there long ago before the dam was built.

Write a report or give a presentation in class on your findings from this investigation. Critically examine your findings and draw your own conclusion as to whether or not dams are only a short term solution to the growing water crisis.

It is important to realise that our hydrosphere exists in a delicate balance with other systems and that disturbing this balance can have serious consequences for life on this planet.

Project: School Action Project

There is a lot that can be done within a school to save water. As a class, discuss what actions could be taken by your class to make people more aware of how important it is to conserve water. Also consider what ways your school can save water. If possible, try to put some of these ideas into action and see if they really do conserve water. During break walk around the school and make a list of all the places where water is being wasted.

How pure is our water?



When you drink a glass of water you are not just drinking water, but many other substances that are dissolved into the water. Some of these come from the process of making the water safe for humans to drink, while others come from the environment. Even if you took water from a mountain stream (which is often considered pure and bottled for people to consume), the water would still have impurities in it. Water pollution increases the amount of impurities in the water and sometimes makes the water unsafe for drinking. In this section we will look at a few of the substances that make water impure and how we can make pure water. We will also look at the pH of water.

In chapter 18 we saw how compounds can dissolve in water. Most of these compounds (e.g. Na^+ , Cl^- , Ca^{2+} , Mg^{2+} , etc.) are safe for humans to consume in the small amounts that are naturally present in water. It is only when the amounts of these ions rise above the safe levels that the water is considered to be polluted.

You may have noticed sometimes that when you pour a glass of water straight from the tap, it has a sharp smell. This smell is the same smell that you notice around swimming pools and

is due to chlorine in the water. Chlorine is the most common compound added to water to make it safe for humans to use. Chlorine helps to remove bacteria and other biological contaminants in the water. Other methods to purify water include filtration (passing the water through a very fine mesh) and flocculation (a process of adding chemicals to the water to help remove small particles).

pH of water is also important. Water that is too basic (pH greater than 7) or too acidic (pH less than 7) may present problems when humans consume the water. If you have ever noticed after swimming that your eyes are red or your skin is itchy, then the pH of the swimming pool was probably too basic or too acidic. This shows you just how sensitive we are to the smallest changes in our environment. The pH of water depends on what ions are dissolved in the water. Adding chlorine to water often lowers the pH. You will learn more about pH in grade 11.

General experiment: Water purity

Aim: To test the purity and pH of water samples

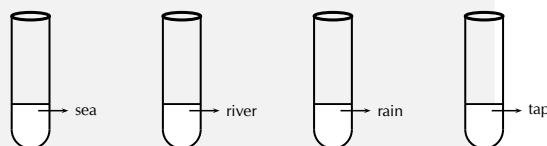
Apparatus:

pH test strips (you can find these at pet shops, they are used to test pH of fish tanks), microscope (or magnifying glass), filter paper, funnel, silver nitrate, concentrated nitric acid, barium chloride, acid, chlorine water (a solution of chlorine in water), carbon tetrachloride, some test-tubes or beakers, water samples from different sources (e.g. a river, a dam, the sea, tap water, etc.).

Method:

1. Look at each water sample and note if the water is clear or cloudy.
2. Examine each water sample under a microscope and note what you see.
3. Test the pH of each of the water samples.
4. Pour some of the water from each sample through filter paper.
5. Refer to chapter 18 for the details of common anion tests. Test for chloride, sulphate, carbonate, bromide and iodide in each of the water samples.

Results: Write down what you saw when you just looked at the water samples. Write down what you saw when you looked at the water samples under a microscope. Were there any dissolved particles? Or other things in the water? Was there a difference in what you saw with just looking and with looking with a microscope?



Write down the pH of each water sample. Look at the filter paper from each sample. Is there sand or other particles on it? Which anions did you find in each sample?

Discussion: Write a report on what you observed. Draw some conclusions on the purity of the water and how you can tell if water is pure or not.

Conclusion: You should have seen that water is not pure, but rather has many substances dissolved in it.

Project: Water purification

Prepare a presentation on how water is purified. This can take the form of a poster, or a presentation or a written report. Things that you should look at are:

- Water for drinking (potable water)
- Distilled water and its uses
- Deionised water and its uses
- What methods are used to prepare water for various uses
- What regulations govern drinking water
- Why water needs to be purified
- How safe are the purification methods

Chapter 23 | Summary

See the summary presentation (📺 Presentation: VPfbj at www.everythingscience.co.za)

- The **hydrosphere** includes all the water that is on Earth. Sources of water include freshwater (e.g. rivers, lakes), saltwater (e.g. oceans), groundwater (e.g. boreholes) and water vapour. Ice (e.g. glaciers) is also part of the hydrosphere.
- The hydrosphere interacts with other **global systems**, including the atmosphere, lithosphere and biosphere.
- The hydrosphere has a number of important **functions**. Water is a part of all living cells, it provides a habitat for many living organisms, it helps to regulate climate and it is used by humans for domestic, industrial and other use.
- Despite the importance of the hydrosphere, a number of factors threaten it. These include **overuse** of water, and **pollution**.
- Water is not pure, but has many substances dissolved in it.

Chapter 23**End of chapter exercises**

1. What is the hydrosphere? How does it interact with other global systems?
2. Why is the hydrosphere important?
3. Write a one page essay on the importance of water and what can be done to ensure that we still have drinkable water in 50 years time.

 More practice  video solutions  or help at www.everythingscience.co.za

(1.) 00b5 (2.) 00b6 (3.) 024v

Quantities used in the book

Physical Quantities		
Quantity	Unit name	Unit symbol
Amplitude (A)	metre	m
Atomic mass unit (amu)	u	u
Average acceleration ($\vec{a}_{av}$)	metre per second squared	$m \cdot s^{-2}$
Average speed (v_{av})	metre	m
Average velocity ($\vec{v}_{av}$)	metre per second	$m \cdot s^{-1}$
Charge (Q)	coulomb	C
Concentration (C)	mol per decimetre cubed	$mol \cdot dm^{-3}$
Current (I)	ampere	A
Density (d)	grams per centimetre cubed	$g \cdot cm^{-3}$
Displacement (Δx)	metre	m
Distance (D)	metre	m
Energy (E)	Joule	J
Frequency (f)	Hertz	Hz
Instantaneous acceleration ($\vec{a}$)	metre per second squared	$m \cdot s^{-2}$
Instantaneous speed (v_{av})	metre per second	$m \cdot s^{-1}$
Instantaneous velocity ($\vec{v}$)	metre per second	$m \cdot s^{-1}$
Intensity (I)	decibel	dB
Magnitude of acceleration (a)	metre per second squared	$m \cdot s^{-2}$
Mass (m)	gram	g
Molar mass (M)	gram per mol	$g \cdot mol^{-1}$
Mole (n)	mole	mol

Table 24.1: Units used in **the book**

Physical Quantities		
Quantity	Unit name	Unit symbol
Period (T)	second	s
Position (x)	metre	m
Potential difference (V)	Volt	V
Pulse speed (v)	metre per second	$\text{m} \cdot \text{s}^{-1}$
Resistance (R)	Ohm	Ω
Temperature (T)	degrees	$^{\circ}$
Volume (V)	decimetre cubed	dm^{-3}
Wavelength (λ)	metre	m
Wavespeed (v)	metre per second	$\text{m} \cdot \text{s}^{-1}$

Table 24.2: Units used in **the book**

Science skills

 ESAHX

Exercise 1-1

 ESAHY

- | | | |
|-------------------------------------|-------------------------------------|-------------|
| 1. (a) $3,63 \times 10^6$ | (c) $5 \times 10^{-7} \text{ m}$ | (e) 59,8 kN |
| (b) 37,83 | (d) $2,50 \times 10^{-7} \text{ m}$ | (f) 2,5 mA |
| (c) $6,3 \times 10^{-4}$ | (e) $3,5 \times 10^{-2} \text{ g}$ | (g) 7,5 km |
| 2. (a) $5,11 \times 10^5 \text{ V}$ | 3. (a) 0,1602 aC | |
| (b) $1,0 \times 10^{-1} \ell$ | (b) 1,992 MJ | |

Exercise 1-2

 ESAHZ

- | | | |
|---|--|---|
| 1. (a) $1,01 \times 10^4 \text{ Pa}$ | (d) megagram and 6 | (b) 1 000 mg |
| (b) $9,8 \times 10^2 \text{ m} \cdot \text{s}^{-2}$ | 3. (a) $1,23 \times 10^{-6} \text{ N}$ | (c) 7,2 Mm |
| (c) $1,256 \times 10^{-6} \text{ A}$ | (b) $4,17 \times 10^8 \text{ kg}$ | (d) 11 nℓ |
| 2. (a) microgram and -6 | (c) $2,47 \times 10^5 \text{ A}$ | |
| (b) milligram and -3 | (d) $8,80 \times 10^{-4} \text{ mm}$ | 5. $234,44 \text{ m} \cdot \text{s}^{-1}$ |
| (c) kilogram and 3 | 4. (a) $1,01 \mu\text{s}$ | 6. 373 K |

Classification of matter

 ESAIA

End of chapter exercises

 ESAIB

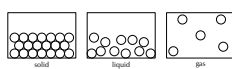
- | | | |
|---------------------------|-----------------------|------------------------|
| 1. a | (b) A | (d) sulphuric acid |
| 2. a | (c) E | 8. (a) FeSO_4 |
| 3. (a) compound | (d) B | (b) BF_3 |
| (b) compound | (e) C | (c) KMnO_4 |
| (c) heterogeneous mixture | 5. (a) use a magnet | (d) ZnCl_2 |
| (d) solution | (b) magnetism | 9. (a) friction |
| (e) solution | 6. (a) Ag | (b) durable |
| (f) compound | (b) KBr | (c) durable |
| (g) element | (c) CO_2 | (d) shiny |
| (h) solution | 7. (a) sodium bromide | (e) easily moulded |
| 4. (a) D | (b) barium sulphate | (f) fibrous |
| | (c) sulphur dioxide | |

States of matter and the KMT

 ESAIC

End of chapter exercises

 ESAID

- | | | |
|-----------------------|--|---|
| 1. (a) sublimation | (b) liquid to gas | (b) carbon |
| (b) evaporation | 3. see definition | (c) helium |
| 2. (a) see definition | 4. (a) solid, solid, gas,
solid, gas, solid | 5.  |

The atom

 ESAIE

End of chapter exercises

 ESAIF

- | | | | |
|---------------------------|--|---|----------------|
| 1. (a) atomic mass number | 4. B | (b) [Ne]3s ² 3p ³ | (c) 27; 32; 27 |
| (b) electron orbital | 5. A | (c) [He]2s ² 2p ² | (d) 3; 4; 3 |
| 2. (a) false | 6. C | (d) [He]2s ² 2p ⁶ | (e) 5; 6; 5 |
| (b) true | 7. B | (e) [Ne]3s ² 3p ⁶ | 11. (a) Rh |
| (c) true | 8. (a) ${}^5_4\text{Be}$ | | (b) 17 |
| (d) false | (b) ${}^{12}_6\text{C}$ | | (c) 9 |
| 3. B | (c) ${}^{48}_{22}\text{Ti}$ | | 12. B |
| | (d) ${}^{19}_9\text{F}$ | | 13. 63, 62 u |
| | 9. (a) [Ne]3s ² 3p ¹ | 10. (a) 78; 117; 78 | |
| | | (b) 18; 22; 18 | |

The periodic table

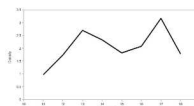
 ESAIG

End of chapter exercises

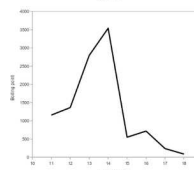
 ESAIH

- | | |
|--------------------------------|-------------|
| 1. (a) false | (c) Cl > Br |
| (b) true | (d) Br > Cl |
| (c) false | |
| (d) false | |
| 2. (a) first ionisation energy | |
| (b) period | |
| (c) halogens | |
| 3. (a) Br > Cl | |
| (b) Cl > Br | |

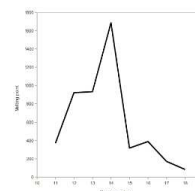
4. (a)



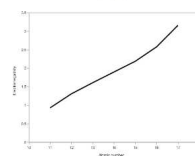
(b)



(c)



(d)



Chemical bonding

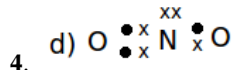
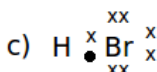
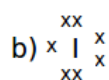
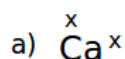


End of chapter exercises



2. B

3. B



5. (a) 1

(b) 3

(c) hydrogen and nitrogen

7. (a) potassium (K^+) dichromate ($\text{Cr}_2\text{O}_7^{2-}$)(b) $\text{K}_2\text{Cr}_2\text{O}_7$

6.

	K^+	Ca^{2+}	NH_4^+
OH^-	KOH	$\text{Ca}(\text{OH})_2$	NH_4OH
O^{2-}	K_2O	CaO	$(\text{NH}_4)_2\text{O}$
NO_3^-	KNO_3	$\text{Ca}(\text{NO}_3)_2$	NH_4NO_3
PO_4^{3-}	K_3PO_4	$\text{Ca}_3(\text{PO}_4)_2$	$(\text{NH}_4)_3\text{PO}_4$

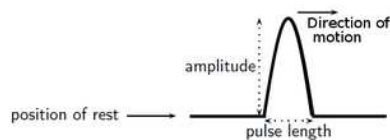
Transverse pulses



End of chapter exercises



1.



2. 15 m

3. $0,3 \text{ m} \cdot \text{s}^{-1}$

4. 0,05 s

Transverse waves

 **ESAIM**

End of chapter exercises

 **ESAIN**

- | | | | |
|--------------|------------|-------------|------------------------------|
| 1. (a) 0,2 m | (b) (i) BD | (c) 0,75 m | (f) 14,4 m · s ⁻¹ |
| (b) 1,33 s | (ii) AB | (d) 0,625 s | |
| 2. (a) 3 | (iii) BD | (e) 1,60 Hz | |

Longitudinal waves

 **ESAIO**

End of chapter exercises

 **ESAIP**

- | | | |
|------|--------------------------------------|-------------------------|
| 1. A | 3. a. 10 m | 4. a. $\frac{1}{256}$ s |
| 2. D | b. $2 \text{ m} \cdot \text{s}^{-1}$ | b. 1,25 m |

Sound

 **ESAIQ**

End of chapter exercises

 **ESAIR**

- | | | | | |
|------------------|------|-------|-------------|----------------------------|
| 1. (a) frequency | 3. D | 8. C | 13. C | 18. 25,8 m |
| (b) amplitude | 4. C | 9. A | 14. 1 700 m | 19. 0,57 m
600 Hz |
| (c) waveform | 5. C | 10. C | 15. 0,15 s | 20. increase
wavelength |
| 2. E | 6. B | 11. B | 16. | 21. 17 mm |
| | 7. E | 12. A | 17. radios | |

22. 34,4 m 25. (a) waves (e) speed
 23. 1 812,5 mm (c) amplitude (h) period
 waves wavelength (f) wave
 (b) speed (g) wave
 24. 172 m transverse frequency

EM radiation

www ESAIS

End of chapter exercises

www ESAIT

- 2.0×10^{-25} J
- 3×10^{-19} J
- $8,6 \times 10^{-21}$ J
- $3,0 \times 10^3$ m
- For the photon with wavelength 532 nm:
 $E = 3,7 \times 10^{-19}$ J
 For the photon with frequency 13 Hz: $E = 8,6 \times 10^{-24}$ J
 The energy of the second photon is: $2,3 \times 10^{-2}$ m.
- Radio, microwave, infrared, visible, ultra-violet, X-ray, gamma ray

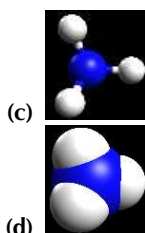
The particles that substances are made of

www ESAIU

End of chapter exercises

www ESAIV

- (a) molecule
(b) empirical formula
- (a) covalent
(b) NH_3
- (a) covalent molecular
- (a) covalent molecular
- (a) carbon dioxide
(b) CO_2
(c) covalent
- (b) metallic network
(c) covalent network
(d) covalent molecular
(e) ionic network



Physical and chemical change

 **ESAIW**

End of chapter exercises

 **ESAIX**

- | | | |
|-------------------------|---------------------|-----------------------|
| (1) (a) physical change | (b) chemical change | (g) physical change |
| (b) chemical change | (c) chemical change | (h) chemical change |
| (c) synthesis reaction | (d) chemical change | (4) (a) decomposition |
| (2) see definitions | (e) physical change | (b) synthesis |
| (3) (a) physical change | (f) physical change | (c) decomposition |

Representing chemical change

 **ESAIY**

End of chapter exercises

 **ESAIZ**

- | | |
|--|--|
| (1) $\text{C}_3\text{H}_8(\ell) + 5\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$ | $6\text{H}_2\text{O}(\ell)$ |
| (2) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$ | (5) $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$ |
| (3) (a) $\text{P}_4(\text{s}) + 5\text{O}_2(\text{g}) \rightarrow 2\text{P}_2\text{O}_5(\text{s})$ | (6) $2\text{H}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{SO}_2 + 2\text{H}_2\text{O}$ |
| (b) same number of atoms on both sides | $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$ |
| (c) phosphorus oxide | (7) $\text{C}_{14}\text{H}_{18}\text{N}_2\text{O}_5(\text{s}) + 16\text{O}_2(\text{g}) \rightarrow 9\text{H}_2\text{O}(\ell) + 14\text{CO}_2(\text{g}) + \text{N}_2(\text{g})$ |
| (d) synthesis | |
| (4) $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) +$ | |

Magnetism

 **ESAJA**

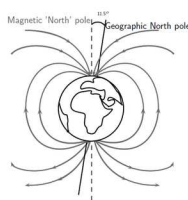
End of chapter exercises

 **ESAJB**

- (4) Each piece will have a N and S pole
 (5) they repel
 (6) they attract

(11) the aurora

(12)



Electrostatics

www ESAJC

End of chapter exercises

www ESAJD

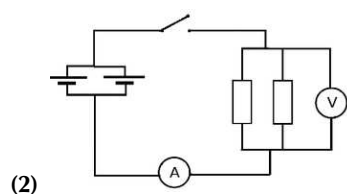
- | | | |
|----------------------------------|-------------------|--|
| (1) positive and negative | of 4 | (13) $-376 \times 10^{-17} \text{ C}$ |
| (2) rubbing two objects together | (5) D | (14) $376 \times 10^{-17} \text{ C}$ |
| (3) repulsive, attractive | (8) B | (15) $7,28 \times 10^{-18} \text{ C}$, 45 electrons |
| (4) (a) repulsive | (9) polarisation | (16) same as at start |
| (b) increase by factor | (12) 60 electrons | (17) neutral, 0 electrons |

Electric circuits

www ESAJE

End of chapter exercises

www ESAJF



- | | | |
|------------------------------------|------------|------------------|
| (4) 1 | (c) B | (11) 1,4 V |
| (5) d | (7) a | (12) 18Ω |
| (6) (a) brighter | (8) 4,25 V | (13) $7,5\Omega$ |
| (b) A and B are the same, C is off | (9) 17 V | (14) 15 V |
| | (10) 7 V | (15) 2 V |

Reactions in aqueous solutions



End of chapter exercises



- | | | |
|----------------------|-------------------|---|
| (1) (a) condensation | (e) C | (e) ionic |
| (b) ion | (f) B | (f) ionic |
| (c) hardness | (g) I | (6) (a) X: carbonate, Y: sulphate, Z: chloride |
| (d) sulphur trioxide | | (b) $\text{CO}_3^{2-} + \text{Ba}^{2+} + \text{Cl}^- \rightarrow \text{BaCO}_3 + \text{Cl}^-$ |
| (2) (a) B | (5) (a) molecular | |
| (b) H | (b) ionic | |
| (c) E | (c) molecular | |
| (d) A | (d) molecular | |

Quantitative aspects of chemical change



End of chapter exercises



- | | | | |
|-----------------------|--------------------|---|------------------|
| (1) (a) molar mass | (b) 0,09 | H: 4, 48 | (14) (b) 0,03 |
| (b) Avogadro's number | (c) 0,04 | (b) Li: 29, 17 | (c) yes |
| | (d) 5,06 | O: 66, 67 | (15) (a) 4 |
| (2) D | (7) (a) 11, 2 | H: 4, 17 | (b) incorrect |
| (3) A | (b) 21, 62 | (9) CF_2Cl_2 | (c) 0,1mols |
| (4) A | (c) 20, 8 | (10) NO_2 | (16) (a) 196, 14 |
| (5) C | (d) 6, 72 | (11) I_4O_9 | (b) 292, 32 |
| | (e) 145, 66 | (12) (a) $\text{C}_2\text{H}_4\text{F}$ | (c) 94, 1 |
| (6) (a) 0,31 | (8) (a) Cl: 31, 49 | (b) $\text{C}_4\text{H}_8\text{F}_2$ | (17) 11,098 |
| | C: 64, 02 | (13) 5 | |

Vectors

 ESAJK

End of chapter exercises

 ESAJL

- (1) B (3) A
(2) D (4) A

Motion in one dimension

 ESAJM

End of chapter exercises

 ESAJN

- | | | |
|----------------------------|--|--|
| (1) (a) displacement | (4) B | (13) (a) $0,6 \text{ m} \cdot \text{s}^{-1}$ $1,5 \text{ m}$ |
| (b) vector | (5) D | $-0,4 \text{ m} \cdot \text{s}^{-2}$ |
| (c) acceleration | (6) A | (b) $11,5 \text{ s}$ |
| (d) reference point | (7) C | (14) (a) $1,11 \text{ m} \cdot \text{s}^{-1}$ |
| (e) velocity | (8) (a) $30 \text{ m} \cdot \text{s}^{-1}$ | (b) $1,8 \text{ s}$ |
| (f) instantaneous velocity | (b) $7,5 \text{ m} \cdot \text{s}^{-2}$ | (c) $16,67 \text{ m} \cdot \text{s}^{-1}$ |
| | (c) 8 s | (d) 30 m |
| (2) (a) displacement | (d) 300 m | (e) determines safe distance |
| (b) acceleration | (9) D | (f) no |
| (c) velocity | (10) A | (g) 60 m |
| (d) velocity | (11) (a) $-1,75 \text{ m} \cdot \text{s}^{-2}$ | (15) (a) 35 s |
| (3) (a) false | (b) $4,8 \text{ s}$ | (b) $15,71 \text{ m} \cdot \text{s}^{-1}$ |
| (b) true | (12) (a) 72 s | |
| (c) true | (b) $27,8 \text{ m} \cdot \text{s}^{-1}$ | |
| (d) false | (c) $100 \text{ km} \cdot \text{h}^{-1}$ | |

Mechanical energy



End of chapter exercises



- | | | |
|-----------------------------|-----------------|---|
| (1) (a) Gravitational force | (b) false | (b) 0,196 J |
| (b) joules | (c) true | (c) $1,4 \text{ m} \cdot \text{s}^{-1}$ |
| (c) free fall | (d) false | (5) (a) $54,22 \text{ m} \cdot \text{s}^{-1}$ |
| (d) mechanical energy | (e) true | (b) no |
| (e) mass | (3) (a) 11,48 m | (6) $7,7 \text{ m} \cdot \text{s}^{-1}$ |
| (2) (a) false | (4) (a) 0,196 J | |

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